# GRAVINER
# Mk 6 OIL MIST DETECTOR

# INSTALLATION, OPERATION AND MAINTENANCE MANUAL

**59812-120-1**

# TABLE OF CONTENTS

# LIST OF ILLUSTRATIONS

## LIST OF ILLUSTRATIONS (continued)

# REVISION HISTORY

| Revision | Date | Details |
|---|---|---|

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# CHAPTER 1

# DESCRIPTION AND OPERATION

## 1.1 INTRODUCTION

High temperatures, in excess of 200°C that occur on bearing surfaces under initial failure conditions, can lead to a rapid generation of oil vapour. When the hot vapour contacts the relatively cooler atmosphere of the crankcase it condenses into a fine mist, with typical particle sizes of around 0.5 to 5 microns in diameter. When the density of these particles reaches between 30 to 50 mg/l (Milligrams per litre), depending upon the type of oil, an explosive condition exists.

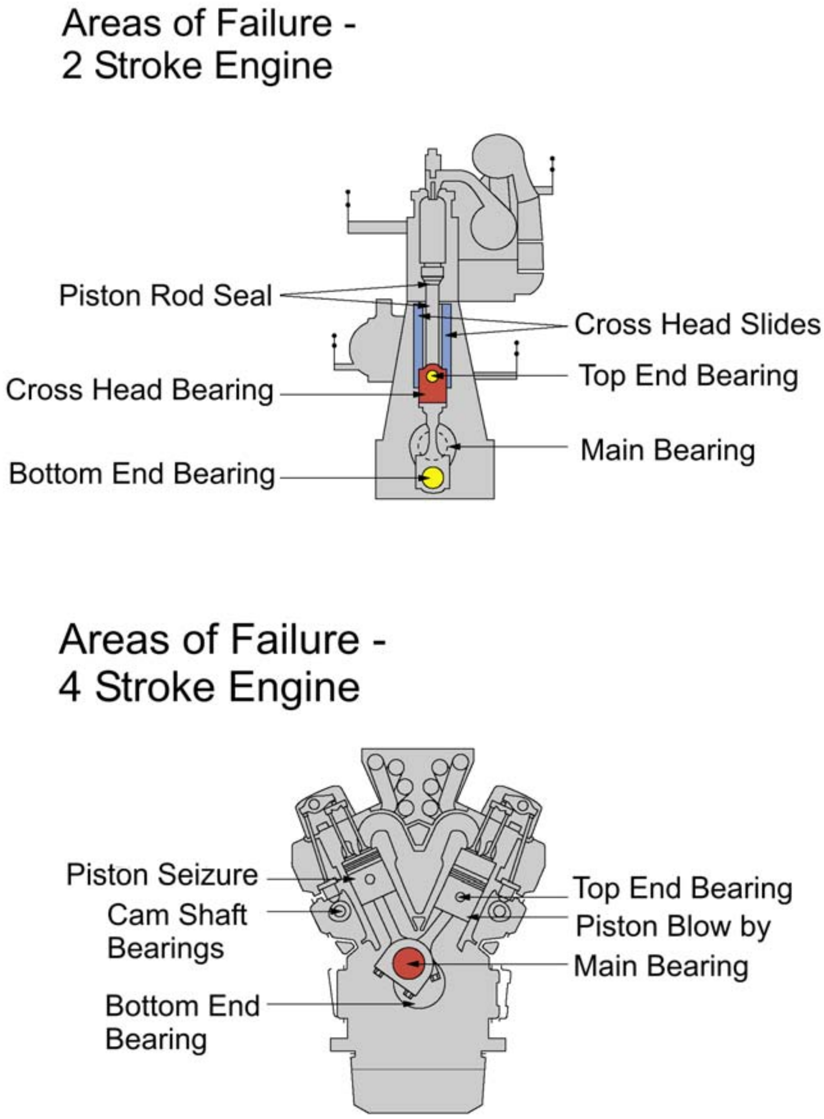



Figure 1: Engine Areas of Failure

A fire or explosion needs three constituents: fuel, oxygen and an ignition source. Remove one of these and no explosion will occur. Similarly, within the crankcase, the three constituents which could cause an explosion are air, oil mist and an ignition source, the "hot spot". Using optical measuring techniques, oil mist density can be measured at levels as low as 0.05 mg/l and give early warning of a rise in oil mist density.

Oil Mist Detection (OMD) techniques have been used to monitor diesel engine crankcases for potential explosive conditions and early detection of bearing failures. The systems available rely mainly on analysing the optical density of oil mist samples drawn from the crankcase compartments, through pipes to the detector. While these systems proved successful in the past, engine design has improved significantly over the years and oil mist detection techniques have improved substantially to maintain adequate protection.

The Graviner Mk 6 OMD provides the following benefits:

Multi engine capability - up to 8 engines on a single system.

Suitable for both 2 stroke and 4 stroke engines.

Elimination of sample pipes - reduced installation costs.

Significant reduction in scanning time - 1.2 seconds for a system of 64 detectors.

Relocating system controls and display to the safety and comfort of control room.

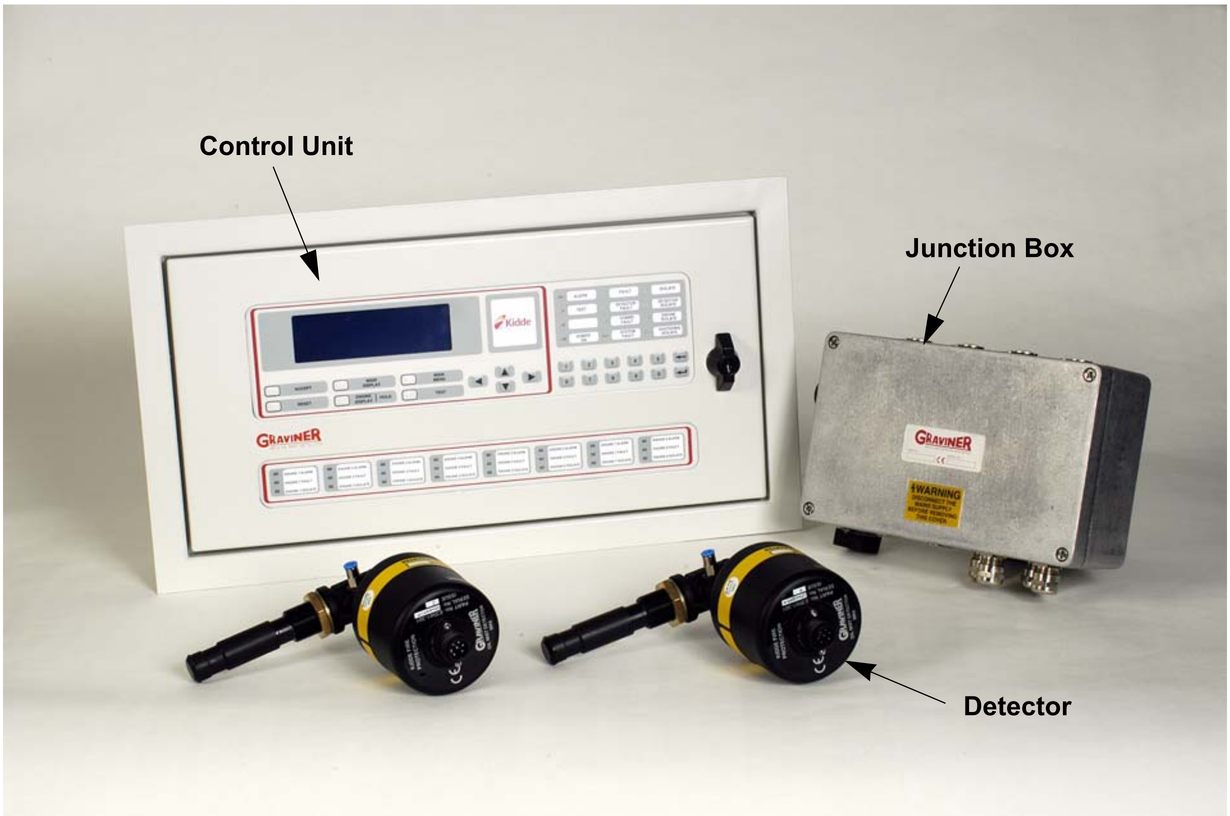



Figure 2: Graviner Mk 6 OMD Main Components

## 1.2 DESCRIPTION

The system comprises three main components (refer to Figure 3):

Detectors
Engine Junction Boxes
Control Unit

The Graviner Mk 6 OMD system can comprise up to 64 detectors directly mounted on the crankcases of up to 8 engines, allowing both main propulsion and auxiliary generators to be monitored at the same time.

Each detector communicates electronically over a serial data link via the engine mounted junction box with the control/display unit which is designed to be mounted within the Engine Control Room. This eliminates the need to enter the machinery space in alarm conditions.

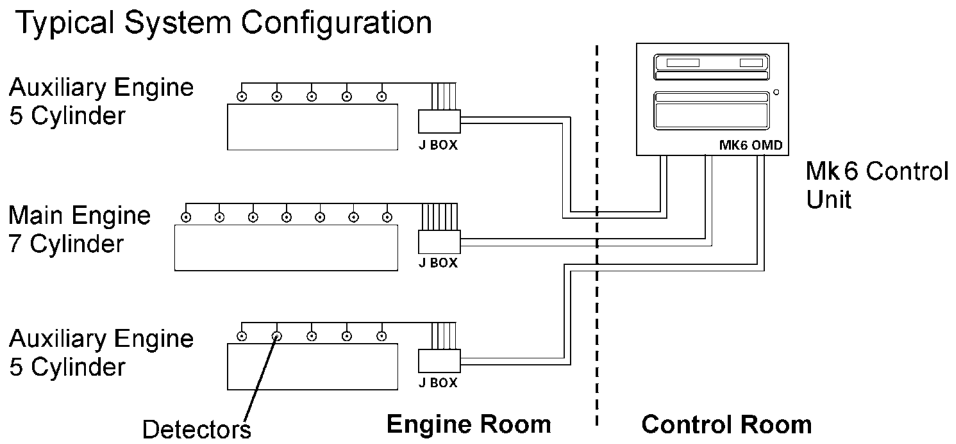



Figure 3: Typical System Configuration

## 1.3 TECHNICAL SPECIFICATION

**Detector**

| | | |
|---|---|---|
| Mounting | ¾ inch BSP | |
| Enclosure Rating | IP65 | |
| Address Switch | 2 x 10 position (0 to 99) | |
| Indicators | Green | Detector On |
| | Red | Alarm |
| | Amber | Detector Fault |

**Electronic Replaceable Micro-Fan**

| | | |
|---|---|---|
| Power Consumption | 2.8 W | |
| Temperature Rating | 0 - 70 ºC | Inlet/Outlet Pipe 0 - 100 ºC |
| Height | 175 mm | |
| Width | 90 mm | |
| Length | 205 mm | |
| Weight | 0.6 kg | |

**Junction Box**

| | | |
|---|---|---|
| Enclosure Rating | IP65 | |
| Max detector inputs | 14 | |
| Fuse Rating | 4.5 A | |
| Power Consumption | Zero | |
| Temperature Rating | 0 - 70 ºC | |
| Dimensions | | |
| Height | 160 mm | (110 mm mounting centres) |
| Width | 260 mm | (240 mm mounting centres) |
| Depth | 90 mm | |
| Weight | 2.3 kg | |

**Control Unit**

| | | |
|---|---|---|
| Enclosure Rating | IP32 | |
| Scanning Time | 1.2 seconds for 64 detectors | |
| Max No. of detectors | 64 | |
| Max No. of engines monitored | 8 | |
| Power Supply | 24 V dc (+30% -25%) | |
| Protection | Self Re-settable fuse inside panel | |
| Insulation | Flash tested to 2000 V for 1 minute | |
| Power Consumption | 5.2 W | |
| Temp Rating | 0 - 55 ºC | |
| Dimensions | | |
| Bulkhead Mounted: | Height | 250 mm (270 mm mounting centres) |
| | Width | 500 mm (445 mm mounting centres) |
| | Depth | 118 mm |
| Panel Mounted: | Height | 309 mm |
| | Width | 559 mm |
| | Depth | 118.5 mm |
| | Weight | 7 kg |
| System Outputs: | Volt-free change over contacts rated at 30V dc1A | |
| Main Alarm | 1 set (energised during normal operation) | |
| Fault Alarm | 1 set (energised during normal operation) | |
| Engine Slow Down | 8 sets, 1 set per engine (De-energised during normal operation) | |
| Alarm Ranges | Average 0.3 mg/l to 1.3 mg/l | |
| | Deviation 0.05 mg/l to 0.5 mg/l | |

## 1.4 OPERATION

The Graviner Mk 6 OMD retains the long established differential measuring system unique to Kidde Fire Protection, which enables high sensitivity to be used while maintaining the maximum false alarm rejection. It still uses optical sensing, but light scatter instead of obscuration. This enables very small detectors to be used. These are rugged and are designed to be engine mounted using standard oil mist detector ports. As they each have their own means of sample acquisition no sample pipes are required. Multiple internal light sources ensure that a single failure will not cause the loss of a detector. Modular construction means that a faulty detector can be replaced in a matter of minutes.

Each detector continually monitors the oil mist density in the crankspace to which it is connected. In addition, it self checks for any internal faults. The control unit interrogates each detector in turn, notes its address and the oil mist density value.

For each engine the average oil mist density is calculated and stored. Each detector signal is then compared in turn with the stored average. A positive difference (the deviation) is then compared with a pre-set, but adjustable reference (the deviation alarm level) for that engine (or detector). If it is greater than the reference a deviation alarm is given.

The stored average level is also compared with a preset reference (the average alarm level) and an average alarm is given if the reference is exceeded.

A full system has a maximum scan time of 1.2 seconds, but with alarm priorities that enable the system to respond if an alarm occurs.

The Control Unit separates the information according to engine groups.

The Control Unit incorporates a Liquid Crystal Display (LCD). The Main Display constantly displays the average oil mist density reading for each engine along with both the average and deviation alarm levels for each engine. It also enables the individual readings of each detector on an engine and the average to be displayed on demand and automatically under alarm conditions.

In the interest of safety, all system controls and alarm displays/outputs are located on the control unit. However to aid fault finding each detector is fitted with 3 indicator lights:

Green - Power on

Red - Alarm

Amber - Fault

Each detector also has an access to its address set switches.

As all detectors operate independently, the loss of one by either failure or the need to clean does not affect the operation of the rest of the system. Individual detectors, or engine groups, can be isolated from the rest of the system for maintenance while the rest of the system remains in operation.

## 1.5 SYSTEM CONTROLS AND DISPLAYS

The software is menu driven and provides a logical route to all functions. It has three operating levels:

User
Engineer
Service

The User level is essentially for read only interrogation only and does not allow any adjustments to be made to alarm settings or system configuration.

The Engineer level is password protected and allows access to most functions and the full range of settings. When selected, a prompt for a password will appear. Enter 012345 then press the ↵ key.

The Service level is also password protected (different from the engineer menu) and allows access to all functions. This is only available to authorised Kidde Fire Protection personnel and authorised service agents.

1.5.1 Controls

**THE ACCEPT KEY** resets the main alarm relay, silences the internal sounder. The display will then enter the individual engine display for the engine(s) in alarm. If there is more than one engine in alarm the display will scroll between engines.

**THE HOLD KEY** holds the display for 15 seconds.

**THE RESET KEY** resets all alarms, faults and returns the system to normal.

**THE TEST KEY** enters the test menu at User level functions only.

The **ENGINE DISPLAY** control calls up the individual engine cylinder readings and the average. It also displays the deviation and average alarm settings. The cursor keys allow each engine in the system to be displayed in turn.

The **MAIN DISPLAY** key always returns the display to this page from anywhere in the software.

The **MAIN MENU** key allows access to all the **USER**, **ENGINEER** and **SERVICE** menus. At the bottom of each displayed page the active navigation keys for that page are shown.

The **MAIN DISPLAY** and **MAIN MENU** keys allow fast return to the normal display or the main menu from anywhere in the software. If this function is carried out from a password protected area ie **ENGINEER** or **SERVICE** menu, then a **RESET** must also be carried out.

The keypad keys are:

| | |
|---|---|
| ▼ | Cursor down |
| ▲ | Cursor up |
| ◀ | Cursor left |
| ▶ | Cursor right |
| ← | Quit page |
| ↵ | Enter |

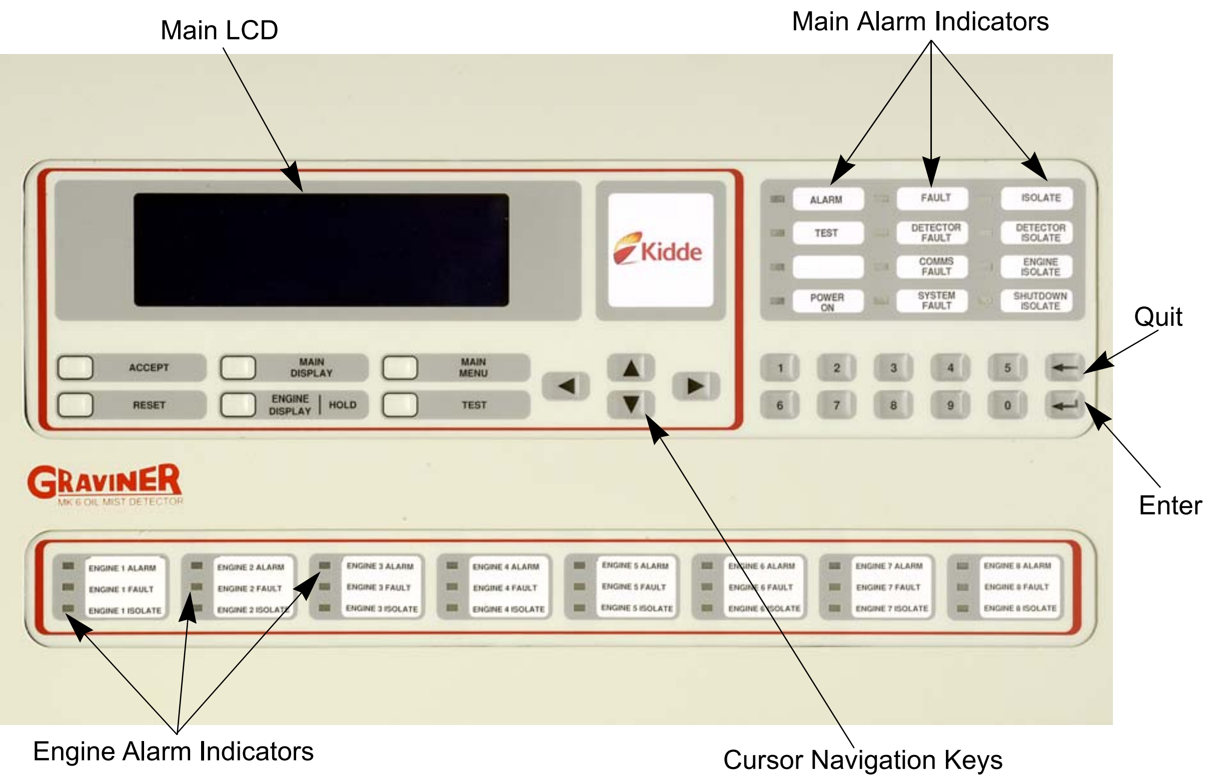



Figure 4: Panel Display

1.5.2 Operation of the navigation keys is accompanied by an audio signal.

1.5.3 Main LCD Display

This provides visual access to all the data required to operate the system and displays the software pages for system configuration and fault analysis. On the main display and the engine display, the left hand vertical scale shows oil mist density in mg/l. In addition the main display shows the average alarm setting for each engine.

On the engine display, both the deviation alarm(s) and the average alarm settings are displayed. Under normal operating conditions, the main display page shows the average oil mist density for all engines and the relevant average alarm settings. It also shows the time, the date and **NORMAL**.

1.5.4 Engine Alarm Indicators

Each of the eight engine alarm indicator sets show the status of that engine ie. **ALARM**, **FAULT** and **ISOLATE**. This display is designed to be a backup to the main LCD.

1.5.5 Main Alarm Indicators

The light display consists of alarm indicators for all individual alarm, fault and isolate conditions. Its function is to provide back up indication in the unlikely event of the loss of the main LCD display.

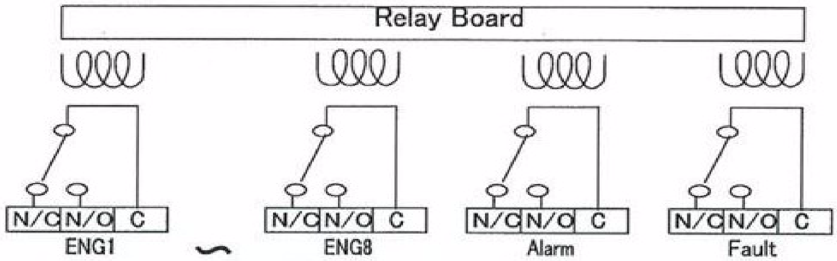
1.5.6 Alarm Outputs

The Control Unit has the following relay outputs:

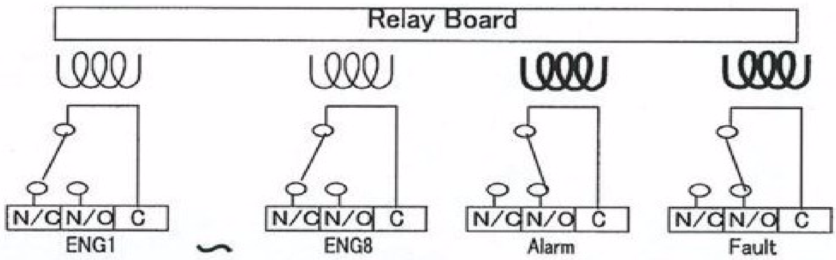
| | |
|---|---|
| Main Alarm. | 1 set of volt free changeover contacts. |
| | Operate on any Average or Deviation alarm from any of the 8 possible engines. |
| Engine Slow down. | 8 sets, 1 per engine. |
| | Operation when the correlating engine has either on Average alarm or Deviation Alarm. |
| Fault Alarm. | 1 Set of volt free contacts. |
| | Operate on any system fault. |

All relays are rated 30 V dc 1 Amp. All relays can be tested for operation in the Test Menu.

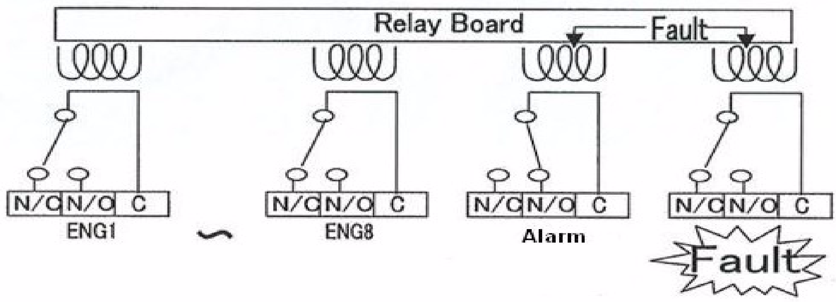
When Power
Source is OFF




Power is On and
in normal
Monitoring Mode




Power is On and
Fault was
detected




Power is On and
Engine 1 Oil Mist
Alarm was
detected

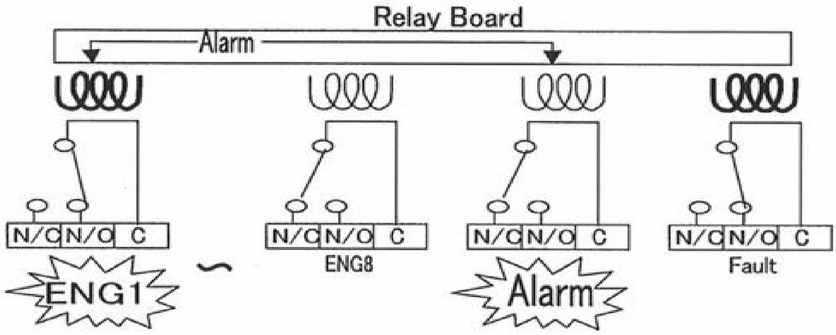



Figure 5: Relay Function Modes

# CHAPTER 2

# INSTALLATION AND COMMISSIONING

## 2.1 INSTALLATION

### 2.1.1 Control Unit

The Control Unit is designed for either bulkhead or flush mounting, and must be installed in a control room or similar environment.

For bulkhead mounting fix to a rigid structure using the four M6 mounting flanges at the rear of the unit.

For flush mounting a bezel, part number 35100-K187 can be supplied (refer to Figure 8).

The position of the control unit must be sited for optimum visibility of the display. Sufficient space must be left around the control unit to allow the fitting of glands and routing of the cables, and to facilitate easy access to all aspects of the control unit. A minimum of 500 mm must be allowed at the front of the control unit to allow the door to be opened.

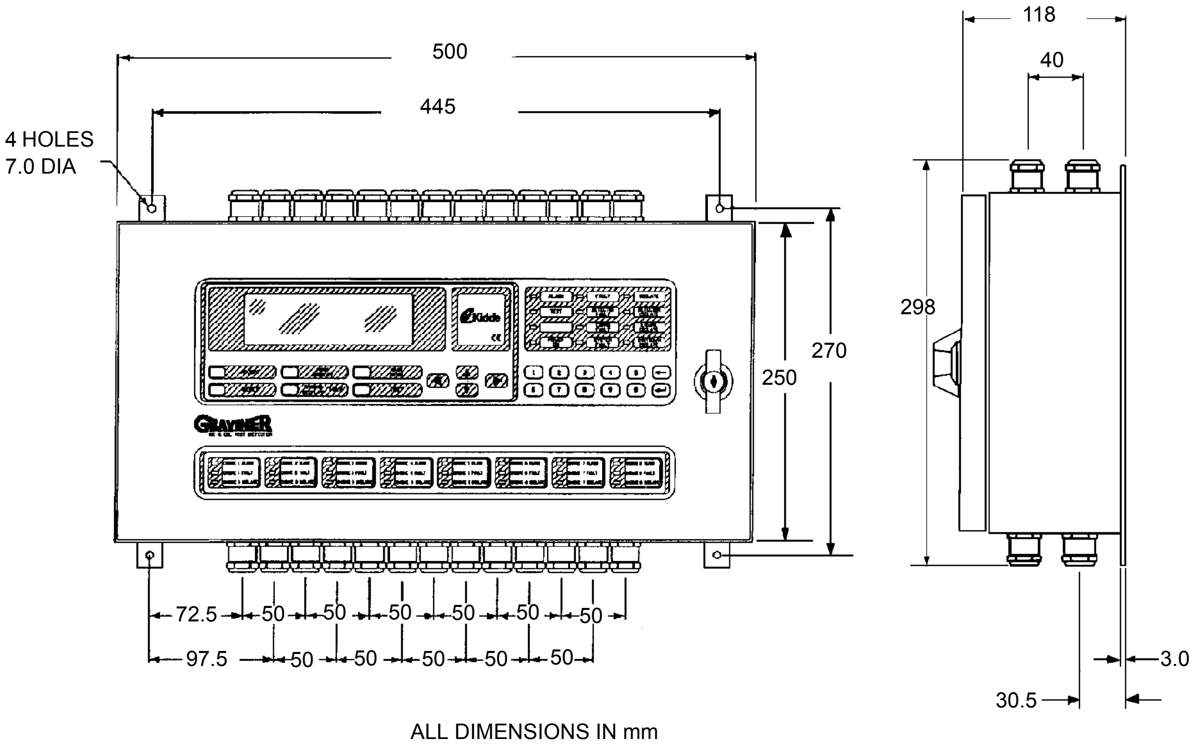
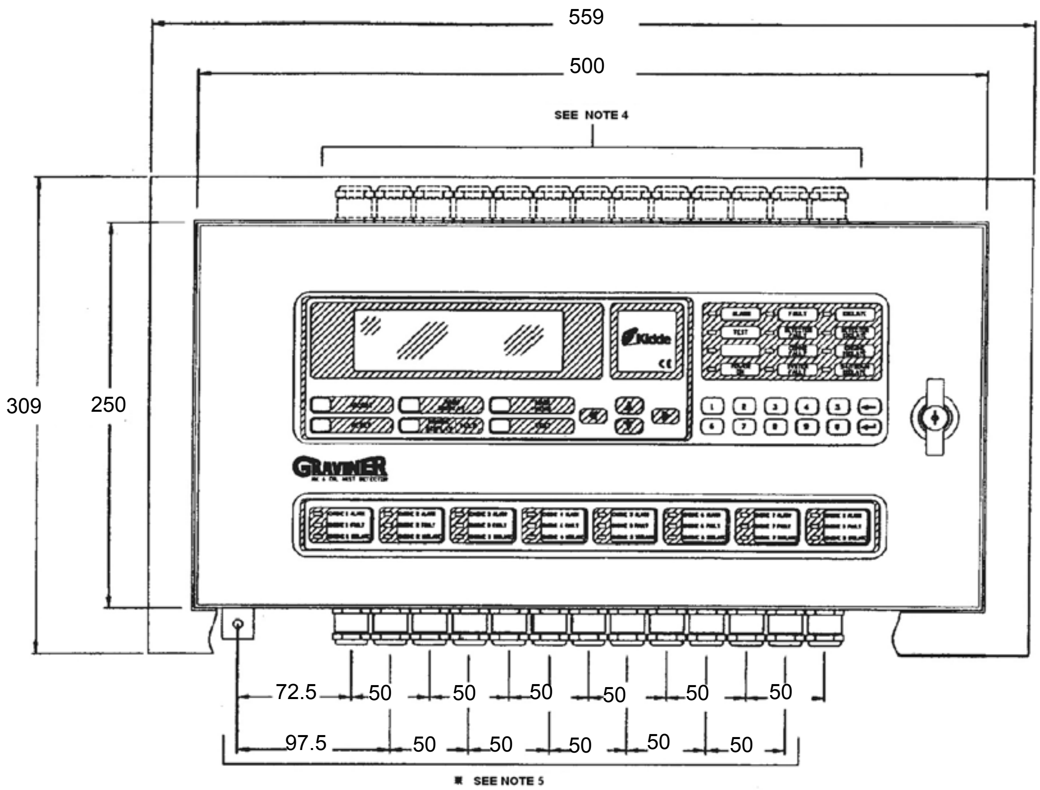
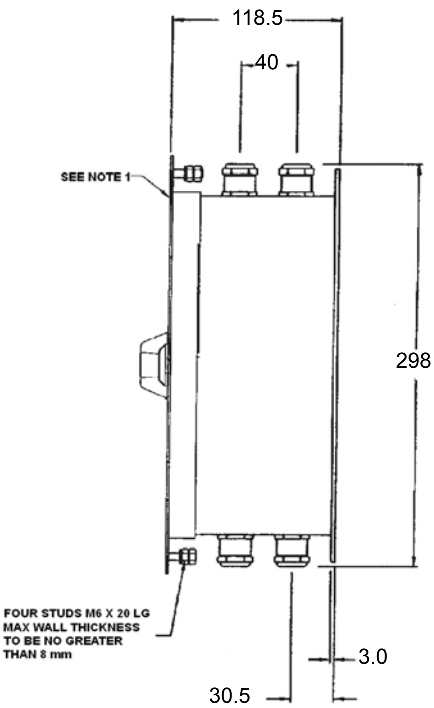



Figure 6: Installation Drawing for Graviner Mk 6 OMD Control Unit Bulkhead Mounting

ALL DIMENSIONS IN mm




| SYSTEM INSTALLATION REQUIREMENTS TABLE | | | |
|---|---|---|---|
| NUMBER OF ENGINES | NUMBER OF JUNCTION BOXES | NUMBER OF RELAY OUTPUTS (SEE NOTE 3) | TOTAL NUMBER OF M20 CABLE GLANDS/LOCKNUTS REQUIRED. |
| 1 | 1 | 3 | 6 |
| 2 | 2 | 4 | 9 |
| 3 | 3 | 5 | 12 |
| 4 | 4 | 6 | 15 |
| 5 | 5 | 7 | 18 |
| 6 | 6 | 8 | 21 |
| 7 | 7 | 9 | 24 |
| 8 | 8 | 10 | 27 |

NOTES

1. IN ORDER FOR THE DOOR TO OPEN FULLY, AN ADDITIONAL 500mm MUST BE ALLOWED BEYOND THIS POINT.
2. SEE TABLE FOR THE NUMBER OF JUNCTION BOXES, RELAY OUTPUTS (ALARM, FAULT AND ENGINE SLOWDOWN) AND CABLE GLANDS/LOCKNUTS.
3. ON A ONE ENGINE SYSTEM ONE ALARM, ONE FAULT AND ONE ENGINE SLOW DOWN RELAY IS REQUIRED. IN CASES WHERE THERE ARE MORE ENGINES AN ADDITIONAL ENGINE SLOWDOWN RELAY WILL BE NEEDED FOR EACH ADDITIONAL ENGINE.
4. PROVISION HAS BEEN MADE FOR 15 CABLE GLAND ENTRIES AT THE TOP OF THE ENCLOSURE AND 15 AT THE BOTTOM. THESE ARE TO BE USED FOR POWER AND COMMUNICATIONS WIRING. THE SUGGESTED MANUFACTURER OF THE CABLE GLAND AND LOCKNUT IS JACOB. THE JACOB PART NUMBERS FOR THESE PARTS ARE 50620MREMVL AND 50220MPOT RESPECTIVELY. THESE EXTERNAL REFERENCE PART NUMBERS ARE FOR REFERENCE ONLY AND NEED NOT BE PURCHASED FROM THE SUGGESTED MANUFACTURER.
5. DIMENSIONS SHOWN FOR MAXIMUM NUMBER OF GLANDS, * INDICATES DIMENSION ROW FOR FRONT GLANDS ON BOX.

Figure 7: Installation Drawing for Graviner Mk 6 OMD Control Unit Flush Mount

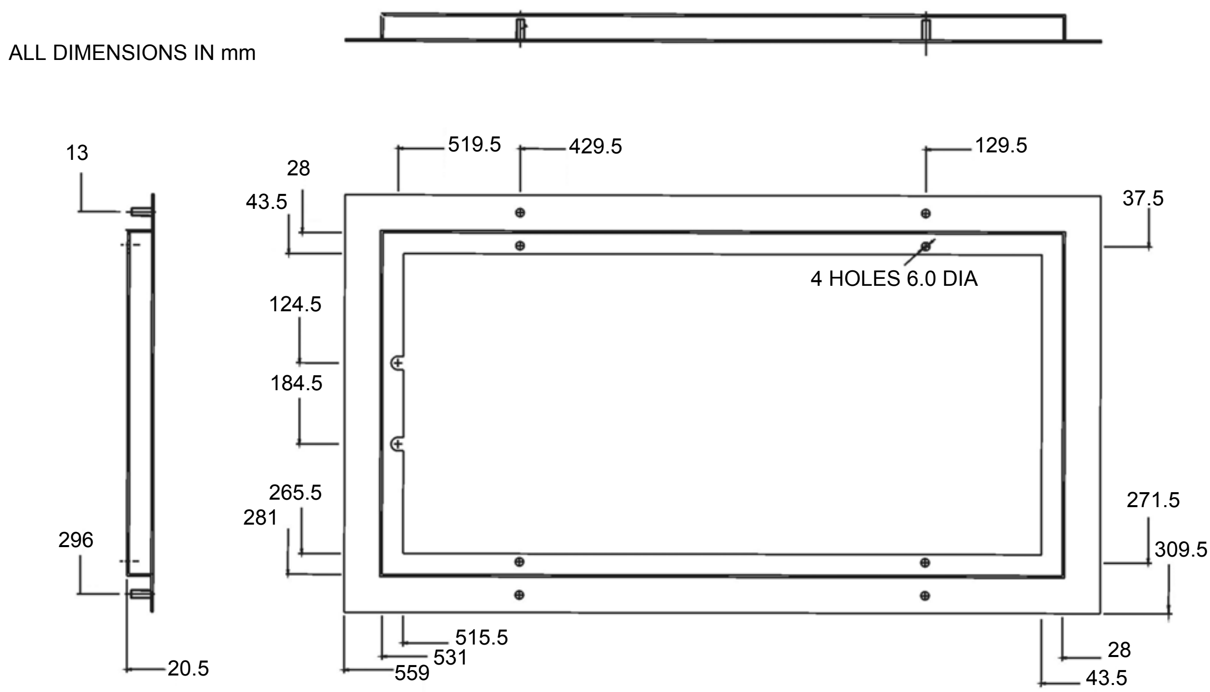
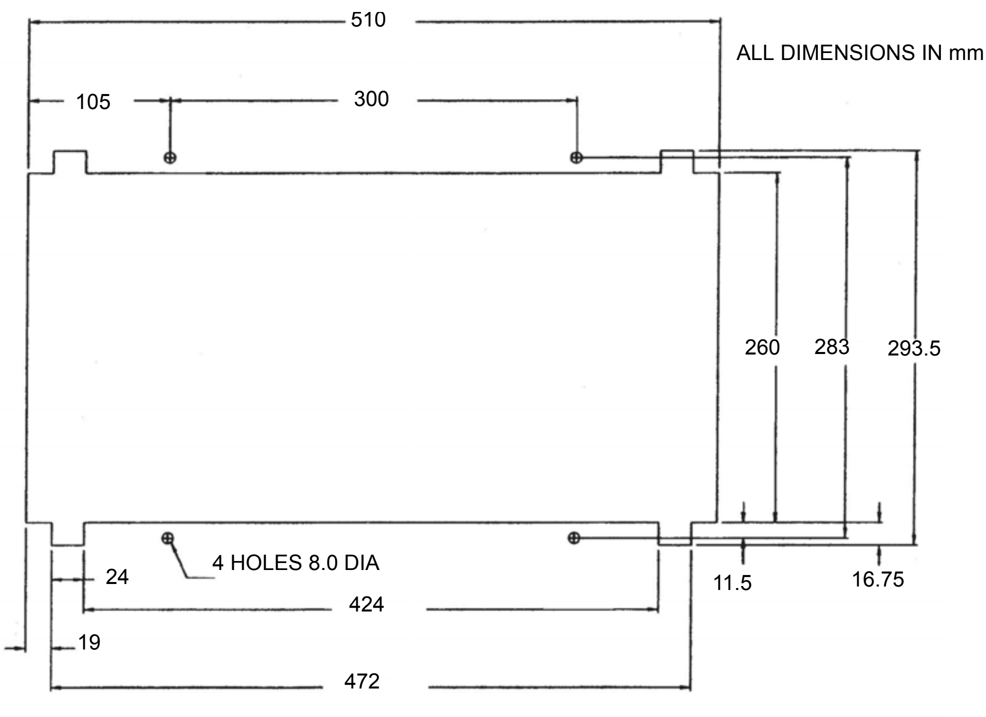



Figure 8: Bezel 35100-K187




NOTE: CUTOUT TO FIT BEZEL 35100-K187

Figure 9: Installation Drawing for Graviner Mk 6 OMD Control Unit Flush Mount Panel Cut Out Details

2.1.2 Control Unit Wiring

The +24 V dc and 0 V dc power input cables for the Control Unit should be terminated onto terminal block TB5.

Note the earth wire must be connected onto the earth stud on the side of the control box. See Figure 11.

Note all terminations should be made using crimped wires.

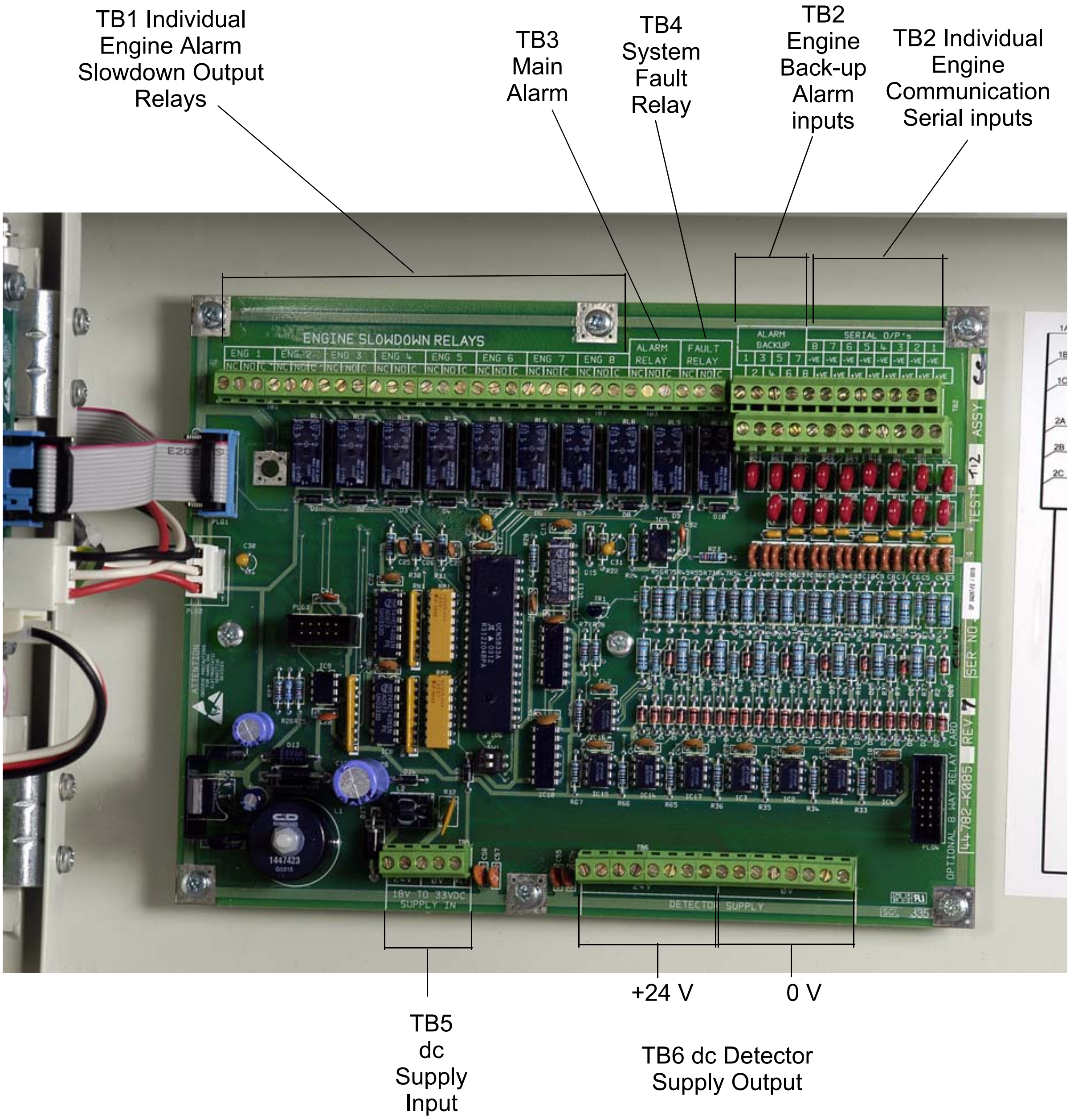



Figure 10: Control Unit Interface Board

Earth Stud

Figure 11: Power Input Cable Earth Stud

2.1.3 Communication & Power Cable from Control Unit

The cable from the Control Unit to each junction box should be of the following types:

Power Cable:

2 Cores + Earth, CSA 2.5 $mm^2$ (50/0.25 mm), flexible stranded bare copper conductors, low smoke halogen-free insulation, cores laid up, braided screen, low smoke halogen-free sheath - grey, outside diameter 9.8 mm, operating temperature 0 ºC to + 80 ºC

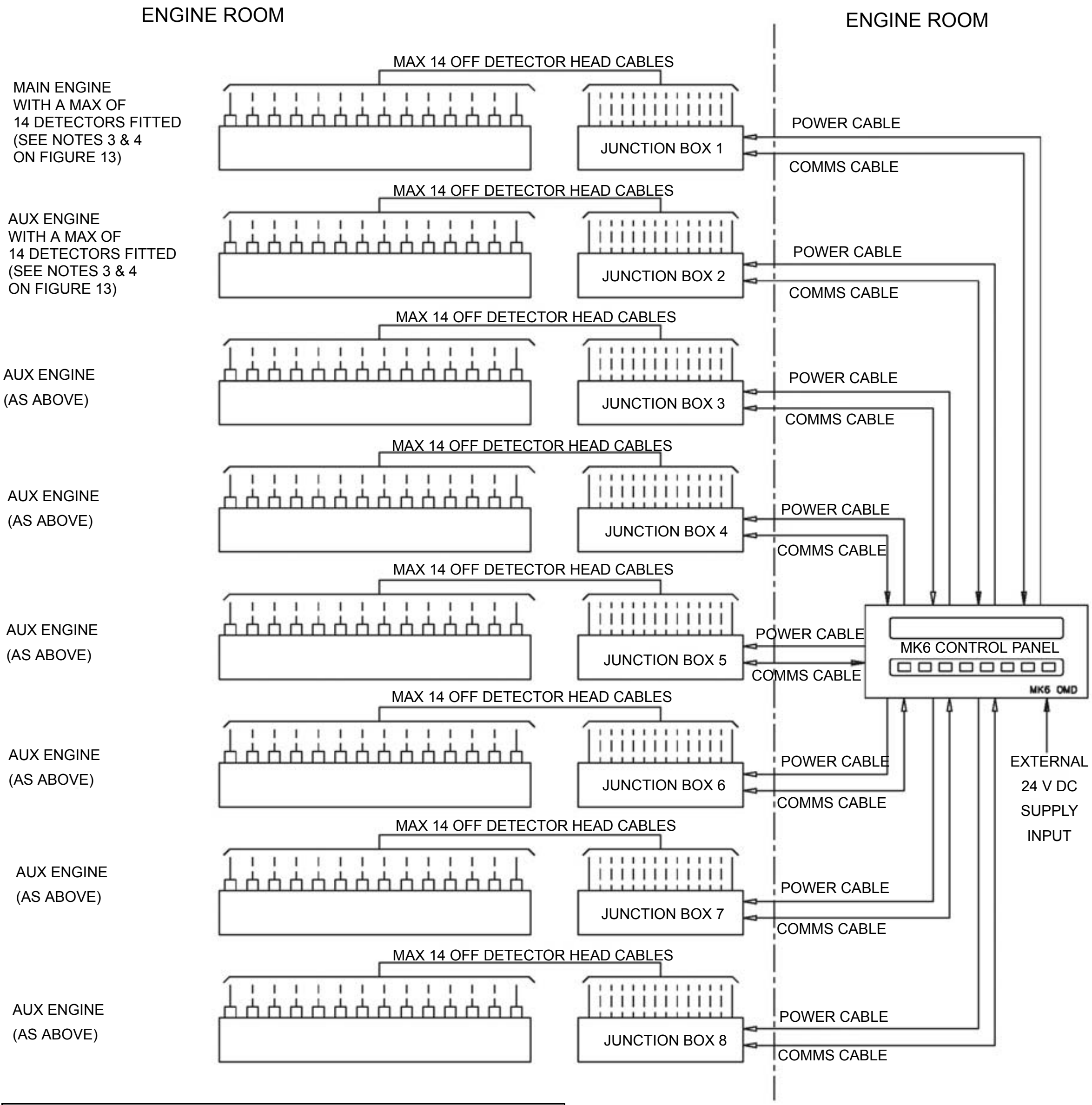
Comms Cable:

2 Twisted low capacitance pairs, 24(7) AWG tinned copper conductors, foam polyolefin insulation, each pair foil screened with a tinned copper drain wire, low smoke halogen free sheath - grey, outside diameter 8.1 mm, operating temperature 0 ºC to 70 ºC

Supplier - FS cables

Order code 2402PIFFH or similar.

Cable screens must be connected to the metal glands of the Control Unit and terminals of TB9 within the junction boxes, refer to Figure 24.




| EQUIPMENT LIST | |
|---|---|
| **PART No.** | **DETECTOR** |
| E3561-301 | MK 6 OMD DETECTOR HEAD |
| 53856-K170 | MK 6 CONTROL PANEL |
| D4720-001-01-16 | MK 6 JUNCTION BOX |
| 43682-K108-00 | CABLE ASSY SKT - 5 m , LONG, DETECTOR HEAD |
| 43682-K108-01 | CABLE ASSY SKT - 7.5 m , LONG, DETECTOR HEAD |
| 43682-K108-02 | CABLE ASSY SKT - 10 m , LONG, DETECTOR HEAD |
| 43682-K108-03 | CABLE ASSY SKT - 12.5 m , LONG, DETECTOR HEAD |
| 43682-K108-04 | CABLE ASSY SKT - 15 m , LONG, DETECTOR HEAD |
| 43682-K108-05 | CABLE ASSY SKT - 17.5 m , LONG, DETECTOR HEAD |
| 43682-K108-06 | CABLE ASSY SKT - 20 m , LONG, DETECTOR HEAD |
| 43682-K108-07 | CABLE ASSY SKT - 22.5 m , LONG, DETECTOR HEAD |
| 43682-K108-08 | CABLE ASSY SKT - 25 m , LONG, DETECTOR HEAD |
| 43682-K109-00 | CABLE ASSY 90 DEG SKT - 5 m , LONG, DET HEAD |
| 43682-K109-01 | CABLE ASSY 90 DEG SKT - 7.5 m , LONG, DET HEAD |
| 43682-K109-02 | CABLE ASSY 90 DEG SKT - 10 m , LONG, DET HEAD |
| 43682-K109-03 | CABLE ASSY 90 DEG SKT - 12.5 m , LONG, DET HEAD |
| 43682-K109-04 | CABLE ASSY 90 DEG SKT - 15 m , LONG, DET HEAD |
| 43682-K109-05 | CABLE ASSY 90 DEG SKT - 17.5 m , LONG, DET HEAD |
| 43682-K109-06 | CABLE ASSY 90 DEG SKT - 20 m , LONG, DET HEAD |
| 43682-K109-07 | CABLE ASSY 90 DEG SKT - 22.5 m , LONG, DET HEAD |
| 43682-K109-08 | CABLE ASSY 90 DEG SKT - 25 m , LONG, DET HEAD |

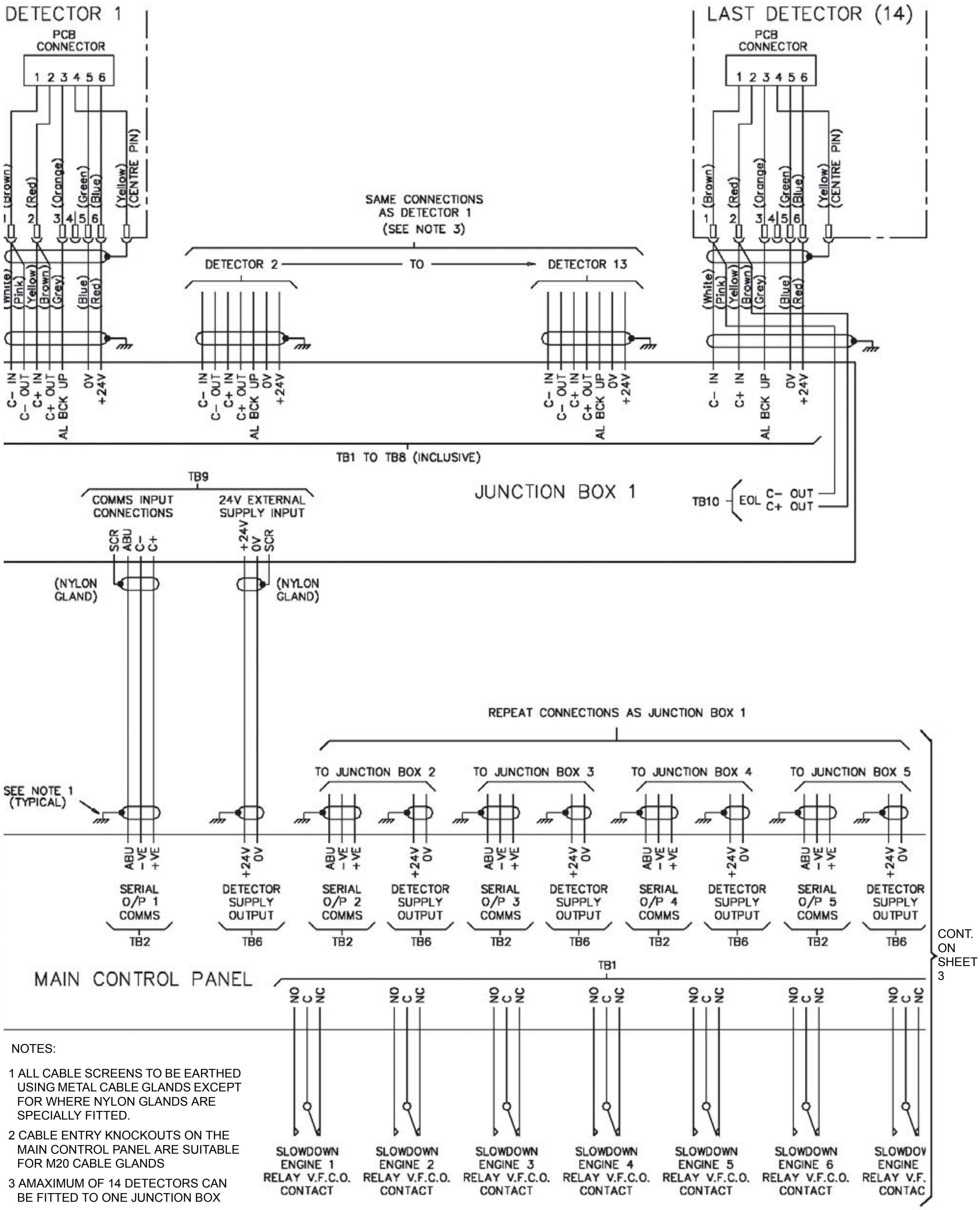
POWER CABLE

2 CORES + EARTH, CSA 25 mm$^2$ (50/0.25 MM)
FLEXIBLE STRANDED BARE COPPER CONDUCTORS,
LOW SMOKE HALOGEN FREE INSULATION,
CORES LAID UP, ALUMINIUM/POLYESTER FOIL
PLUS A TINNED COPPER BRAID.
LOW SMOKE HALOGEN FREE SHEATH - GREY
OUTSIDE DIAMETER 9.8 mm.
OPERATING TEMPERATURE 0 TO 80 $^o$C
SUPPLIER - FS CABLES, ORDER CODE 42302503 OR SIMILAR.

COMMS CABLE

2 TWISTED LOW CAPACITANCE PAIRS 24 (7) AWG
TINNED COPPER CONDUCTORS.
FOAM POLYOLEFIN INSULATION, EACH PAIRFOIL SCREENED
WITH A TINNED COPPER DRAIN WIRE.
LOW SMOKE HALOGEN FREE SHEATH - GREY
OUTSIDE DIAMETER 8.1 mm.
OPERATING TEMPERATURE 0 TO 70 $^o$C
SUPPLIER - FS CABLES, ORDER CODE 2402PFFH OR SIMILAR.

Figure 12 (Sheet 1 of 3): Graviner Mk 6 OMD System E9261-002




NOTES:

1 ALL CABLE SCREENS TO BE EARTHED USING METAL CABLE GLANDS EXCEPT FOR WHERE NYLON GLANDS ARE SPECIALLY FITTED.

2 CABLE ENTRY KNOCKOUTS ON THE MAIN CONTROL PANEL ARE SUITABLE FOR M20 CABLE GLANDS

3 AMAXIMUM OF 14 DETECTORS CAN BE FITTED TO ONE JUNCTION BOX

4 A MAXIMIUM TOTAL OF 64 DETECTORS CAN BE CONNECTED OVER EIGHT JUNCTION BOXES FOR ONE CONTROL PANEL

5 LAST DETECTOR ON EACH ENGINE MUST HAVE END OF LINE (EOL) CONNECTIONS TERMINATED ON TO TB10

Figure 12 (Sheet 2 of 3): Graviner Mk 6 OMD System E9261-002

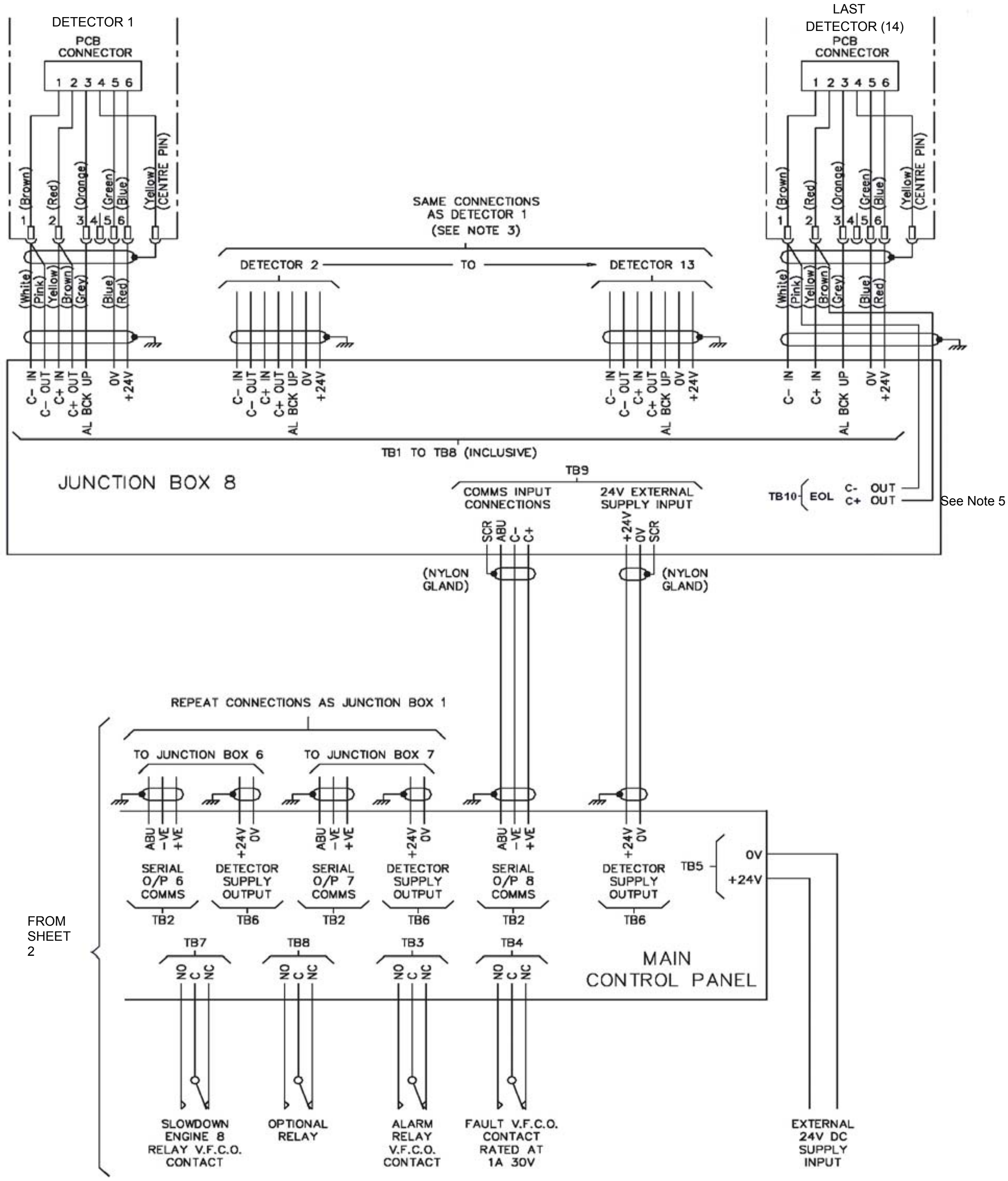



Figure 12 (Sheet 3 of 3): Graviner Mk 6 OMD System E9261-002

2.1.4 Detector

Each detector is mounted to an individual crankcase via a ¾ inch BSP threaded hole.

Ensure all detectors fitted to the engine are locked tightly in place by means of the lock nut supplied.

The detector should be located at the upper part of the crankcase wall NOT in the direct line of the oil throw. On smaller engines it is permissible to mount the detector on the crankcase door if desired or as installation dictates.

The detector should be mounted such that any oil throw comes from above the sample pipe.

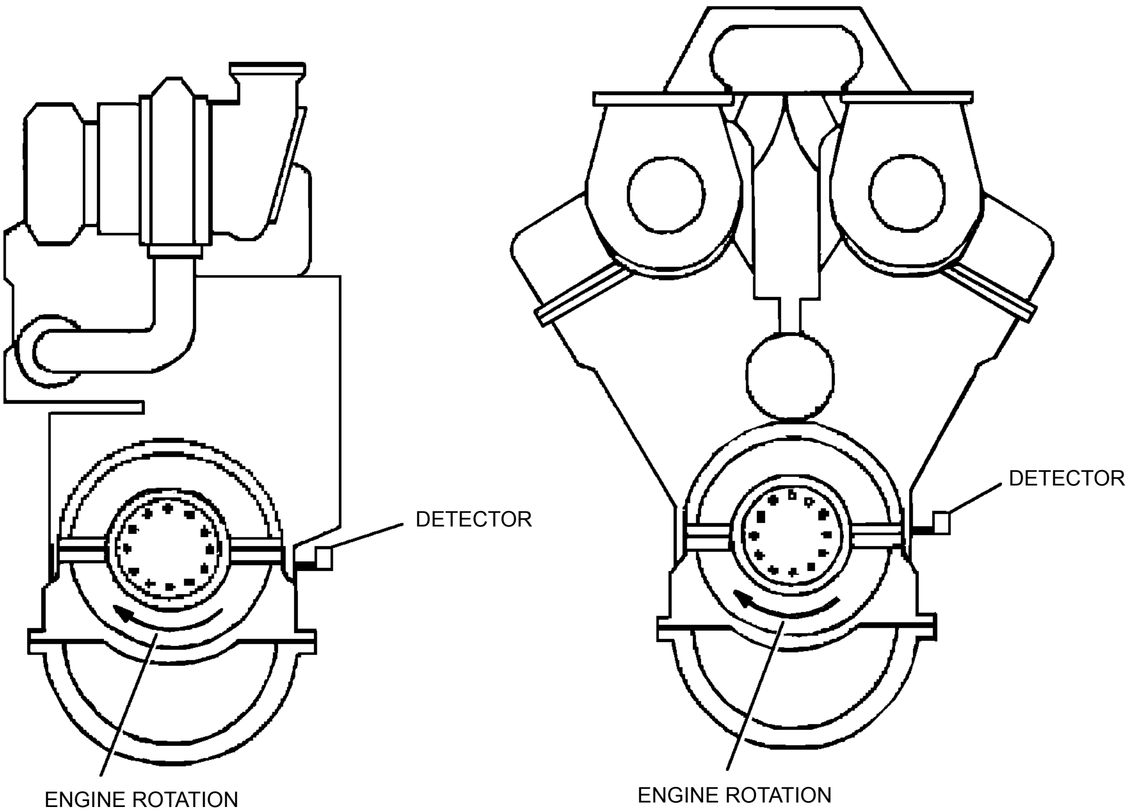



Figure 13: Typical Mounting Position

The detector must be fitted at a maximum of plus or minus 20 degrees from the vertical. Horizontally the detector must be mounted level or with the detector body inclined towards the engine to ensure oil drainage. Refer to Figure 13.

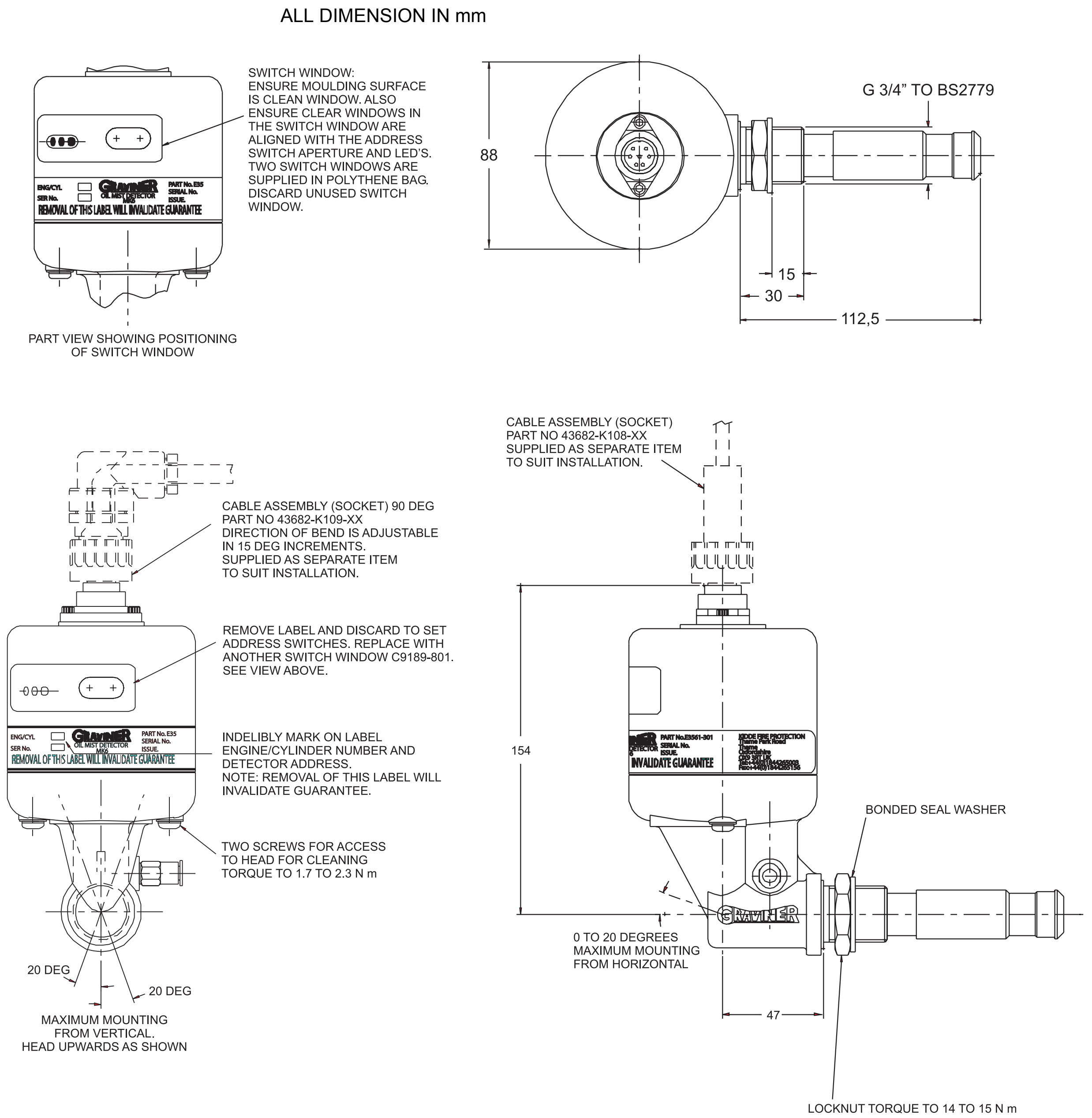



Figure 14: Graviner Mk 6 OMD Detector Head Installation E3561-301

Setting Detector Address

Correct operation of the system depends on all detector heads being correctly addressed. This is carried out after installation (refer to Figure 14).

1. Remove the adhesive label covering the access port to the address switches.

2. Use an instrument screwdriver to set the switches.

3. The left-hand switch sets the TENS, the right hand switch sets the UNITS.

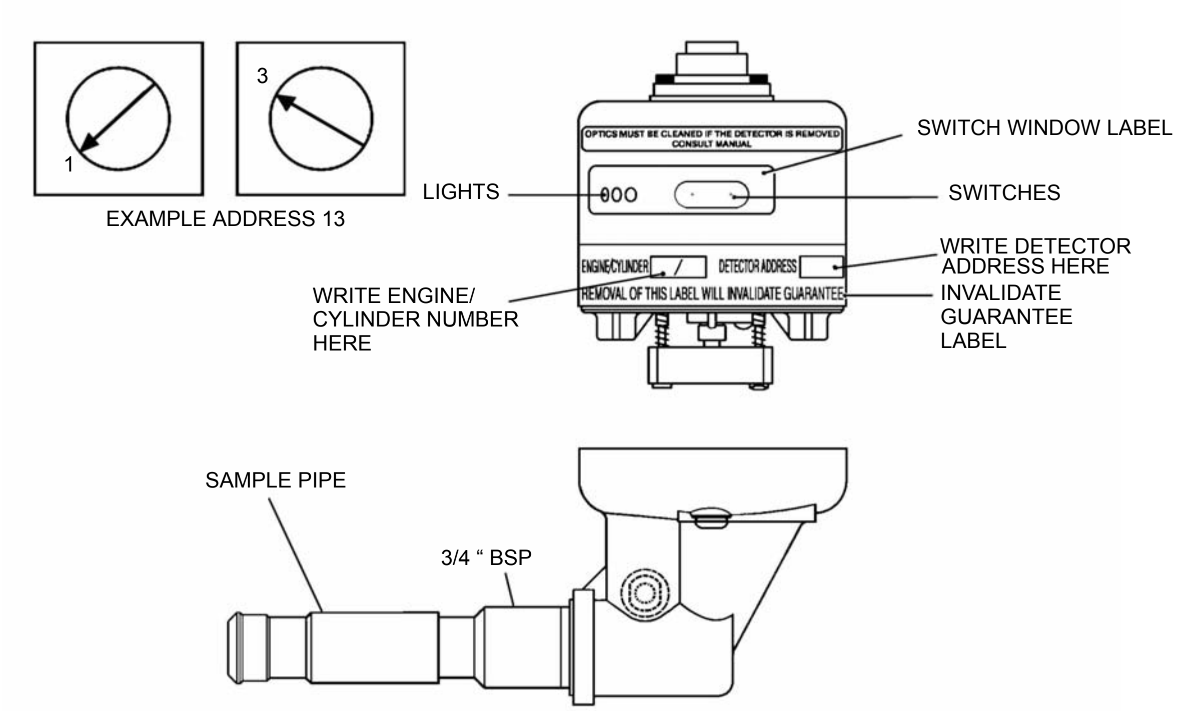



Figure 15: Detector Head E3561-301

4. Detectors are supplied with the switches set to 00 as factory default.

5. The detector addresses must be sequential and should run in sequence from engine to engine, ie if the last detector head on the first engine is address 08, then the first detector head on the second engine must be 09. **It is essential that if detector heads are removed for overhaul they are returned to their original position or they are re-addressed**.

6. Clean the detector head in the area around the address switches and indicator lights with wet and dry wipes to ensure any oil or grease is removed. Attach the switch window label so that both the switches and the indicator lights are visible.

7. The engine/cylinder number and address should be written on the invalidate guarantee label in the position shown.

2.1.5 Detector Cables Assemblies

Each detector is then connected via a straight or 90º bend cable assembly to its relevant junction box.

Ensure all cable screens are properly made off with the junction box glands.

It is recommended that all cable terminations are made using crimp terminals.

| Cable Length m | Straight Connector Part No. | 90º Connector Part No. |
|---|---|---|
| 2.5 | 43682-K108 | 43682-K109 |
| 5.0 | 43682-K108-00 | 43682-K109-00 |
| 7.5 | 43682-K108-01 | 43682-K109-01 |
| 10.0 | 43682-K108-02 | 43682-K109-02 |
| 12.5 | 43682-K108-03 | 43682-K109-03 |
| 15.0 | 43682-K108-04 | 43682-K109-04 |
| 17.5 | 43682-K108-05 | 43682-K109-05 |
| 20.0 | 43682-K108-06 | 43682-K109-06 |
| 22.5 | 43682-K108-07 | 43682-K109-07 |
| 25.0 | 43682-K108-08 | 43682-K109-08 |

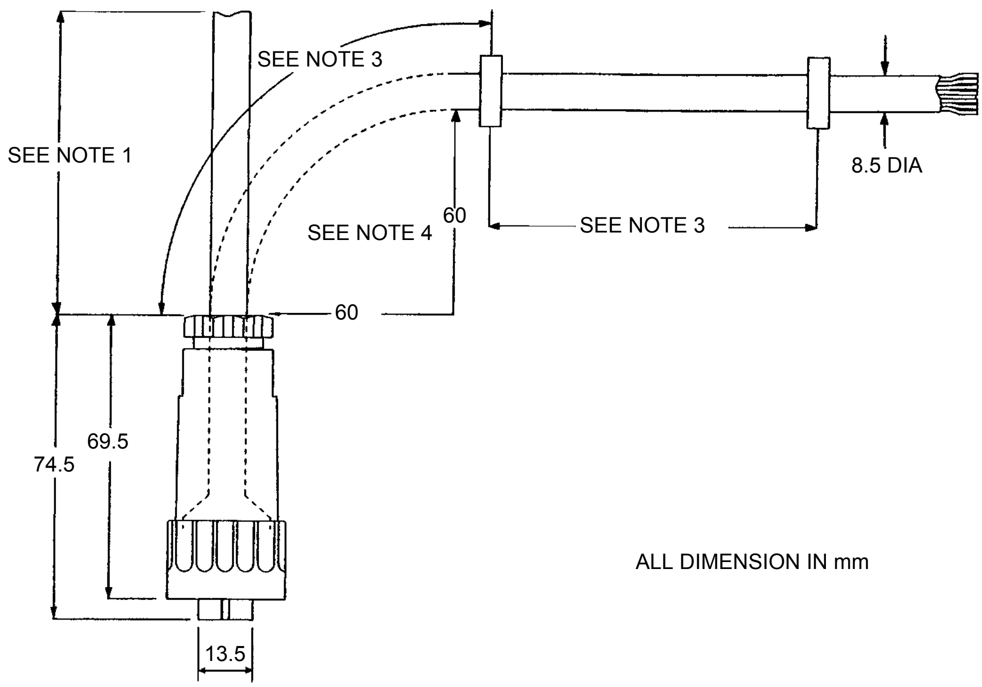



Figure 16: Detector Cable Assemblies

Notes:

1. Cable Spec 8-core screened (90 $^{o}$C) Halogen free & oil resistant.
2. Cable lengths are 'straight lengths'.
3. Clips to be secured every 0.5 m (clips are not supplied by Kidde Fire Protection).
4. Do not exceed the minimum bend radius as shown.

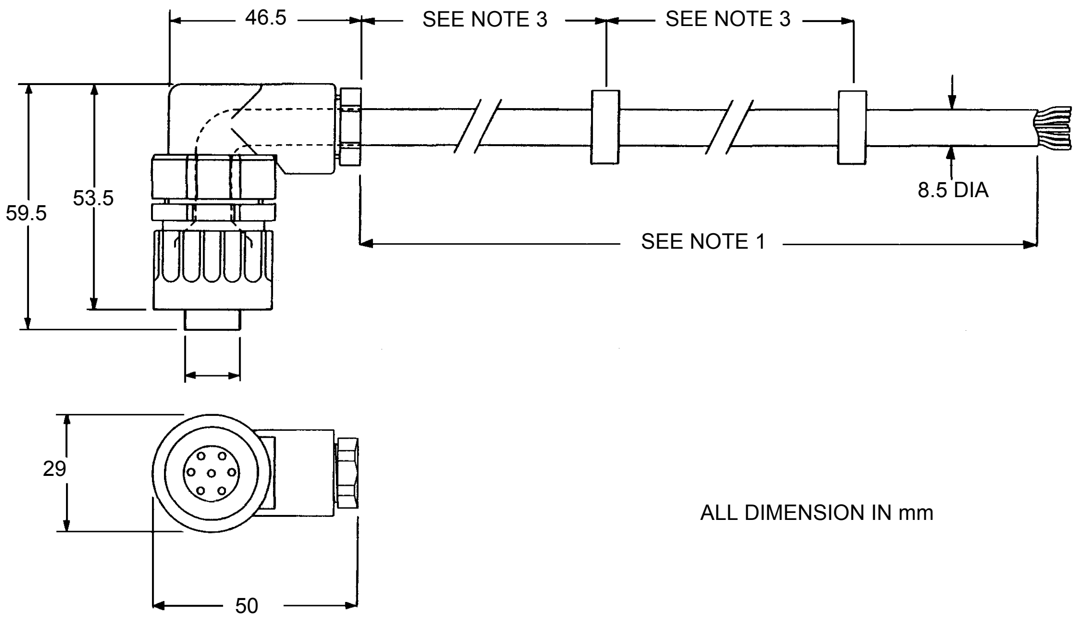



Figure 17: Detector Cable Assemblies

Pin Out Table

| Connector Pin No. | Colour of Wire | Function | Last Detector Function |
|---|---|---|---|
| 1 | Pink | C -out | TB10 EOL - |
| 1 | White | C -in | C -in |
| 2 | Yellow | C +in | C +in |
| 2 | Brown | C +out | TB10 EOL + |
| 3 | Grey | Alarm Backup | Alarm Back-up |
| 4 | Not used | Not used | Not used |
| 5 | Blue | 0 V | 0 V |
| 6 | Red | +24 V dc | +24 V |
| Centre | Screen | | |

Figure 18: Junction Box Cable Gland Components

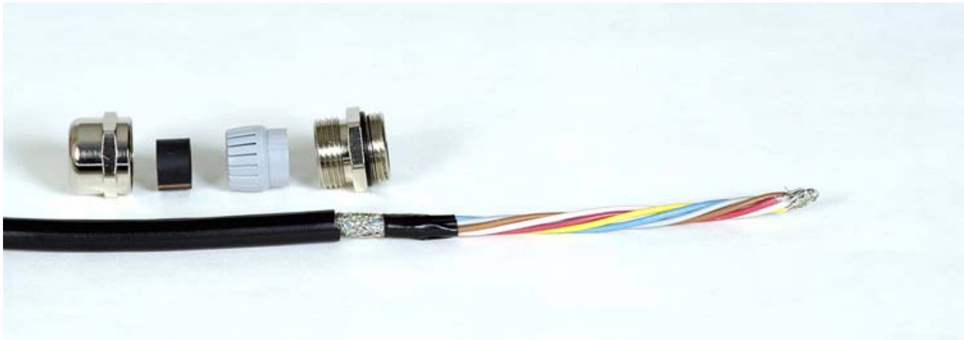


Figure 19: Junction Box Cable Gland Components with Detector Cable

Figure 20: Junction Box Cable Gland Components in Order of Fitting to Cable

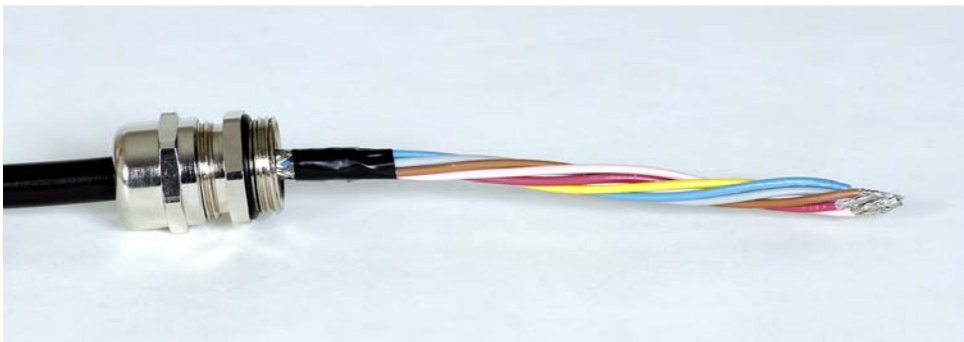


Figure 21: Junction Box Cable Gland Components Fitted to Detector Cable and Screen Made Off

### 2.1.6 Junction Box

The Junction Box is designed for on-engine mounting and it is recommended that the Box is installed as near to the centre of the engine as possible to minimise detector cable lengths. Mounting is via the four M6 locating holes in the box. Sufficient space must be left around the Junction Box to allow access to the cable glands and the routing of the cables and to facilitate easy access to all aspects of the Junction Box.

| No of entries | Junction Box Part No. with 20mm output glands | Junction Box Part No. with 25mm output glands | Junction Box Part No. vertically mounted |
|---|---|---|---|
| 1 | D4720-001-01 | 53836-K224-01 | 53836-K233-01 |
| 2 | D4720-001-02 | 53836-K224-02 | 53836-K233-02 |
| 3 | D4720-001-03 | 53836-K224-03 | 53836-K233-03 |
| 4 | D4720-001-04 | 53836-K224-04 | 53836-K233-04 |
| 5 | D4720-001-05 | 53836-K224-05 | 53836-K233-05 |
| 6 | D4720-001-06 | 53836-K224-06 | 53836-K233-06 |
| 7 | D4720-001-07 | 53836-K224-07 | 53836-K233-07 |
| 8 | D4720-001-08 | 53836-K224-08 | 53836-K233-08 |
| 9 | D4720-001-09 | 53836-K224-09 | 53836-K233-09 |
| 10 | D4720-001-10 | 53836-K224-10 | 53836-K233-10 |
| 11 | D4720-001-11 | 53836-K224-11 | 53836-K233-11 |
| 12 | D4720-001-12 | 53836-K224-12 | 53836-K233-12 |
| 13 | D4720-001-13 | 53836-K224-13 | 53836-K233-13 |
| 14 | D4720-001-14 | 53836-K224-14 | 53836-K233-14 |

With larger 2 stroke engines, eg. 12K98MC, it may become necessary after the engine tests are complete that due to the size and weight the engine is dismantled in 2 or 3 parts for transportation purposes.  To facilitate this it is possible with the Graviner Mk 6 OMD to use two Junction Boxes for one engine zone.

For connection details and the different engine types, refer to Figure 24.

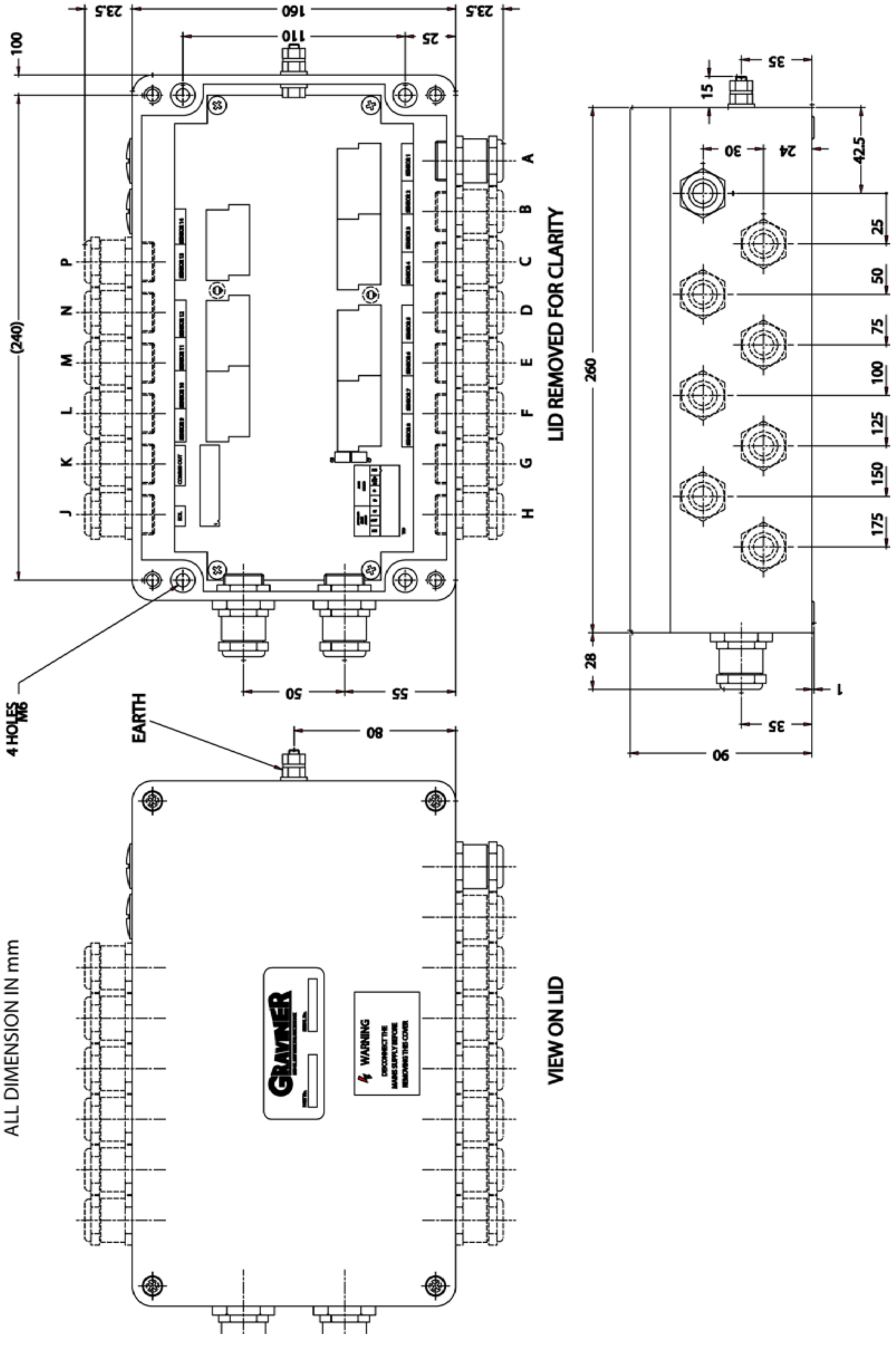



Figure 22: Junction Box Installatiion

It is recommended that all cable terminations within the junction box are made using crimp terminals.

The Screens for both the communications cable & power cable from the control unit must be terminated on to TB9 (refer to Figure 23).

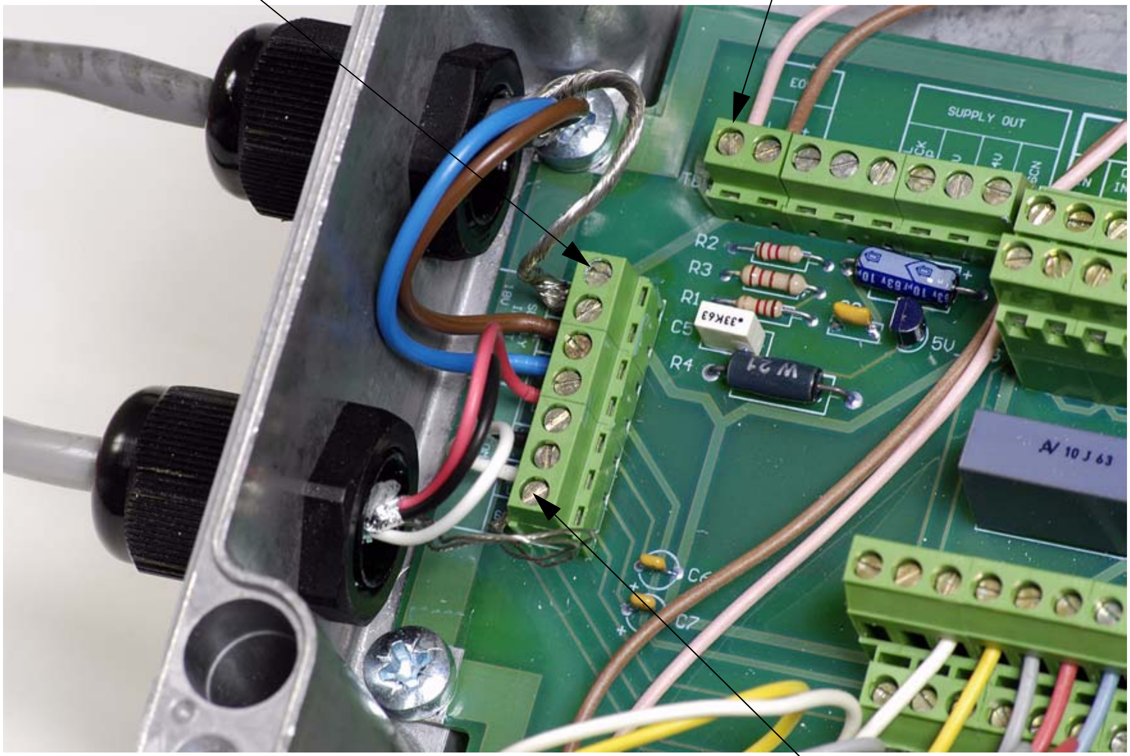
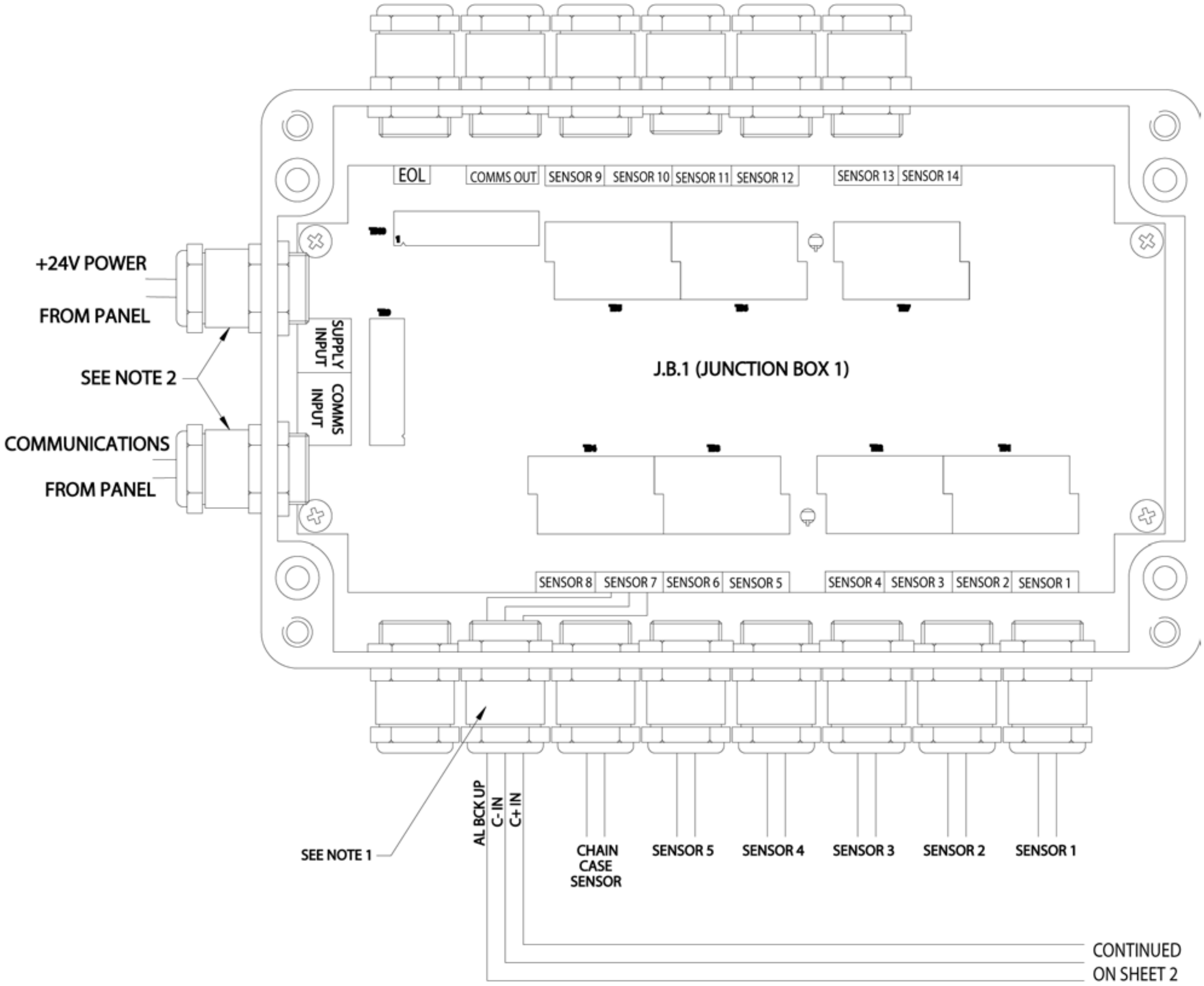



Figure 23: Cable Screen Terminations

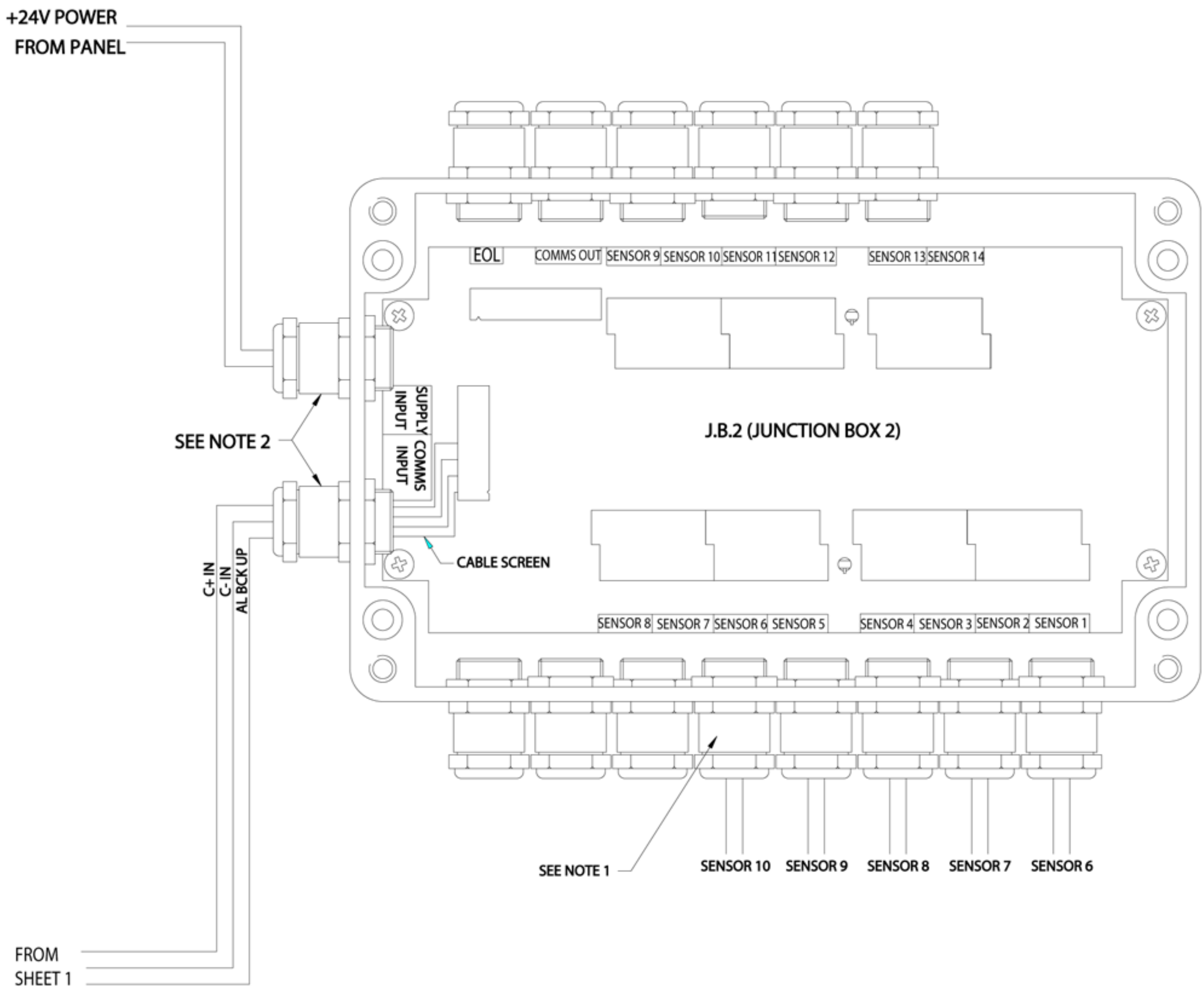
CONNECTION DETAIL FOR ENGINE TYPE 10K80MC




NOTES

1. CABLE SCREENS TO BE MADE OFF IN METAL GLANDS.
2. CABLE SCREENS TO BE CONNECTED AT TERMINAL BLOCKS.

Figure 24 (Sheet 1 of 2): Engine Connection Details




| ENGINE TYPE | JUNCTION BOX 1 | | | | | | | | JUNCTION BOX 2 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | SENSOR 1 | SENSOR 2 | SENSOR 3 | SENSOR 4 | SENSOR 5 | SENSOR 6 | SENSOR 7 | SENSOR 8 | SENSOR 1 | SENSOR 2 | SENSOR 3 | SENSOR 4 | SENSOR 5 | SENSOR 6 |
| 10K80MC (AS SHOWN ABOVE) | SENSOR 1 | SENSOR 2 | SENSOR 3 | SENSOR 4 | SENSOR 5 | CHAIN CASE | JB2 COMMS I/P | | SENSOR 6 | SENSOR 7 | SENSOR 8 | SENSOR 9 | SENSOR 10 | |
| 10K90MC | SENSOR 1 | SENSOR 2 | SENSOR 3 | SENSOR 4 | SENSOR 5 | CHAIN CASE | JB2 COMMS I/P | | SENSOR 6 | SENSOR 7 | SENSOR 8 | SENSOR 9 | SENSOR 10 | |
| 12K90MC | SENSOR 1 | SENSOR 2 | SENSOR 3 | SENSOR 4 | SENSOR 5 | SENSOR 6 | CHAIN CASE | JB2 COMMS I/P | SENSOR 7 | SENSOR 8 | SENSOR 9 | SENSOR 10 | SENSOR 11 | SENSOR 12 |
| 10K98MC | SENSOR 1 | SENSOR 2 | SENSOR 3 | SENSOR 4 | SENSOR 5 | CHAIN CASE | JB2 COMMS I/P | | SENSOR 6 | SENSOR 7 | SENSOR 8 | SENSOR 9 | SENSOR 10 | |
| 11K98MC | SENSOR 1 | SENSOR 2 | SENSOR 3 | SENSOR 4 | SENSOR 5 | SENSOR 6 | CHAIN CASE | JB2 COMMS I/P | SENSOR 7 | SENSOR 8 | SENSOR 9 | SENSOR 10 | SENSOR 11 | |
| 12K98MC | SENSOR 1 | SENSOR 2 | SENSOR 3 | SENSOR 4 | SENSOR 5 | SENSOR 6 | CHAIN CASE | JB2 COMMS I/P | SENSOR 7 | SENSOR 8 | SENSOR 9 | SENSOR 10 | SENSOR 11 | SENSOR 12 |

Figure 24 (Sheet 2 of 2): Engine Connection Details

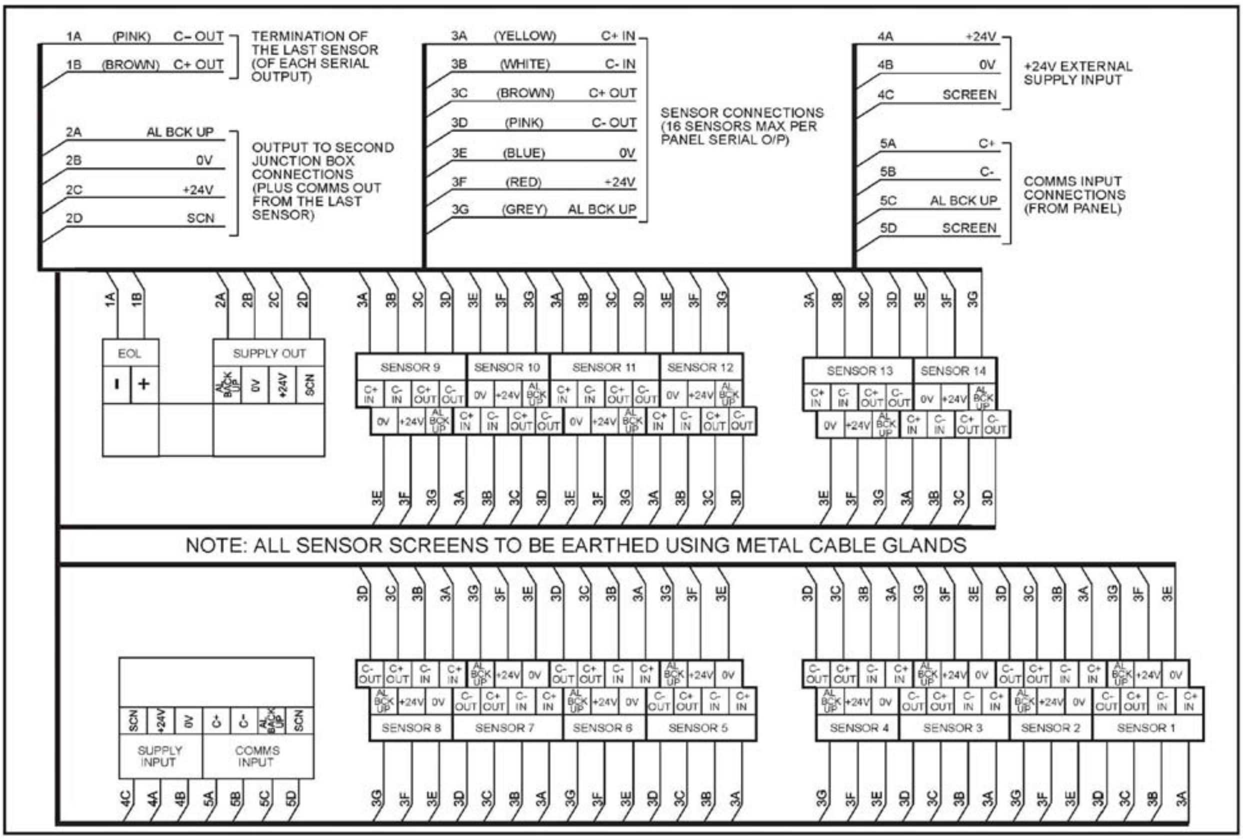



Figure 25: Junction Box External Connections Label

## 2.2 SYSTEM CHECKS PRIOR TO SWITCHING ON

2.2.1 Check that the detector cables are correctly terminated in the Junction Box and that the screens are made off correctly in the glands, e.g. the detector that is addressed 01 is connected to the detector 1 position in the Junction Box, and the detector addressed 02 is connected to the detector 2 position in the Junction Box etc. **Ensure last detector for each Junction Box is also terminated on TB10 EOL**

2.2.2 Ensure that the Communication and Junction Box power supply cables are connected correctly in the junction box (refer to Figure 12).

2.2.3 Check the cable run of the Communication and Junction Box power supply cables back from the junction box to the Control Unit to ensure that they are not damaged.

2.2.4 Ensure that the Communication and Junction Box power supply cables are connected correctly in the Control Unit (refer to Figure 12 and properly screened).

2.2.5 Ensure that the Engine Slowdown, Main Alarm and Fault Alarm relays are connected correctly in the Control Unit (refer to Figure 12).

2.2.6 Ensure that the supply input cable is connected correctly to the Control Unit (refer to Figure 12).

2.2.7 Ensure that the input voltage to the Control Unit is 24 V dc +30%, - 25%

2.2.8 Check the location and function of the main controls on the front of the Control Unit (refer to Figure 4).

2.2.9 When all of the above have been checked and are satisfactory the system is ready to switch on.

## 2.3 SYSTEM CONFIGURATION AND COMMISSIONING

### 2.3.1 Initial Actions and Settings

1. After switch on, the control unit display shows the message **SCANNING FOR DETECTORS**. Followed by a flashing **COMMS FAULT** message. The green LEDs on the detectors illuminate.
2. Press **ACCEPT** to silence the audible alarm. The **COMMS FAULT** continues to flash. Select **MAIN MENU** use the cursor to highlight **ENGINEER**. Press ↵ .
3. The display calls for password. Enter the default password **012345** press ↵. Display shows **MAIN MENU ENGINEER**. Use the cursor to highlight option **1 CONFIGURE SYSTEM**. Press ↵

### 2.3.2 Setting Engine Details

1. Select **ENGINEER MAIN MENU**, followed by **CONFIGURATION SYSTEM** and **ENGINE/DETECTOR**.
2. In **ENGINE/DETECTOR CONFIGURATION** enter number of engines. Press ↵.
3. Select each engine in turn using the ▲ and ▼ navigation keys.
4. For each engine enter the number of detectors. Press ↵ after each entry.
5. Press 🡐 to return to configuration menu.
6. Select **SET ENGINE NAME**. Select engine 1 (2, 3, 4, etc) press ↵
7. Enter engine description letter by letter using the ▲/▼ keys to sequence through the alphabet and the ◀/▶ keys to move to the next letter. Press ↵to store the name. Press 🡐 to return to engine description page and select **NEXT ENGINE**.
8. Repeat this section to name all configured engines.

| **Attribute** | **Default** | **Range** |
|---|---|---|
| Number of engines | 1 | User selectable - 1 to 8 |
| Number of detectors | 4 | User selectable - 1 to 64 |
| Detectors per engine | 4 | User selectable - 1 to 14 |

### 2.3.3 Clean Air Offset (For systems supplied before 2004)

Once the system is configured to the engine installation, the detectors must be set to zero. As the detectors are electro-optical devices it is normal for each of them to exhibit a small offset when first switched on. This is referred to in the configuration menu as **CLEAN AIR OFFSET**.

Note: The detectors must be set to zero prior to engine start with a cold engine and should only be carried out at the initial commissioning stage or when a replacement unit is fitted.

As each detector is set to zero the words 'Saving Configuration Data' will appear on the display for a few seconds, wait for this to disappear before proceeding to the next detector.

To zero the detectors proceed as follows:

1. Select **ENGINEER MAIN MENU** and call up **CONFIGURATION**.
2. Select **DETECTOR OFFSETS**. **CLEAN AIR OFFSET-ZERO DETECTORS** is displayed. Press ↵.
3. **DETECTOR OFFSET MENU** is displayed.

4. Type in engine number and detector number, press ↵.

5. Repeat this for each detector on each engine.

6. All detectors will now be set to zero.

2.3.4 Setting Alarm Levels

The system is supplied with default alarm settings for both average and deviation. These are based on past experience and allow the system to operate initially and gather data from the engines being monitored. To enable the alarm settings to be matched to the individual engines, the actual oil mist density readings should be taken from each engine after it has been operating at full load and reached maximum operating temperature. This could take up to 2 hours. Also clear '**MAXIMUM ACTUAL AVERAGE**' for each engine to ensure all test readings are erased. This can be done in the **ENGINEER MAIN MENU** then **CONFIGURATION** and then '**CLR MAXIMUM AVERAGE**'. The test readings for the detectors should also be cleared. This is done also in the **ENGINEER MAIN MENU** then **CONFIGURATION** and then '**CLR PEAK AND AVERAGE**'. The above should be carried out for all engines being monitored. This should also be carried out after any smoke testing.

| **Alarm Type** | **Default Settings** | **Range** |
|---|---|---|
| Deviation | 0.3 mg/l | 0.05 to 0.5 mg/l |
| Average | 0.7 mg/l | 0.3 to 1.3 mg/l |

2.3.5 Average Alarm

1. Enter **MAIN MENU** and select **ENGINEER/SYSTEM STATUS**. Select **ENGINE 1** (**2, 3, 4,** etc). **SYSTEM STATUS ENGINE 1** (**2, 3, 4**, etc) shows.

2. Read maximum actual average value (retain this value for deviation alarm setting).

3. Reset average alarm level to a maximum of twice the max actual average, as follows:

4. Return to **CONFIGURATION MENU**, select **3, ALARM LEVELS**, Select **SET AVERAGE ALARM**. Select engine number, press ↵ Enter new alarm level, as calculated from above.

5. Repeat steps 1 to 3 for all engines.

2.3.6 Deviation Alarm

Have the max actual average for each engine to hand as used in setting the average alarm (from step 2. in para 2.3.5 above).

1. Enter **MAIN MENU** and select **ENGINEER MAIN MENU**. Enter password, Press ↵

2. Select **SYSTEM STATUS** followed by **DETECTOR**, Select **DETECTOR STATUS**, Select engine 1 then select detector 1 (2, 3, 4, etc.) in turn and read the maximum peak level on each detector. Subtract the maximum average level from each of the detector peak levels and multiply each result x 2. This becomes the deviation value entered into the Control Unit. Deviation value = 2 x (Detector maximum peak level minus engine maximum average level).

   If the deviation value calculated is less than 0.05, enter 0.05.

   If the deviation value calculated is greater than 0.5, enter 0.5

3. Repeat for each engine by using ← key to return to **DETECTOR STATUS** menu.

4. Use ← key to return to **CONFIGURATION MENU**. Select **ALARM LEVELS**.

5. Select **DEVIATION ALARM**. Select engine 1, (2, 3, 4, etc). Enter new deviation alarm level for each detector for engine 1. (2, 3, 4, etc)

6. It may become necessary over a period of time that due to changes in the engine's characteristics, the alarm levels need to be readjusted.

2.3.7 Detector Smoke Alarm Test

1. Commence the smoke alarm test as follows: with the detector fitted, electrically connected, functioning and configured as described in the instruction manual,

   a. Cut a length of wick approximately 30 mm long. Assemble the smoke tester by pushing the wick into the wick holder fitted with the pipette bulb. Press the nylon pipe into the Pipe connector (refer to Figure 26).

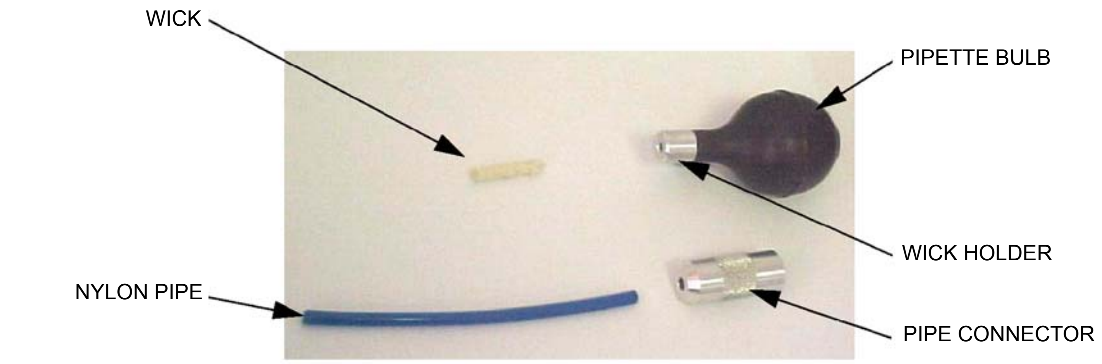



Figure 26: Smoke Tester

   b. Push the nylon pipe of the smoke tester into the connector on the side of the detector base body (refer to Figure 27).

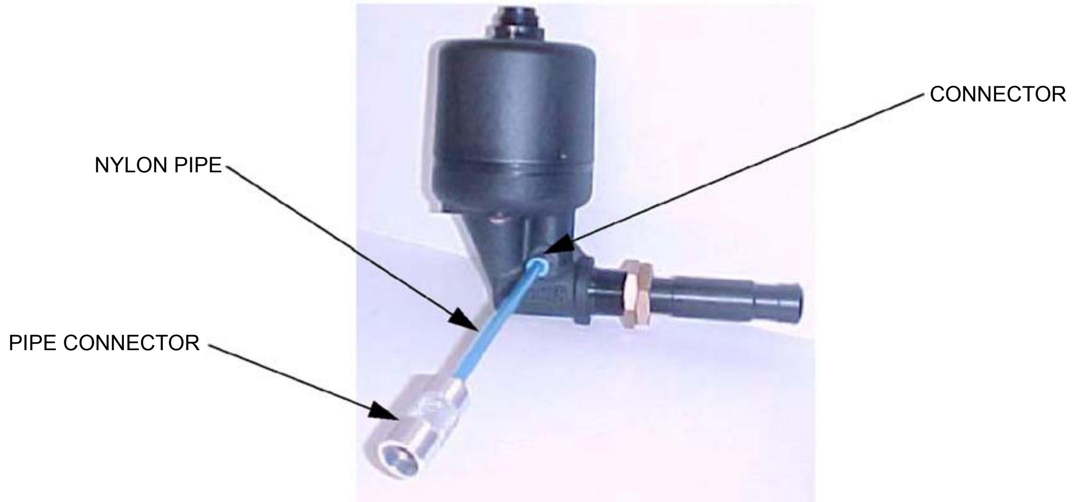



Figure 27: Smoke Test

   c. Dip the wick into the bottle of smoke oil and reseal the bottle firmly

   Note: Only a small quantity of oil is required.

   d. Ignite the wick of the smoke tester and blow out the flame. Squeeze the pipette bulb to keep the wick smoking.

   Note: Care to be taken with this activity at all times.

   e. While the wick is still smouldering, insert it into the pipe connector and squeeze the pipette bulb.

f. Observe the wick is still smouldering, insert nylon pipe into the pipe connector of the detector and squeeze the pipette bulb.

g. After tests are completed the Maximum Actual Average readings should be erased. This can be done in the Engineer Main Menu then configuration and then CLR. Maximum Average.

h. To release the pipe from the connector, press in the blue plastic collar on the end of the connector at the same time as pulling the pipe out.

i. Remove the nylon pipe from the pipe connector for stowage purposes.

j. The wick is reusable and can be left in the wick holder. Fully extinguish the wick after use at all times.

k. Refer to the Material Safety Data Sheet in the event of health or safety issues.

2.3.8 Back Up Alarm

The back up alarm is a hard wired link from each detector installed on the system. This facility will allow any detector which is in a fault condition and sees an oil mist level of 1.6mg/l or greater produce a 'Back Up Alarm'. This will produce operation of both the shutdown and main alarm relays, the back up alarm will also override any detector or detectors that are isolated.

**Whilst any detectors are in a back up alarm condition the keypad keys are inoperative until the oil mist level drops below 1.6mg/l.**

It is possible for a healthy detector to produce a back up alarm if the level of oil rises very rapidly.

2.3.9 System Access Password

Access to the system is at three levels:

User

Engineer

Service

The User access level allows for READ only. No adjustments are possible but all data and settings can be read.

The Engineer access level is password protected and allows the system to be configured and read. It also allows the alarm settings to be adjusted. The system is supplied with a default password, but it can be altered to suit individual operators. The default password will always remain active for emergency.

The Engineer access level, when selected, a prompt for a password will appear. Enter 0.1,2,3,4,5, then press ↵

2.3.9.1 To change the password proceed as follows:

1. Press **MAIN MENU** and then select **ENGINEER**. Enter default password, select **1, CONFIGURE SYSTEM**.
2. In configuration menu select **5, SET PASSWORD**.
3. In set **PASSWORD MENU**, select **SET ENGINEER PASSWORD**.
4. Enter new password, (min 2 digits, max 6 digits) press ↵ enter new password again to confirm and press ↵ key.
5. The new password is now active.

2.3.9.2 The Service access level is also password protected and access is for authorised service agents of Kidde Fire Protection only.

## 2.4 SYSTEM OPERATION

### 2.4.1 Action on Alarm

On receipt of either a Deviation or Average alarm the engine should, unless connected to a shutdown relay, be stopped if safe to do so and allowed to cool down so that the background oil mist levels reduce before entering the engine room. Investigations can then be carried out to find the cause of the alarm and rectify. Once the fault in the engine has been rectified the OMD system can be re-set and the display returns to the normal mode.

When a system fault alarm is received, the information on the display should be noted and then the appropriate fault finding procedure in Chapter 4 of the manual should be consulted to enable the fault to be rectified.

Note: When the engine is started from cold in Arctic/Antarctic conditions, a water mist can be produced that could give a false alarm.

Alarm and fault messages have an associated priority. These are detailed below:

| **Message** | **Priority** |
|---|---|
| Alarm | Highest |
| Comms. fault | 2nd highest |
| Sensor fault | 3rd highest |
| System fault | 4th highest |

All events are stored in the alarm/fault queue in order of occurrence. The user can scroll through the queue by use of the arrow keys.

Once an event is active the event is displayed on the LCD in the appropriate format. To clear the display press the **ACCEPT** key. Once the **ACCEPT** key is pressed, the display shows the **ENGINE AVERAGES** display.

When an alarm condition exists, the following warning message is displayed regardless of other information on the display (ie alarm events have the highest priority): **DEVIATION ALARM**.

### 2.4.2 System Status

#### 2.4.2.1 Engine Status

This menu will show the average alarm level as set in the engineer configuration menu, the maximum average level that has been reached, if the engine slow down relays and engine are isolated, if the engine system has any general faults and finally the number of detectors configured for this engine.

#### 2.4.2.2 Detector Status

This menu permits the following information for each detector to be read. Detector status including peak level of oil mist in mg/l.

The Deviation Alarm level

If the detector is isolated

If the detector has a Communication Fault, Fan Fault, LED Fault watch dog or address Fault. Plus the zero offset values per detector.

### 2.4.3 Isolation

If required it is possible to isolate any individual detector, a complete engine or the engine slow down relays. This allows maintenance to be carried out without affecting the remaining system. Isolation inhibits all alarms and faults for the isolated item except the Back up Alarm, which is a fixed level.

Engine Isolation

To isolate all of the detectors on one particular engine, access the **MAIN MENU** and select **ENGINEER**. Enter either the default password or own unique password if this option has been used.

1. When in the engineer menu, press key number 3 to highlight **ISOLATE** and then press the ↵key.
2. Once in the isolate menu, select **1 ENGINE**, press the ↵key and the display will show the number of engines configured with the word **DE-ISOLATED** beside each one.
3. Using the ▲ and ▼ keys, highlight the engine required and press the ↵key.
4. The display will show the engine selected with the words **DE-ISOLATION** - **ISOLATE** beside it.
5. Press the ▶ key to highlight **ISOLATE** and press the ↵key.
6. The display reverts to showing the number of engines configured, but the selected engine will have the word **ISOLATED** flashing beside it instead of **DE-ISOLATED**.
7. The three lights marked **ISOLATE**, **DETECTOR ISOLATE** and **ENGINE ISOLATE** all come on.
8. To de-isolate, repeat the above until you reach the display that shows **DE-ISOLATION** - **ISOLATE** (step (4)) beside the selected engine and as **DE-ISOLATED** is highlighted press the ↵key and the detectors on the engine return to normal operation.
9. To return to the normal menu press the ← key once and then the **MAIN DISPLAY** key.

Detector Isolation

To isolate a detector, access the engineer menu and then the isolate menu as above.

1. When in the isolate menu press the ▼ key and press the ↵key.
2. The display will show the number of engines configured.
3. Using the ▲ and ▼ keys highlight the engine on which the detector is to be isolated and press the ↵key.
4. The display changes to show the number of detectors configured on the selected engine with the word **DE-ISOLATED** beside each detector.
5. Using the ▲ and ▼ keys highlight the detector to be isolated and press the ↵key.
6. The words beside the detector selected will change to **DE-ISO ISO**.
7. By using the ▶ key, highlight ISO and press the ↵ key.
8. The display returns to show all detectors with the word **ISOLATED** flashing by the selected detector. All other detectors still have the word **DE-ISOLATED** beside them.
9. The **ISOLATE** and **DETECTOR ISOLATE** lights will come on.
10. To de-isolate the detector ensure this detector is highlighted and press the ↵key to return to the display showing **DE-ISO ISO** and, as **DE-ISO** is highlighted, press the ↵ key again and the screen returns to showing all detectors de-isolated and the lights will go off.
11. To return to the normal screen, press the ← key once and press the **MAIN DISPLAY** key.

Slow Down Relay Isolation

To isolate a slow down relay, access the engineer menu and the isolate menu as above.

1. In the isolate menu select **3 RELAY** and press the ↵key.
2. The display shows the number of engines configured with the word **DE-ISOLATED** beside each one.
3. Highlight the engine of which the relay is to be isolated by use of the ▲ and ▼ keys and press the ↵key.
4. The display shows DE-ISOLATION ISOLATE beside the engine selected.
5. Press the ▶ key to highlight **ISOLATE**.
6. Press the ↵key and the display will show **ENGINE 1** (**2 - 8**) with the word **ISOLATED** flashing beside it.
7. The **SHUTDOWN ISOLATE** light and the **ENGINE ISOLATE** lights come on.
8. To return the relay to normal operation, press the ↵key whilst the display is showing the engine number and the word **ISOLATED** is flashing.
9. The display shows **ENGINE 1** (**2 - 8**) **DE-ISOLATION ISOLATE**. Press ↵ The display shows **ENGINE 1** (**2 - 8**) **DE-ISOLATED** and the two lights will go out.
10. Press the **MAIN DISPLAY** key to return to the **NORMAL** display.

## 2.5 TEST MENU

Press the TEST key pad on the control unit. The TEST MENU appears, with this menu selected eight more menu options appear on screen as follows: (the eighth option is only available through the Engineer Menu.)

1. Alarm Relay — Checks the correct operation of the main alarm relay.
2. Fault Relay — Checks the correct operation of the fault relay.
3. System Test — Checks the correct operation of the system software (not used after version 109 software).
4. LED/LCD — Test Checks all of the lights and all segments of the LCD on the control unit and also the internal sounder.
5. Back-up Alarm — Checks the Back-up alarm connection from each detector by switching the internal sounder on & off once for each detector.
6. Optics — Checks the output from the detector light array to ensure it is within limits.
7. Detector Alarm — Simulates a deviation alarm without operating the slowdown relay.
8. Slowdown Relay — Checks the correct operation of the slowdown relay. (Only available in password protected screen).

**Warning: Carrying out this test with the engine operating will cause the engine shut/slow down system to operate if fitted.**

### 2.5.1 Alarm Relay

1. Press the **TEST** key pad on the control unit. The **TEST MENU** is displayed, defaulting to No.1 **ALARM RELAY**.
2. Press ↵key.

3. Display changes to:

**ALARM RELAY TEST**

**TEST ALARM RELAY**

**DISABLE** **ENABLE**

Note:**DISABLE** will be highlighted.

4. Press ▶ key to highlight **ENABLE**.

5. Press ↵key and observe:

   a. The display returns to the test menu.

   b. The **TEST** light on the control unit comes on.

   c. The main alarm relay operates. This is confirmed by any audible and visual alarms operating that are connected to this relay.

6. To cancel the alarm:

   a. Ensure **ALARM RELAY** in the test menu is highlighted.

   b. Press ↵key.

   c. Display changes to that shown in para 2.5.1 step (3) above with **DISABLE** highlighted.

   d. Press ↵key again and the alarm will reset and the display will return to the test menu.

7. When testing of the main alarm is complete either; Press the **MAIN DISPLAY** key on the control unit to return to the normal display, or select another test option.

2.5.2 Fault Relay

1. Press the **TEST** key pad on the control unit. The **TEST MENU** is displayed, defaulting to No.1 **ALARM RELAY**.

2. Press key pad No. 2 or the ▼ key to highlight No. 2 **FAULT RELAY**.

3. Press the ↵key.

4. The display changes to:

**FAULT RELAY TEST**

**TEST FAULT RELAY ?**

**DISABLE** **ENABLE**

Note:**DISABLE** will be highlighted.

5. Press the ▶ key to highlight **ENABLE**.

6. Press ↵and observe:

   a. The display returns to the Test Menu.

   b. The **TEST** light on the control unit comes on.

   c. Fault relay changes state. This is confirmed by any audible and visual alarms operating that are connected to this relay.

   d. To cancel the alarm repeat para 2.5.2 steps 2 & 3 above and then press the key again.

7. When testing of the fault relay is complete, either; Press the **MAIN DISPLAY** key to return to the normal display or select another test option.

2.5.3 System Test

1. Press the **TEST** key pad on the control unit. The **TEST MENU** is displayed, defaulting to No.1 **ALARM RELAY**.
2. Press key pad No. 3 or the ▼ key twice to highlight No. 3 **SYSTEM TEST**.
3. Press ↵and observe:
4. The words **MAIN PROCESSOR POWER ON** appear for 2-3 seconds and then the display returns to the test menu.
5. This test is now complete. The operator can now either: Press the **MAIN DISPLAY** key to return to the normal display or select another test option.

2.5.4 LED/LCD Test

1. Press the **TEST** key pad on the control unit. The **TEST MENU** is displayed, defaulting to No.1 **ALARM RELAY**.
2. Press key pad No. 4 or the ▼ key 3 times to highlight No 4 **LED/LCD TEST**.
3. Press ↵and observe:
   a. All of the lights on the control unit come on.
   b. Horizontal bars scroll down the display.
   c. The following appears on the display after the bars scroll down

      **LCD AND LED TEST**
      **OIL MIST DETECTOR MK6**
      **VERSION: P57100-10***

      (*Will depend on the software version fitted)
   d. The internal sounder operates for approximately 5 seconds.
   e. All lights except for the green **POWER ON** light go out.
   f. The display returns to the test menu.
4. This test is now complete. The operator can now either: Press the **MAIN DISPLAY** key to return to the normal display, or select another test option.

2.5.5 Backup Alarm

1. Press the **TEST** key pad on the control unit. The **TEST MENU** is displayed, defaulting to No.1 **ALARM RELAY**.
2. Press key pad No. 5 or the ▼ key four times to highlight No. 5 **BACK-UP ALARM**.
3. Press ↵and check that the internal sounder operates the same number of times as there are detectors fitted to the system and the display shows **BACK-UP ALARM ACTIVE**.

   Note:If the system has 6 detectors fitted then the internal sounder operates 6 times.
4. When the test is complete the display will return to the test menu.
5. This test is now complete. The operator can now either: Press the **MAIN DISPLAY** key and return to the normal display or select another test option.

2.5.6 Optics

**Up to version 107 software**

1. Press the **TEST** key pad on the control unit. The **TEST MENU** is displayed, defaulting to No.1 **ALARM RELAY**.

2. Press key pad No. 6 or the ▼ key five times to highlight No. 6 **OPTICS**.

3. Press ↵and the display changes to:

**DETECTOR OPTICS TEST**

**ENABLE OR DISABLE ?**

**DISABLE** **ENABLE**

4. Press the ▶ key to highlight **ENABLE**.

5. Press ↵and observe:

   a. The words **OPTIC TEST ENABLED** appear on the display underneath what is shown in step (3) above.

   b. The internal sounder operates once to indicate that the test is complete and the display reverts back to the test menu.

6. This test is now complete. The operator can now either: Press the **MAIN DISPLAY** key and return to the normal display or select another test option.

**After version 107 software**

1. Press the **TEST** key pad on the control unit. The **TEST MENU** is displayed, defaulting to No.1 **ALARM RELAY**.

2. Press key pad No. 6 or the ▼ key five times to highlight No. 6 **OPTICS**.

3. Press ↵and the display changes to:

**TESTING OPTICS - PLEASE WAIT**

4. Approximately 3 seconds after the test is initiated the internal sounder will operate for 1 second.

5. Approximately 25 seconds after the test is initiated the display will change to:

**OPTICS TEST PASSED**

6. and then the display will return to the Test Menu.

7. This test is now complete. The operator can now either: Press the **MAIN DISPLAY** key and return to the normal display or select another test option.

2.5.7 Detector Alarm

1. Press the **TEST** key pad on the control unit. The **TEST MENU** is displayed, defaulting to No.1 **ALARM RELAY**.

2. Press key pad No. 7 or the ▼ key 6 times to highlight No. 7 **DETECTOR ALARM**.

3. Press ↵and the display changes to:

**DETECTOR ALARM TEST**

**ENGINE 1**

**ENGINE 2**

Note: The number of engines shown is dependant on how many are configured up to a maximum of 8. However, **ENGINE 1** is highlighted as the default setting.

4. Press ↵again and the display changes to:

**DETECTOR ALARM TEST**

| | | |
|---|---|---|
| **ENGINE 1** | **DET. 1** | **TEST OFF** |
| **ENGINE 1** | **DET. 2** | **TEST OFF** |

Note: The number of detectors shown will depend on how many are configured up to a maximum of 14 per engine. However, **ENGINE 1 DET. 1** is highlighted as the default setting.

5. Press ↵again and check the following:

   a. The internal sounder comes on immediately.

   b. The display briefly shows the number of engines configured as in step (3) above.

   c. The red **ALARM** and yellow **TEST** lights come on as does the **ENGINE ALARM** and **FAULT** lights. Both the red lights are flashing. The main alarm relay goes into an alarm condition.

   d. The display shows:

**DEVIATION ALARM**

**ENGINE NO DET**

**TIME DATE**

6. Press the **ACCEPT** keypad and observe:

   a. The display changes to the toggling graphics mode.

   b. The red **ALARM** lights both change to a steady state.

   c. The main alarm relay resets.

7. Press the **RESET** key and check:

   a. The display returns to normal.

   b. All lights except the **POWER ON** go out.

8. Repeat the complete test for each detector that is to be tested. Select the detector to be tested by use of the ▲, ▼, ◀, and ▶ keypads.

2.5.8 Slowdown Relay

**Warning: Carrying out this test while the engine is running WILL cause it to stop or slow down.**

1. Press the Main Menu key on the control unit and select **ENGINEER** by using the ▼ down key or the number 2 key.

2. Press the ↵enter key and enter the password when prompted, when entered press the ↵enter key again.

3. The display will show 7 menu headings defaulted to No. 1 Configure System. Press the No.4 key or the ▼ down key 3 times to highlight No.4 **TEST**.

4. Press the ↵enter key and 8 test menus will show on the display defaulted to No.1 Alarm Relay. Press the No.8 key or the ▼ down key 7 times to highlight **SLOWDOWN RELAY**.

5. Press the ↵enter key and the display shows the following defaulting to Engine 1 which will be highlighted:

**RELAY TEST MENU**

**ENGINE 1 TEST OFF**

Note: The number of engines shown is dependent on how many are configured up to a maximum of 8. However, **ENGINE 1** is highlighted as the default setting.

6. Press the ↵enter key and the display shows the following, **DE-ACTIVATE** will be highlighted:

**RELAY TEST MENU**

**ENGINE 1 DE-ACTIVATE ACTIVATE**

7. Press the ▶ right key to highlight Activate and the press ↵enter key, the display will show the following with the words **TEST ON** flashing.

**RELAY TEST MENU**

**ENGINE 1 TEST ON**

8. The slow down relay changes state and all alarms connected to the relay operate, also the test light on the control unit comes on.

9. To cancel the alarm press the ↵enter key and the display will be as step (6) above with **DE-ACTIVATE** highlighted.

10. Press the ↵enter key again and the display will be as step (5) above and the test light on the control unit will go out.

11. Press the 🡐 quit key to return to the test menus or the Main Display and **RESET** to return to the normal display.

2.6 EVENT LOG

2.6.1 The Event Log enables the user to interrogate the past 256 events and can be accessed via the user menu separately.

1. To access the event log press the **MAIN MENU** and when the choice of access levels appears on the screen, **USER** will be highlighted. Press the ↵key and press 3 **EVENT LOG** and press the ↵key. The display will show the last event that has occurred. With ▲ and ▼, scroll through the complete event log until the required entry is found.

2. To speed up the search for the required event it is also possible to do so by event type eg. deviation alarm or, if the approximate date of the event is known, a search from time and date is also available in the Event log sub-menu.

3. To return to the normal display press 🡐 key once and then the **MAIN DISPLAY** key.

2.6.2 The event log may also be downloaded onto a computer using the serial link connector on the main processor PCB.

1. Plug serial lead into Com1 on PC and plug in to round serial connector on the Control Unit.

Hyperterminal Setup

2. From the Windows™ start button choose:-

Start -> Programmes -> Accessories -> Hyperterminal

3. This should bring up the following box or something similar depending on Windows™ version, Windows 95 is shown.

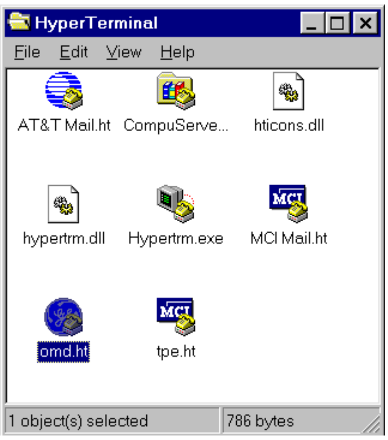



Figure 28: Hyperterminal Screen

4. Then double click on Hypertrm.exe this should start the hyperterminal program and give the following screen.

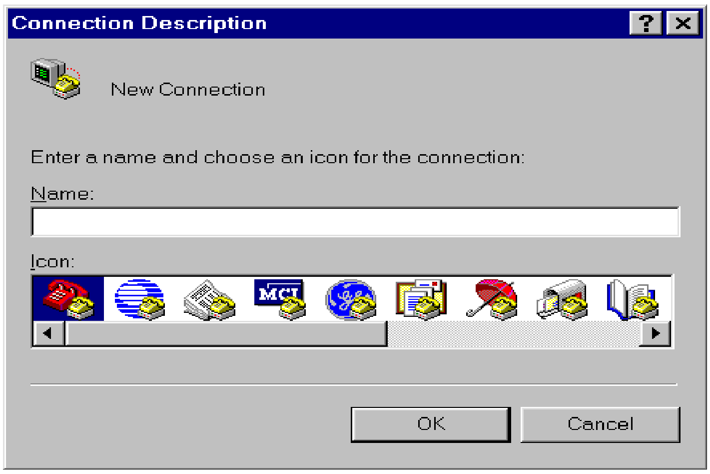



Figure 29: Connection Description Screen

5. Type in a name, eg omd6, and choose one of the icons then click ok.
6. In the next option screen select direct to Com 1 and click ok (note if com 1 is being used for another application Com 2 can be selected if the pc has 2 comms ports).
7. This should bring up the following settings screen.

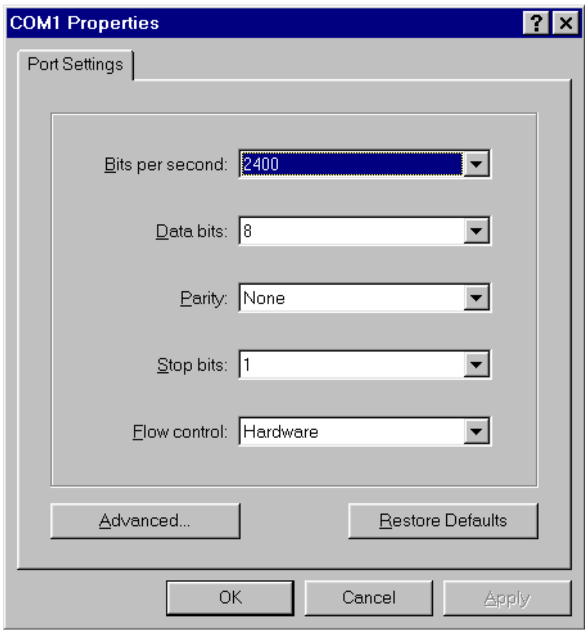



Figure 30: COM1 Properties Screen

a. In bits per second select - 9600

b. Data bits - 8

c. Parity - None

d. Stop bits - 1

e. Flow control - None

8. Then click OK, this connects hyperterminal to com 1 and will take you back to the main screen.

9. Main Screen which should say connected in the bottom left hand corner.

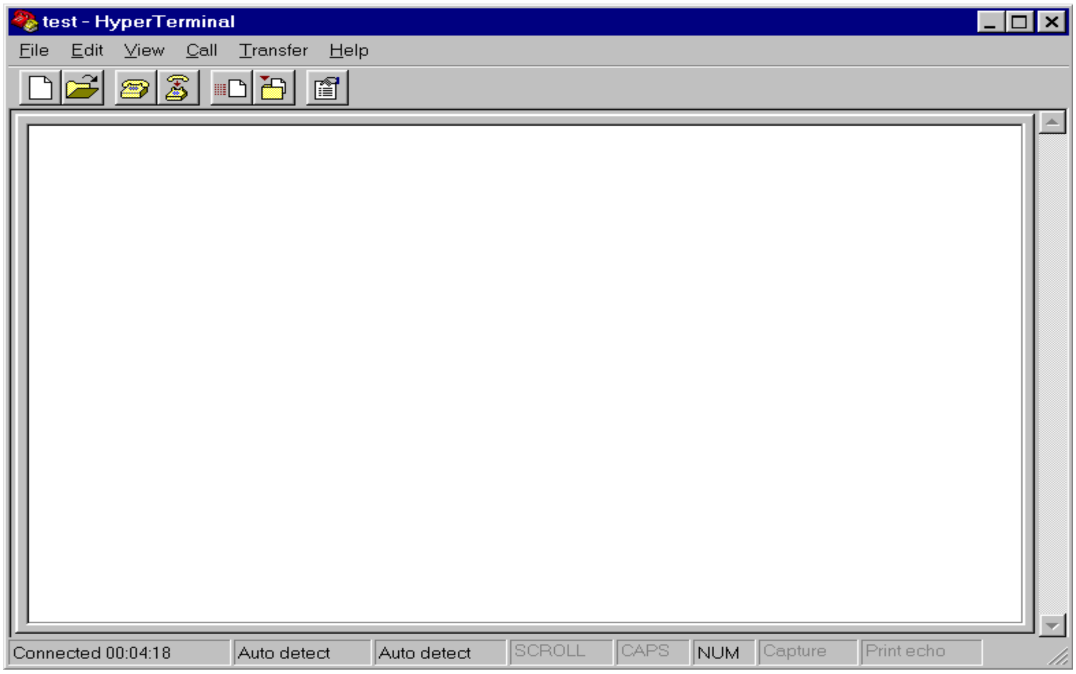



Figure 31: Hyperterminal Screen

10. From the top level menu select Transfer -> Capture text and in the file box type where you want the file to be saved and what name e.g. C:\omd.txt. then click on the start button.

11. At the Control Unit, go through the menu structure and choose Download Event Log, then select enter. You should now see the event log downloading into the white panel of the above screen.

12. When the event log has finished downloading select Transfer -> capture text -> stop.

13. Then using Word/WordPad/NotePad you can examine the event log by loading up C:\omd.txt.

If you want to do another event log download in the future, launch hyperterminal from the start bar as normal and in the first screen there should be a new icon called omd6.ht (if that is what you called it). Simply double click on this icon and hyperterminal will be launched with all the correct settings and the connection to Com port 1 already established. After that the only thing that needs setting up is the transfer capture to save the event log.

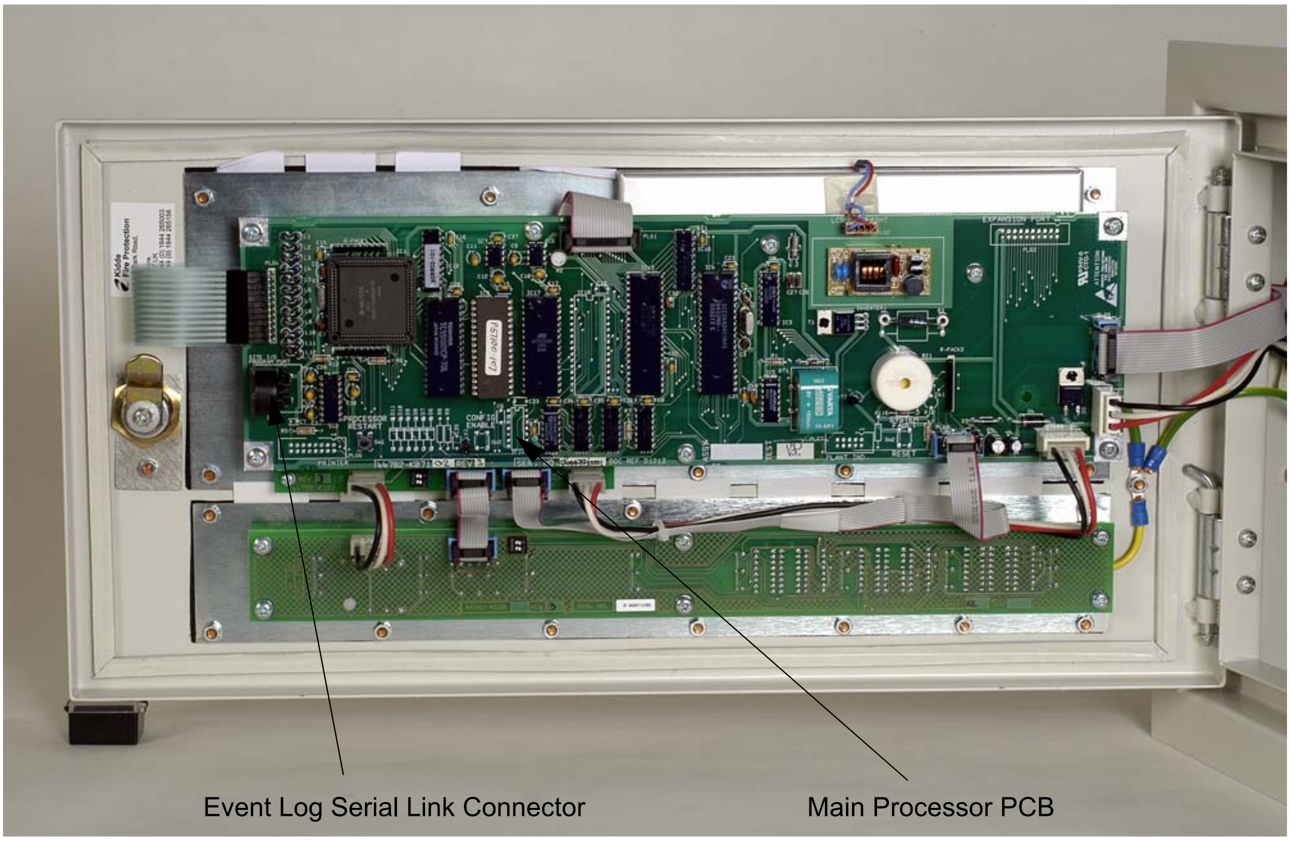
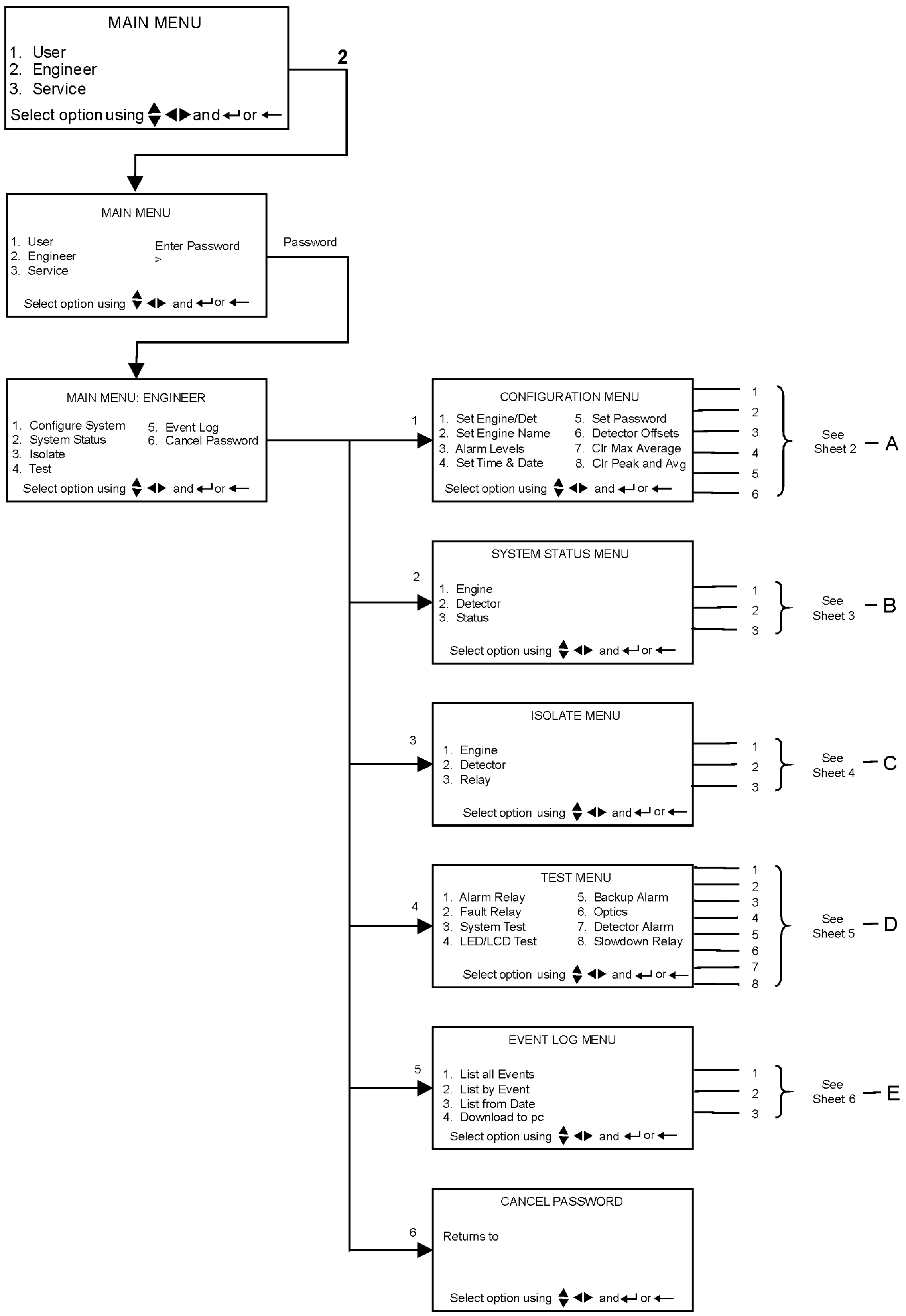



Figure 32: Event Log Connector Location

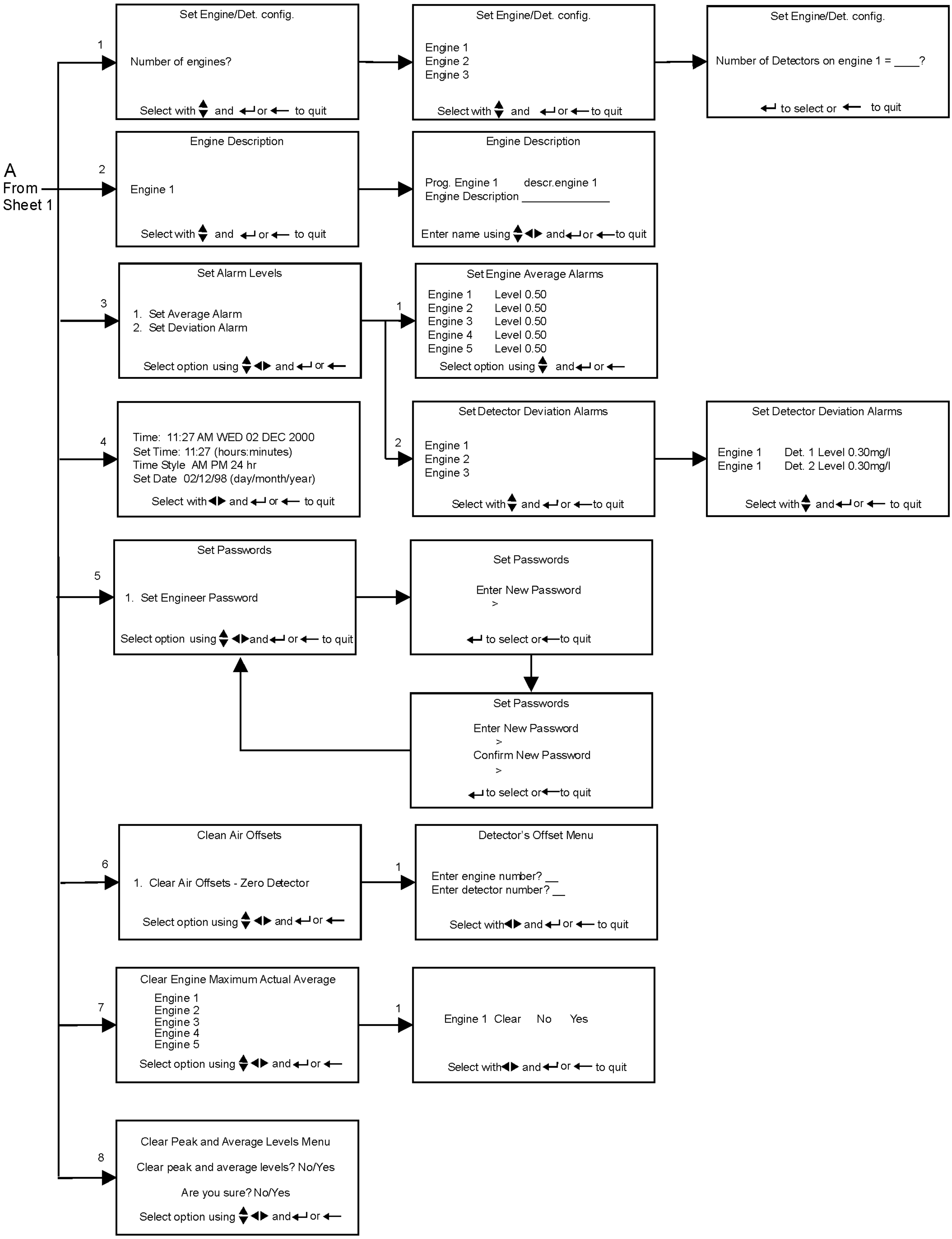
## 2.7 ENGINEER & USER MENU FLOWCHARTS

The following charts show the Engineer and User Menus.

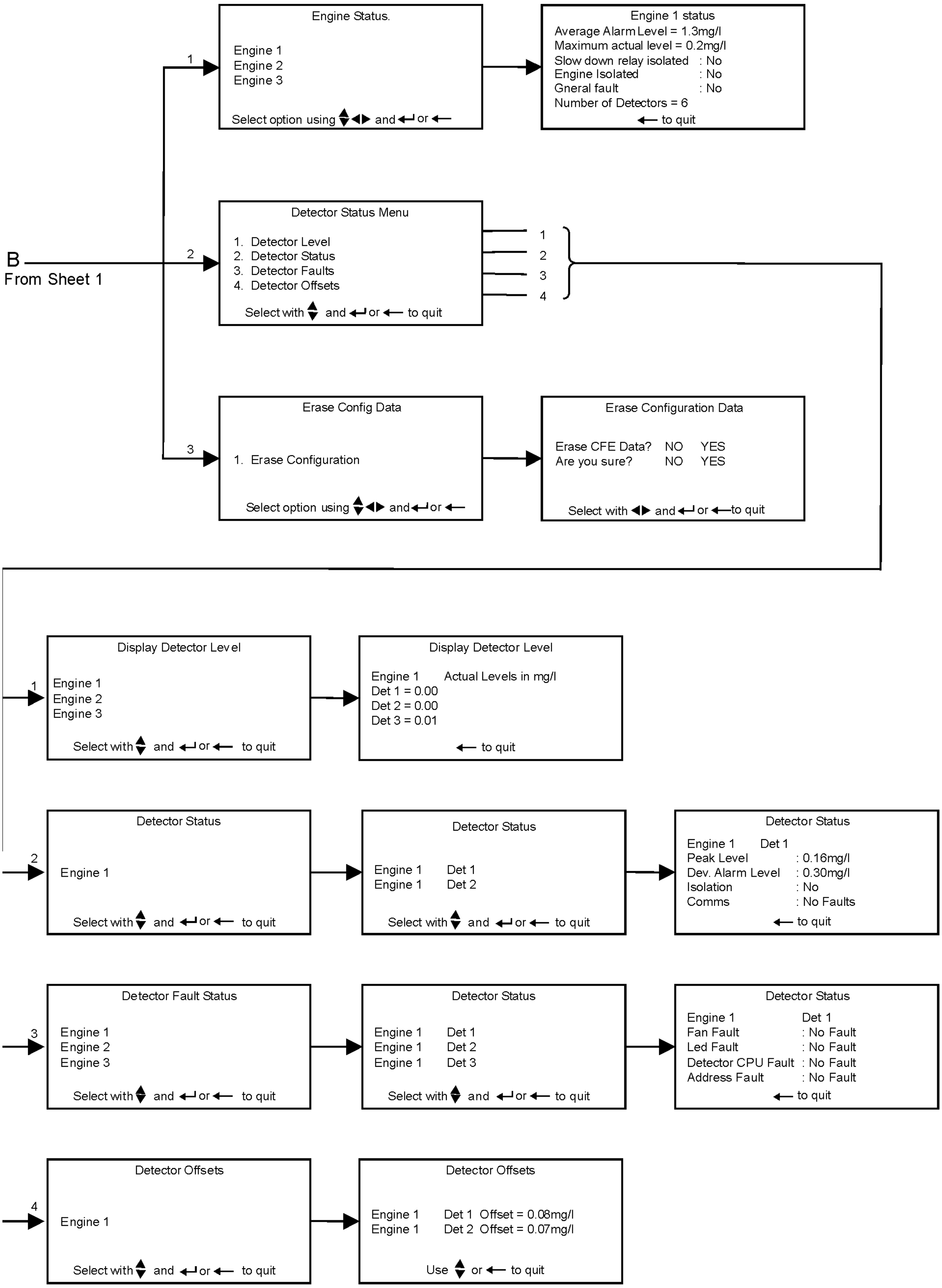
# Engineer Menu Sheet 1



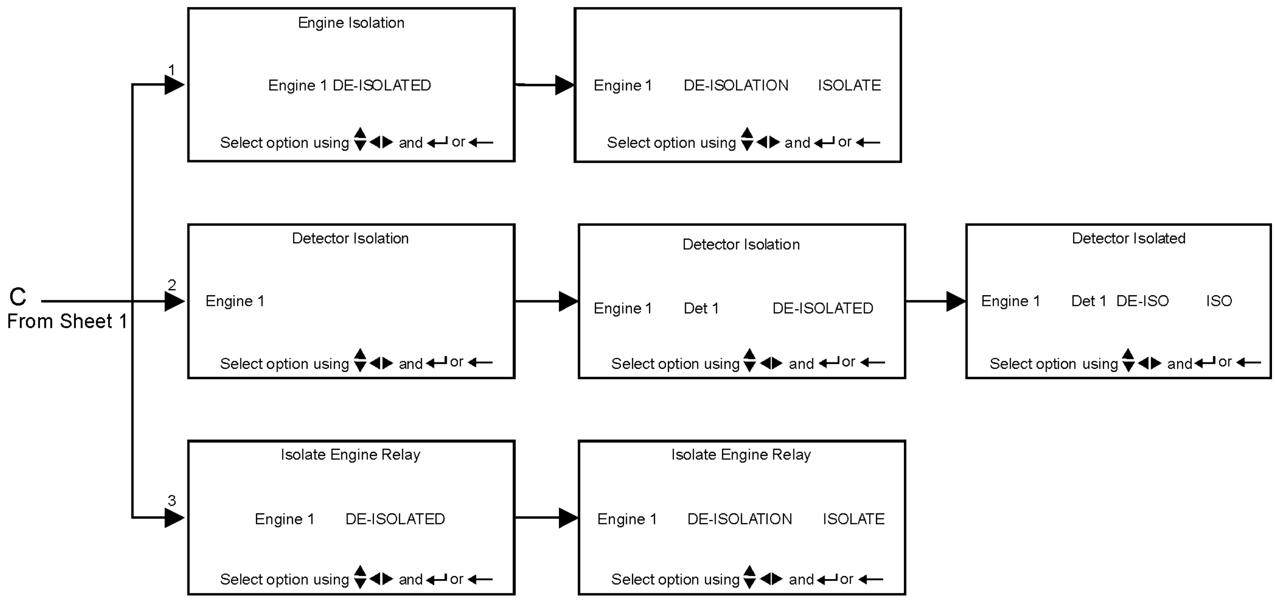
# Engineer Menu Sheet 2



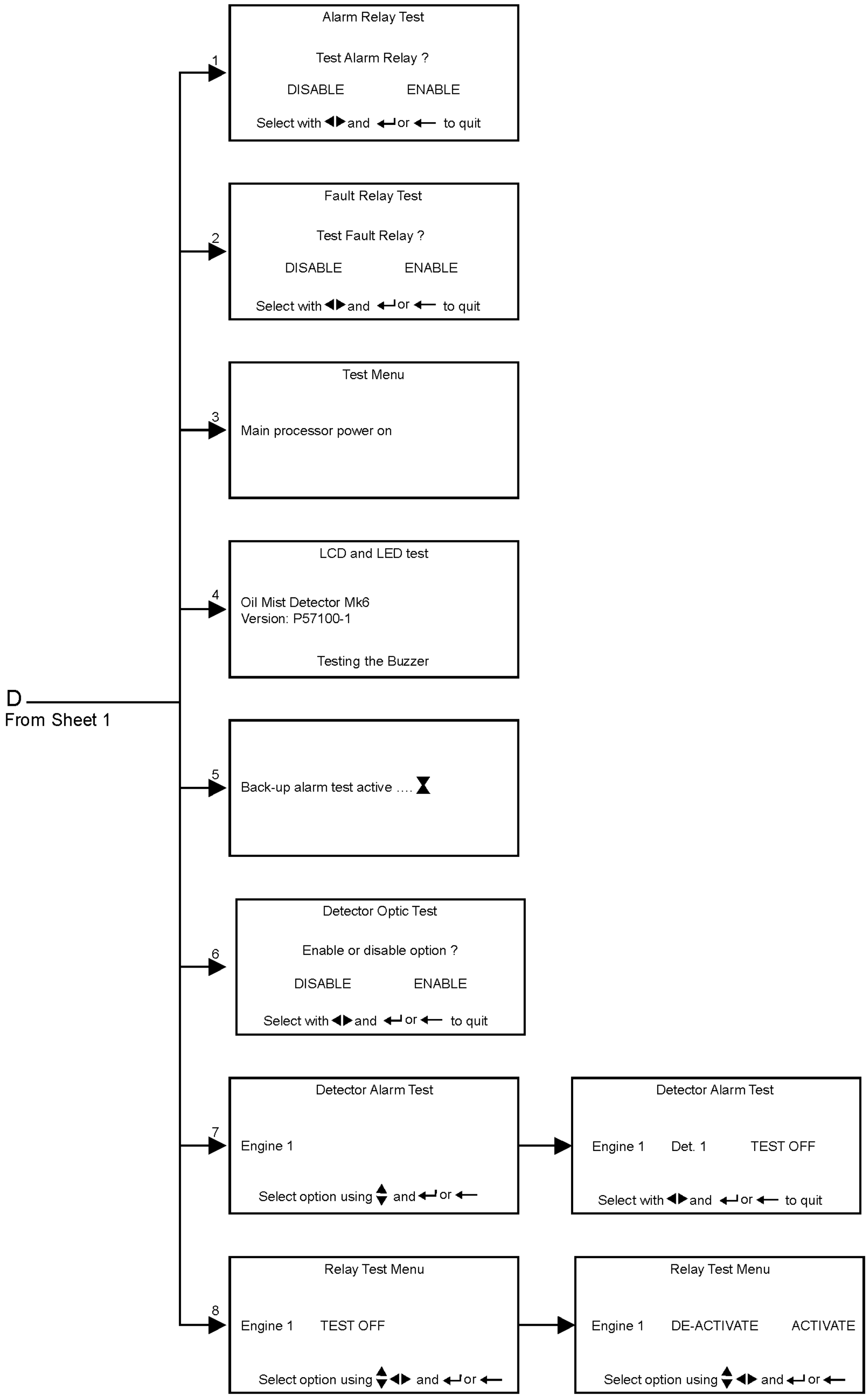
# Engineer Menu Sheet 3



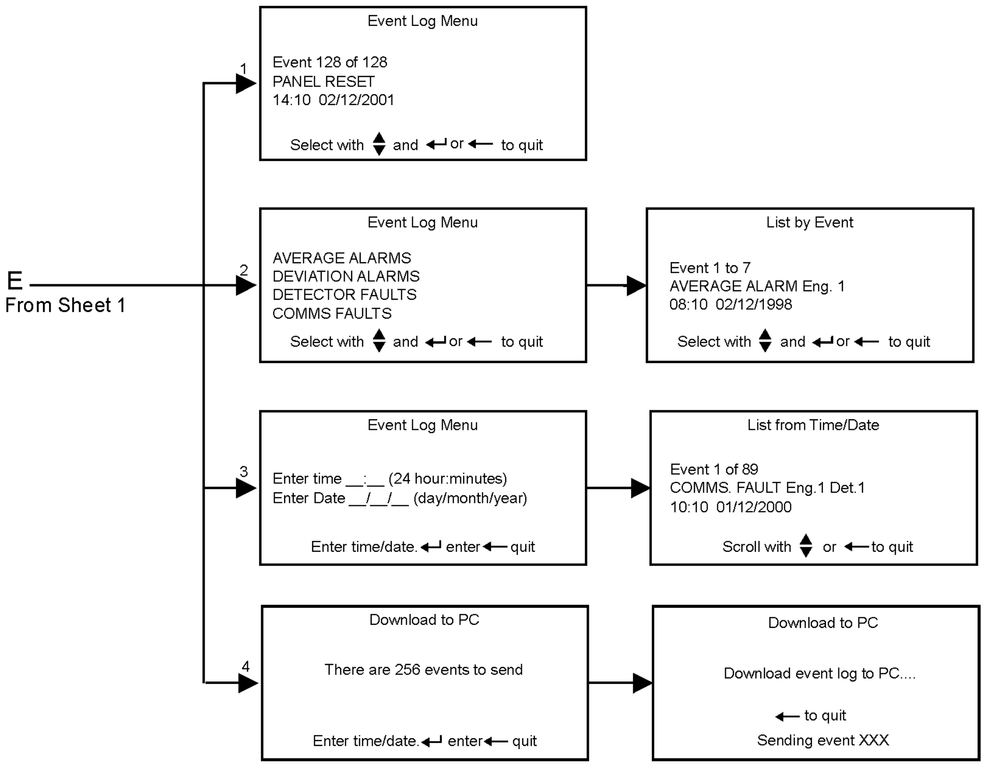
## Engineer Menu Sheet 4



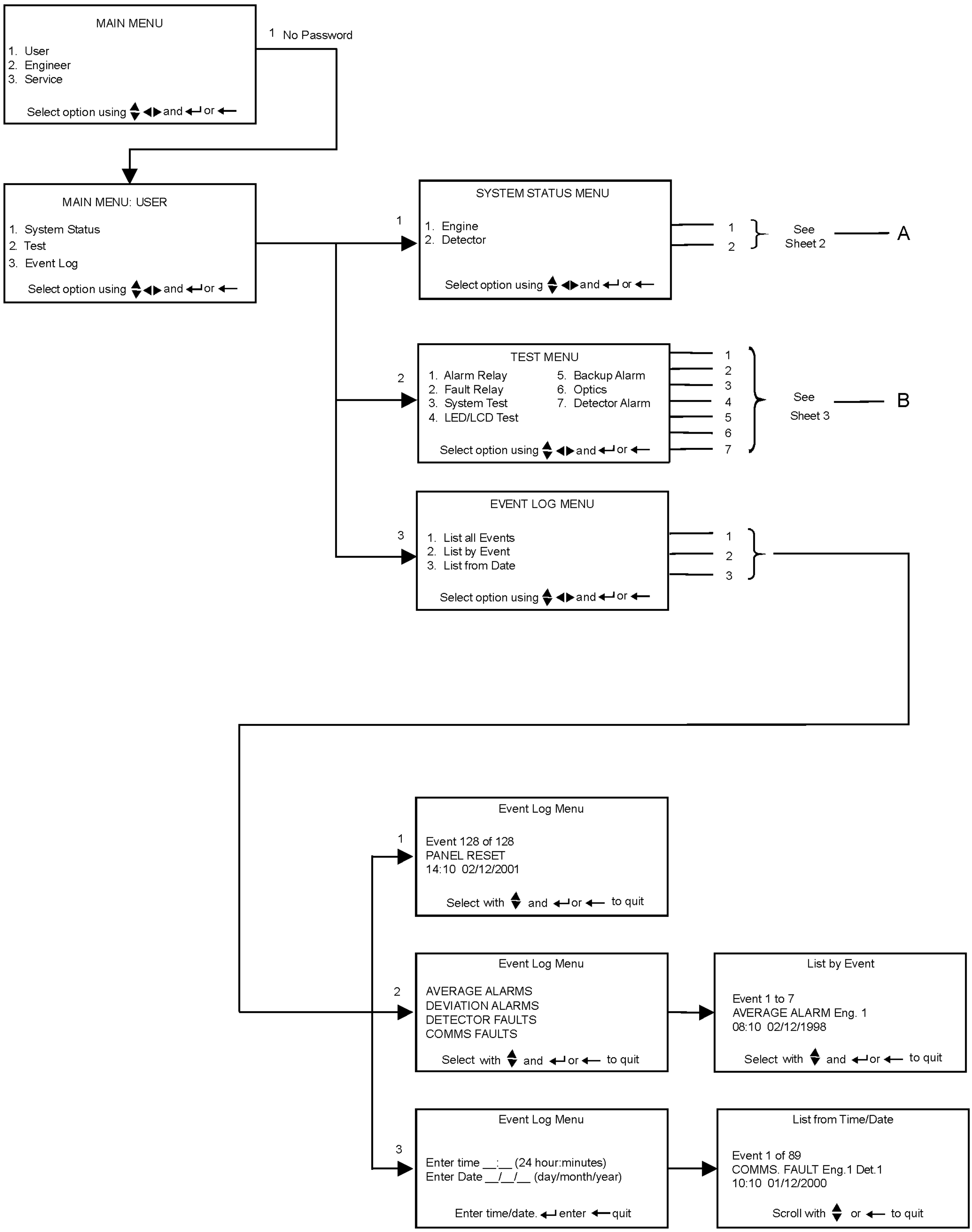
# Engineer Menu Sheet 5



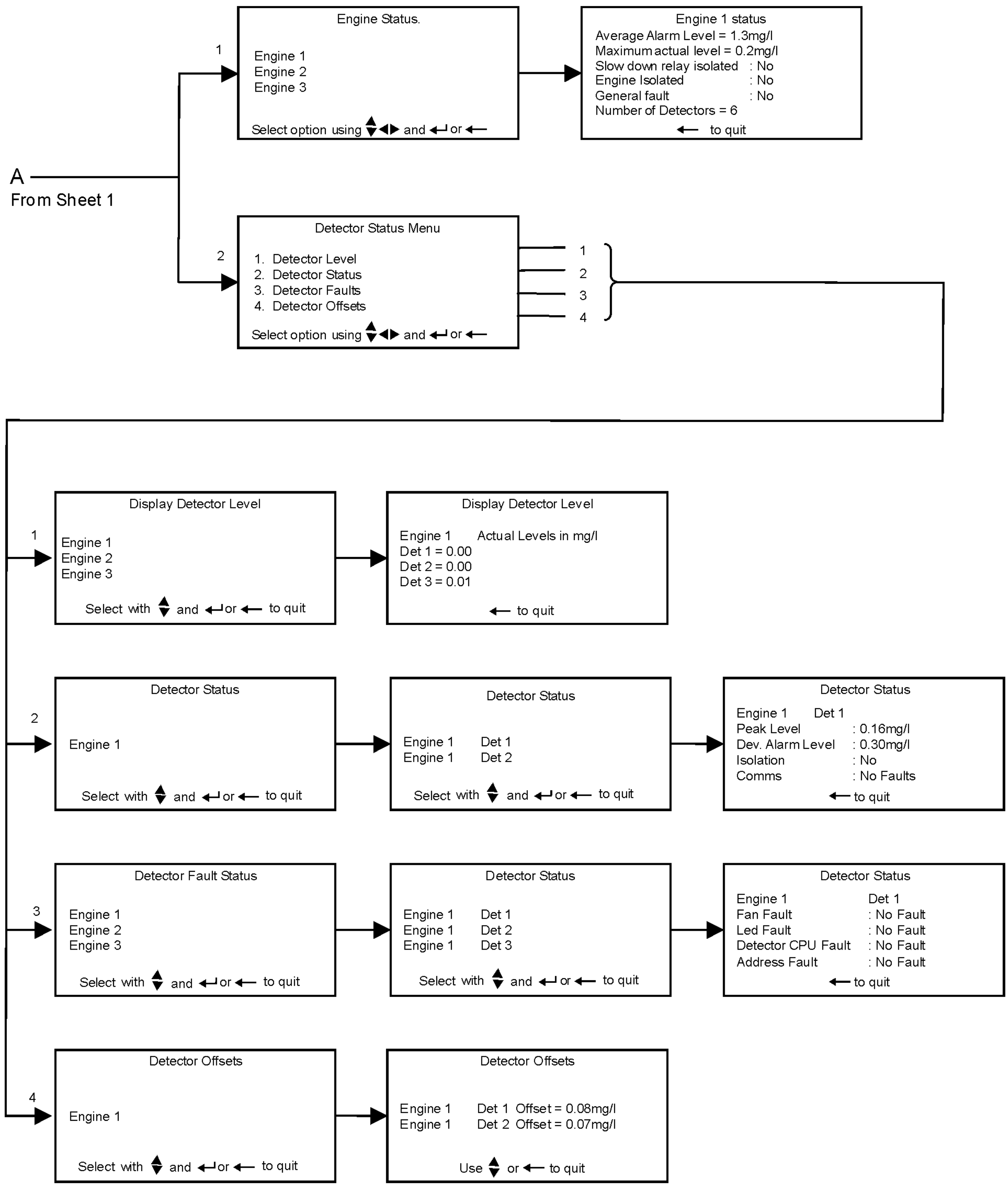
# Engineer Menu Sheet 6



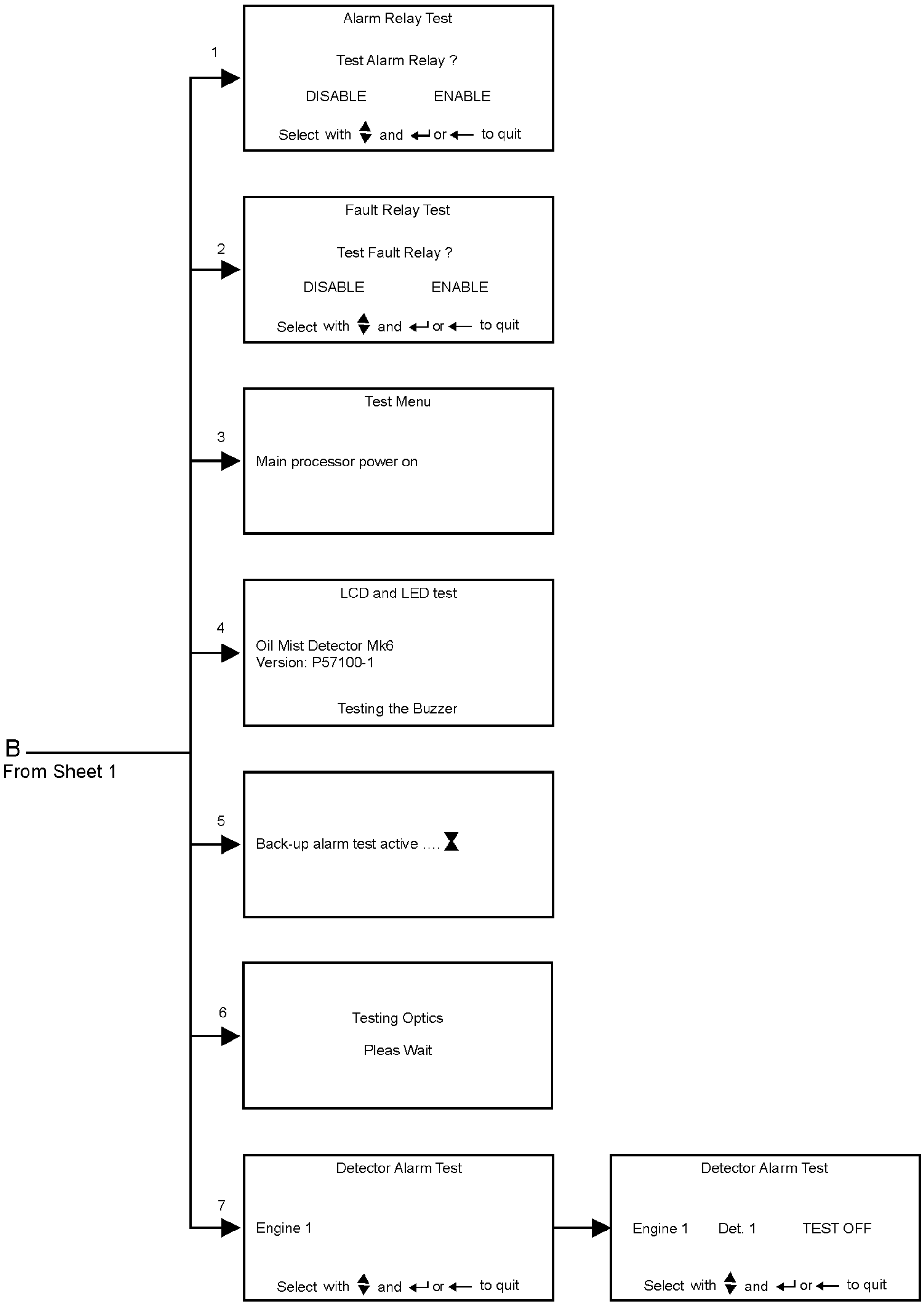
# User Menu Sheet 1



# User Menu Sheet 2



# User Menu Sheet 3



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# CHAPTER 3

# MAINTENANCE

## 3.1 ROUTINE MAINTENANCE

**Warning: Do not work on the system unless the power is switched off or isolated.**


**Caution: Ensure that anti - static handling procedures are observed where appropriate.**

The following checks are recommended to be carried out every 6 months, with the system switched off. The checks should be carried out by competent personnel with suitable skill levels.

### 3.1.1 Control Unit

1. Ensure that all glands are tight to prevent ingress of oil and moisture.
2. Check the sealing strip between the door and box is not damaged preventing a seal being made.

### 3.1.2 Junction Box

1. Ensure that all glands are tight to prevent ingress of oil and moisture.
2. Ensure that the lid fixing screws are tight, to prevent ingress of oil and water.

### 3.1.3 Cables

1. Ensure all connections in both the control unit and junction box(es) are tight.
2. Check all cables. Replace any that are found to be damaged.

### 3.1.4 Detectors

1. Ensure that the detector base is screwed tight into the crankcase.
2. Remove the cable connector from the detector and check for damage.

## 3.2 DETECTOR HEAD REPLACEMENT

**Warning: Do not remove the detector base from the crankcase whilst the engine is in operation. This operation should be carried out while the engine is stopped to avoid the possibility of hot oil coming out of the base fixing hole.**

If an in-service detector head is removed for any reason, the detector optics must be cleaned before reassembling and replacing it.

### 3.2.1 To replace the detector:

1. Switch off the system (if safe to do so) or isolate the detector.
2. Remove the cable connector fitted to the top of the detector.
3. Using a 4mm hexangonal key, loosen the two fixing screws in the assembly base.


Figure 33: Base Fixing Screw Removal

4. Lift out the detector head and note its address.
5. Set the address on the new detector head in accordance with para 2.1.1. The selected address must be the same as that of the detector head that has been removed.
6. Fit the detector head onto its base and tighten up the fixing screws. Affix the new switch window label.
7. Re-fit the cable to the detector head.
8. If the system was switched off, switch back on and allow the system to initialise.
9. If isolated, then de-isolate, return to main display by pressing the **MAIN DISPLAY** keypad and press **RESET**. Allow the system to initialise.

3.3 DETECTOR HEAD REFURBISHMENT

1. Isolate the detector as described in paragraph 2.4.3 of the instruction manual. If all detectors on one engine are to be serviced, then isolate the engine as described in paragraph 2.4.3 of the instruction manual.
2. Disconnect the cable from the top of the detector unit and remove the unit from the engine casing (refer to Figure 34).


Figure 34: Cable Disconnect

3. Using a 4mm hexagon key, unscrew 2 off screws from the underside of the detector head (refer to Figure 35). The screws are self-retaining.

4. Remove and invert the top part of the detector head. Examine the base moulding seal and replace if damaged or perished (refer to Figure 36).


Figure 35: Screw Removal

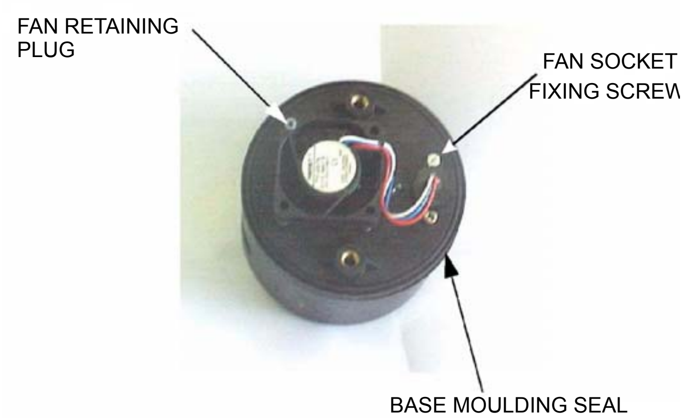



Figure 36: Base Moulding Seal

**Caution: Do not press the fan, only handle the outer housing.**

5. Using the Pulling Tool (refer to Figure 37), remove the Fan Retaining Plug by capturing the shoulder and pulling. Carefully remove the Fan from its mountings (refer to Figure 38).

Figure 37: Pulling Tool

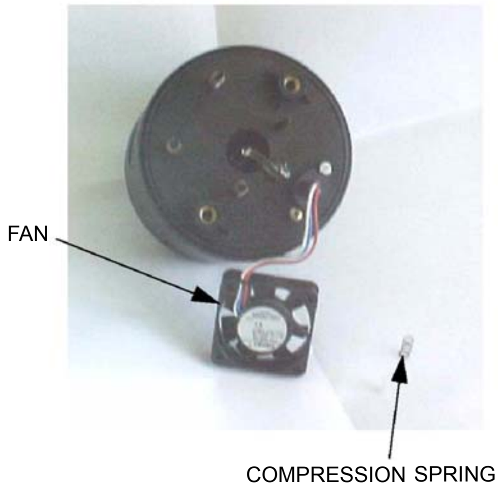



Figure 38: Fan Removal

6. Examine the 4 off compression springs and the fan retainer plug; replace any bent or damaged items from the spares.

7. Examine the fan for free running and clogging due to oil residues. If damaged, refer to paragraph 3.4 in the instruction manual. Although not necessary for service cleaning, spares of the M3 screw and the fan connector seal are included in the kit.

8. Using a foam bud with glass cleaner applied, wipe carefully around the inside of the smoke detecting orifice and the end of the light guide (refer to Figure 39).


Figure 39: Cleaning Light Guide

9. Use the Air Duster to blow away solution residue and dry the inside of the unit.

10. Examine the base body cavity and sampling tube, and wipe clean where necessary.

11. Reassemble the fan to the detector using a fan retaining plug.

**Caution: Do not press the fan, only handle the outer housing.**

12. Reassemble the detector head and base body. Replace the detector and then de-isolate.

13. Repeat the procedure for all detectors to be cleaned.

Refer to the Materials Safety Data Sheet in the event of health or safety issues.

3.4 FAN REPLACEMENT

**Warning: Do not remove the detector base from the crankcase whilst the engine is in operation. This operation should be carried out while the engine is stopped to avoid the possibility of hot oil coming out of the base fixing hole.**

3.4.1 To replace the fan:

1. Switch off the system (if safe to do so), or isolate the associated detector.
2. Remove the cable connector from the top of the detector.
3. Using a 4mm Allen key, loosen the detector fixing screws on the base.
4. Remove the detector from its base and turn it upside down to reveal the fan.
5. Remove the fixing screw holding the fan socket to the mounting plate.
6. Using a pulling tool (Part Number D9131-002 available with service kit Part No.D9221-027), remove the fan retaining plug holding the fan onto its mounting legs.
7. Lift the fan off of its mounting legs, ensuring the springs under the fan are not lost.

**Caution: Ensure a fan of the same make as that removed is fitted. If not, then a fan failure fault will occur. The correct fan can be selected by matching the detector serial number suffix (M or P) to the fan type. I.E. Papst or Micronel. See spare parts list.**

8. Discard the failed fan and fit a replacement fan in reverse order of disassembly.

3.5 CABLE REPLACEMENT

If changing a detector cable, isolating that detector will be sufficient (see section 2.4.3 as isolation)

3.5.1 Detector Cable

1. Isolate the detector with the damaged cable.
2. Remove the cable connector on top of the detector.
3. Remove the lid from the junction box and identify the cable to be removed.
4. Disconnect the wires in the junction box, undo the cable gland and pull the cable out of the junction box. Discard the damaged cable.
5. Push the new cable through the cable gland, ensuring the gland seal and lock nut have first been put onto the cable. Strip back the outer covering of the cable as required. Make-off ends ready to connect to the terminals in the junction box.
6. Connect the wires to the terminals and make-off the screen in the gland and then tighten to ensure a good seal.
7. Connect the cable connector to the detector.
8. When satisfied that all connections are satisfactory, re-fit the junction box lid.
9. De-isolate the detector, return to the main display and press the **RESET** switch and allow the system to initialise.

   Note: If the cable being replaced is the last in any zone, then the Pink and Brown wires must be left longer to reach the end of line terminals.

3.5.2 Junction Box Power Supply Cable Replacement

1. Switch off the system.
2. Open the door of the control unit and disconnect the damaged power supply cable. Undo the cable gland and remove the cable from the control unit.

3. Remove the lid from the relevant junction box and disconnect the wires and screen from the terminals inside. Undo the cable gland and remove the cable from the junction box.

4. Starting at the junction box, using the correct specification cable, pass it through the gland into the box. Make-off ends and connect the 2 cores and screen as indicated on the label in the junction box lid.

5. At the control unit, pass the cable through the gland and make-off the wire cable ends. Connect the two wires as indicated by the label inside of the box. Make-off the screen inside of the gland and then tighten the gland nut to ensure the gland is sealed.

6. When satisfied that all connections are correct, close and lock the door of the control unit and replace the lid of the junction box.

7. Switch the system on and allow to initialise.

#### 3.5.3 Communications Cable Replacement

1. Switch off the system.

2. Open the door of the control unit and identify the connections of the damaged cable and disconnect from the relative terminals. Undo the cable gland and remove the cable from the control unit.

3. Remove the lid from the relevant junction box and disconnect the wires from the terminals. Undo the gland nut and pull the cable out.

4. Push the correct specification cable through the gland of the junction box. Make-off the ends of the wires and connect to the relevant terminals. Tighten the gland nut.

5. Push the other end of the cable through the cable gland of the control unit. Make-off the ends and connect to the relevant terminals ensuring that the same wire colours have been used for the Comms positive, negative and also for the alarm back up. Make-off the screens inside of the gland and then tighten the gland nut.

6. When satisfied that all connections are satisfactory, close and lock the control unit door and replace the junction box lid.

7. Switch on the system and allow to initialise.

### 3.6 CONTROL UNIT PCBs

#### 3.6.1 Main Control Processor PCB - Part No. 44782-K071-02 (refer to Figure 40)

1. Switch off the system and replace the Main PCB as follows:

2. Remove all connectors, PLG1, PLG2, PLG4, PLG5, PLG8, PLG 9 and PLG10 noting the orientation.

3. Remove 8 off fixing screws. Remove the PCB.Remove EPROM from the failed PCB and fit to the new PCB. Remember to follow anti static precautions.

4. Fit the replacement PCB with the 8 off screws, re-fit all connectors.

5. Switch on system and allow to initialise.

Note: After changing the Processor PCB, it maybe necessary to adjust gain control R2 to ensure the clarity of the LCD.

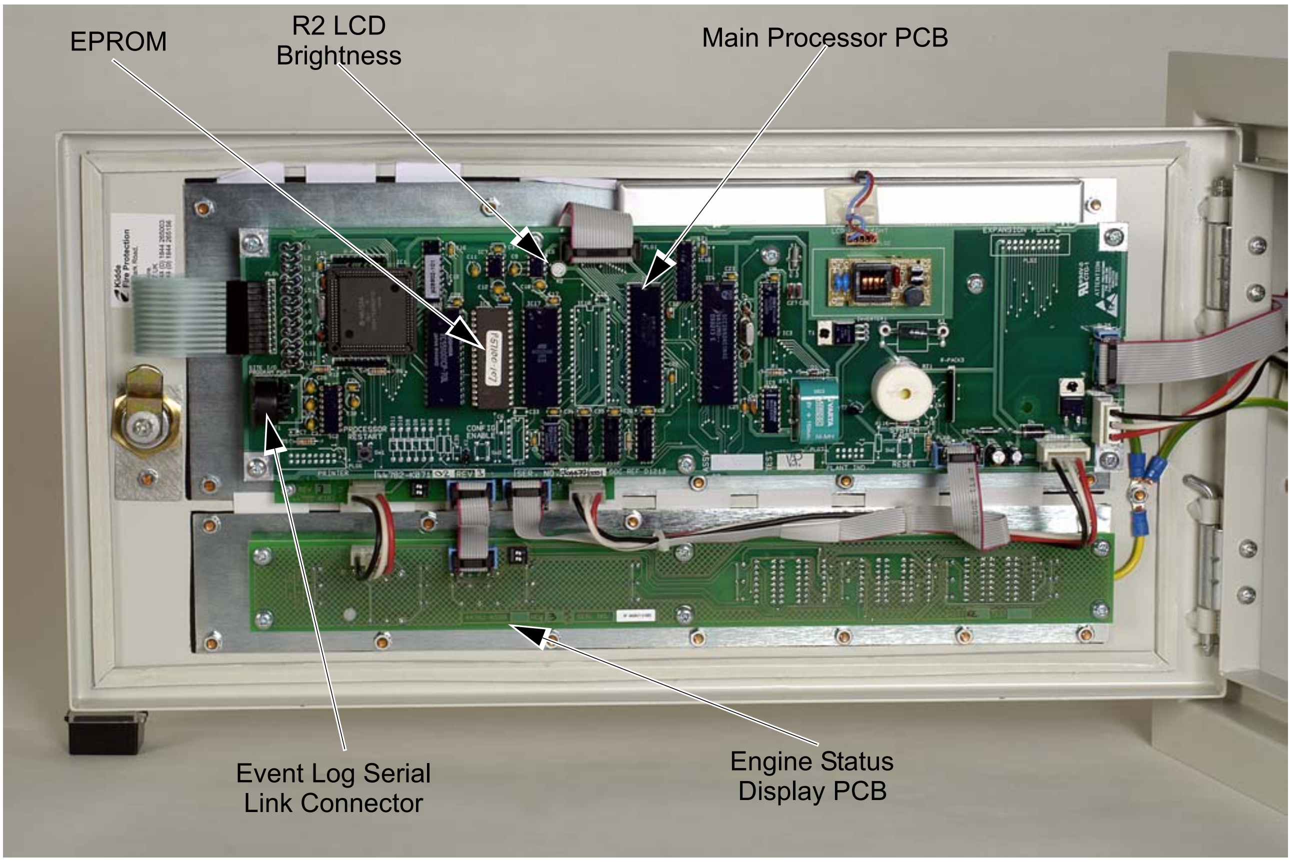



Figure 40: Control Unit PCBs

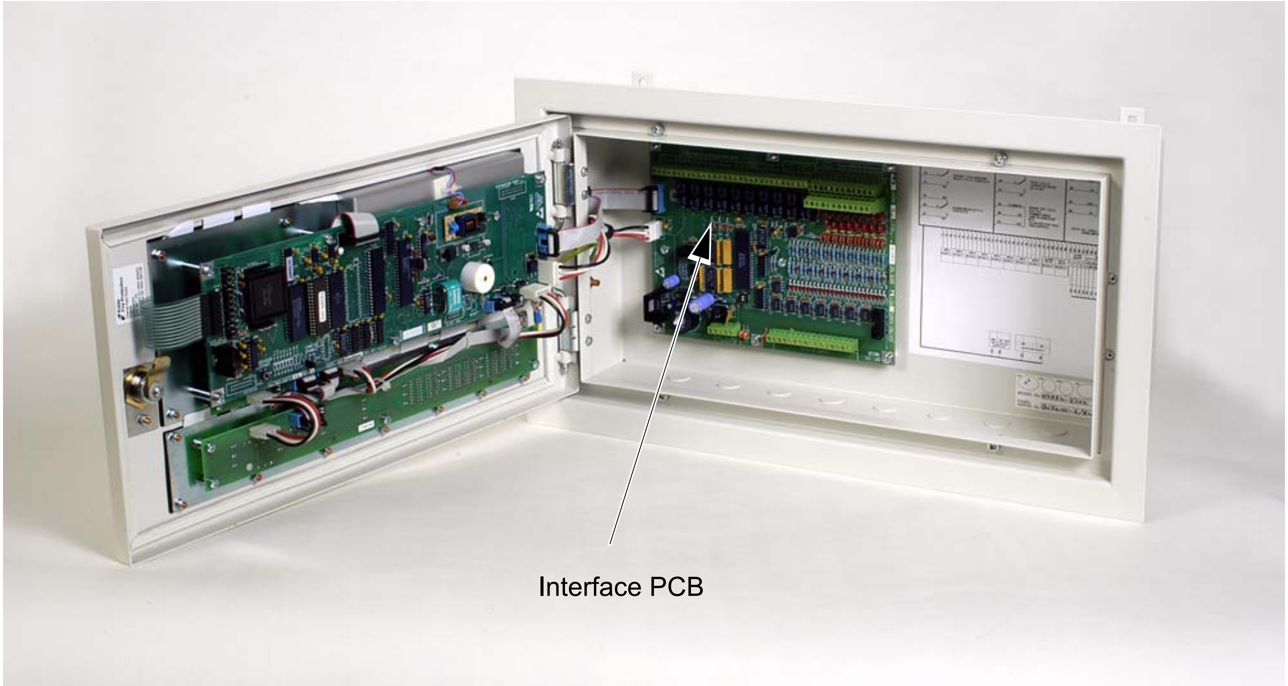



Figure 41: Control Unit PCBs

3.6.2 8 Engine Status Display PCB Replacement - Part No. 44782-K098 (refer to Figure 44)

1. Switch off the system and remove the 8 Engine Status Display PCB as follows:
2. Remove the two connectors located on the top left hand side of the board, noting the orientation.
3. Remove 6 off fixing screws and remove the PCB.
4. Fit the new PCB with the 6 off screws and replace the two connectors.
5. Switch the system on and allow to initialise.

3.6.3 Interface PCB Replacement - Part No. 44782-K085 (refer to Figure 42)

1. Switch off the system and replace the PCB as follows:
2. Remove and identify all external wiring.
3. Remove connectors PLG1 and PLG2 - noting the orientation.
4. Remove 8 off retaining screws and remove the board.
5. Fit the replacement board in place with the 8 off screws.
6. Re-fit connectors PLG1, PLG2 and all external wiring.
7. Switch the system back on and allow to initialise.

3.6.4 Status Display PCB - Part No. 44782-K102-01 (refer to Figure 45)

1. Switch off the system and remove the Status Display PCB as follows:
2. Remove the Main Control Processor PCB. (Refer to para 3.6.2)
3. Remove the connectors from the Status Display PCB, noting the orientation.
4. Remove the 4 off fixing screws and remove the PCB.
5. Fit the replacement PCB with the 4 off screws and re-fit the connectors.
6. Re-fit the Main control processor PCB.
7. Switch on the system and allow to initialise.

3.6.5 Junction Box PCB Replacement - Part No. 44782-K117 (refer to Figure 46)

1. Switch off the system and remove the Junction Box PCB as follows:
2. Remove the lid from the junction box.
3. Disconnect the detector, power supply and comms cables.
4. Remove the 4 off screws securing the board in the box and remove the board, noting the position of detector 1.
5. Fit the new board ensuring that detector 1 is in the same position on the new board as the removed board and secure with the 4 off screws.
6. Reconnect the detector, power supply and comms cables and replace the lid.
7. Switch the system on and allow to initialise.

## 3.7 REPLACEMENT OF 8 ENGINE STATUS DISPLAY MEMBRANE

3.7.1 Replace the 8 Engine Status Display Membrane as follows:

1. Remove the 8 Engine Status Display PCB as above. (Refer to 3.6.3)
2. Remove the seven M4 nuts and washers holding the membrane to the door unit.
3. Remove the membrane.

4. Fit the replacement 8 Engine Status Display membrane in reverse order of disassembly.

## 3.8 REPLACEMENT OF LCD DISPLAY

3.8.1 Replace the LCD Display as follows:

1. Remove the Main Control Processor PCB as in 3.6.2.
2. Undo the 4 off M3 screws in the LCD cover.
3. Gently lift up the cover and unplug the LCD connection cable and feed through the slot in the end of the cover.
4. Undo and remove the 2 off studs on the left of the display and loosen the 2 off studs on the right of the display and then slide the LCD assembly out.
5. Refit in the reverse order of disassembly. Ensure the white wires from the LCD display that connect into PLG2 are located correctly in the top of the cover. Ensure the wires are not trapped when the cover is replaced.

## 3.9 REPLACEMENT OF LCD DISPLAY MEMBRANE

3.9.1 Replace the LCD Display Membrane as follows:

1. Remove the LCD Display as above (Refer to 3.8)
2. Remove the membrane.
3. Refit the replacement LCD Display membrane in the reverse order of disassembly.

## 3.10 DECOMMISSIONING

All the components of the Graviner MK 6 OMD system may be disposed of as electrical/electronic equipment waste. i.e. using waste disposal methods in accordance with current local waste disposal regulations.

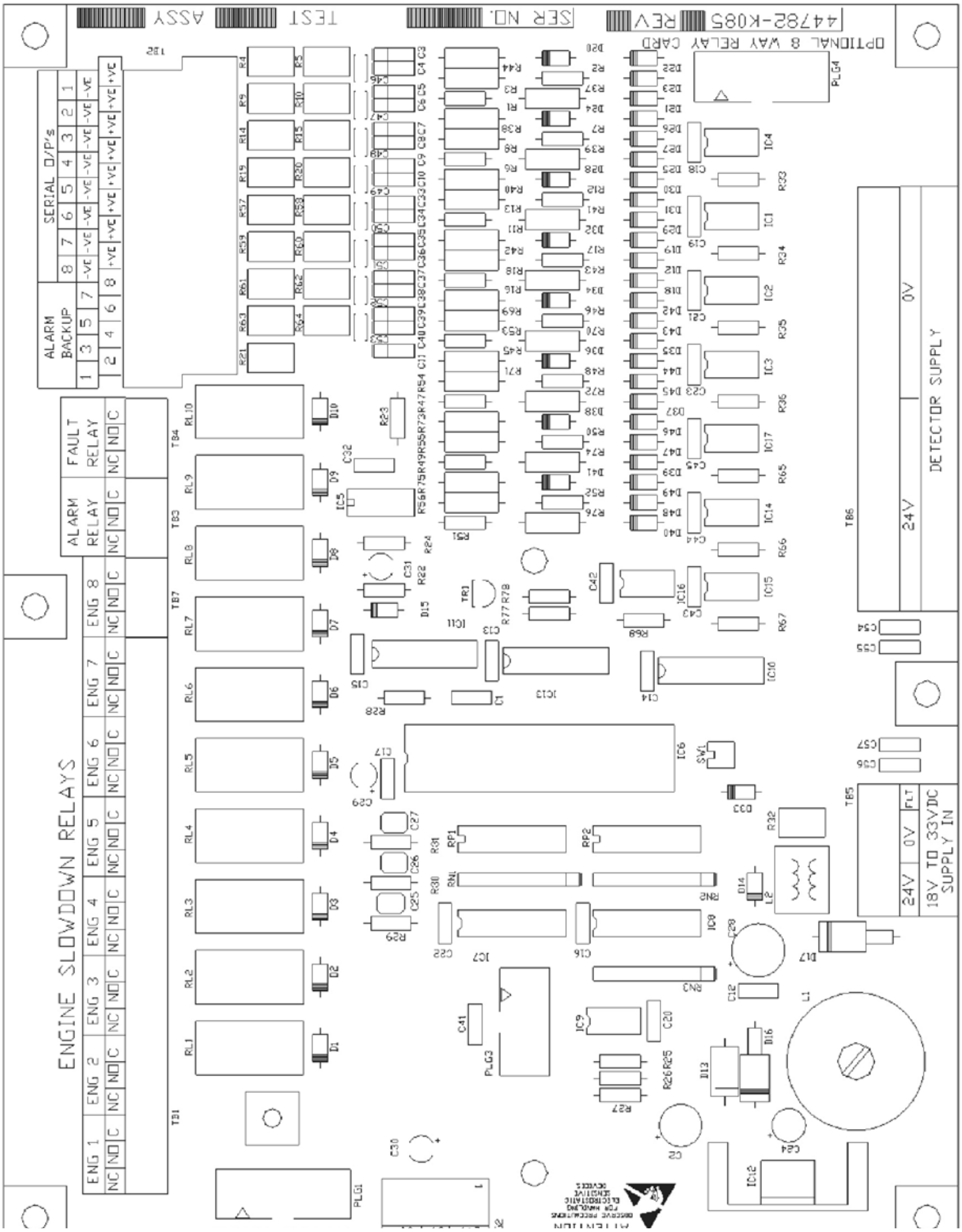



Figure 42: Interface PCB

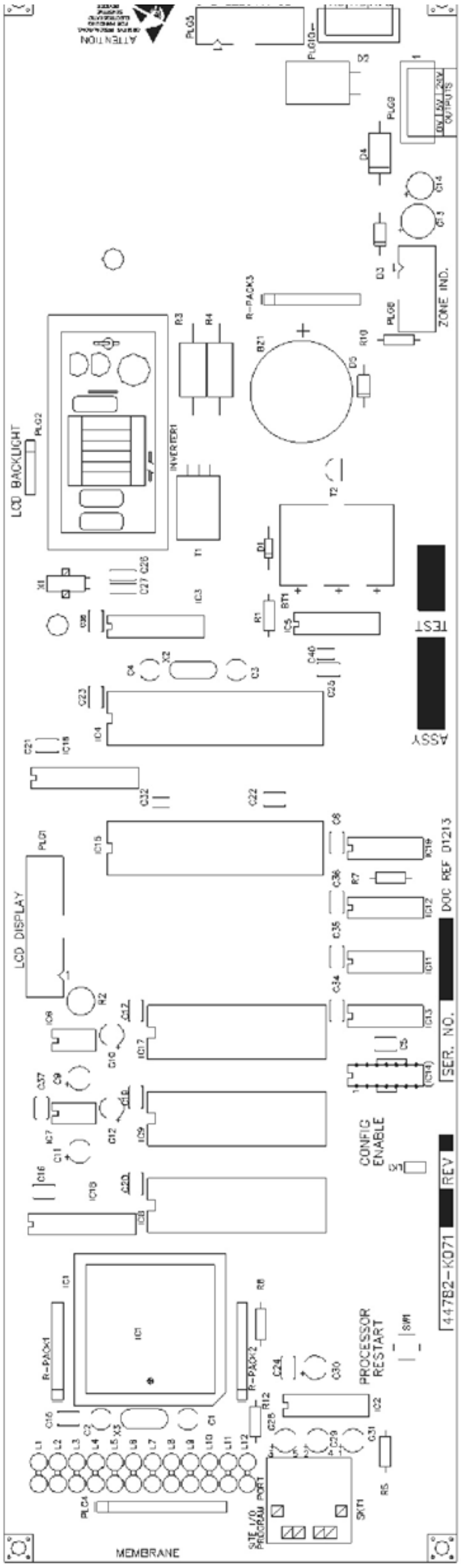


Figure 43: Main Control Processor PCB

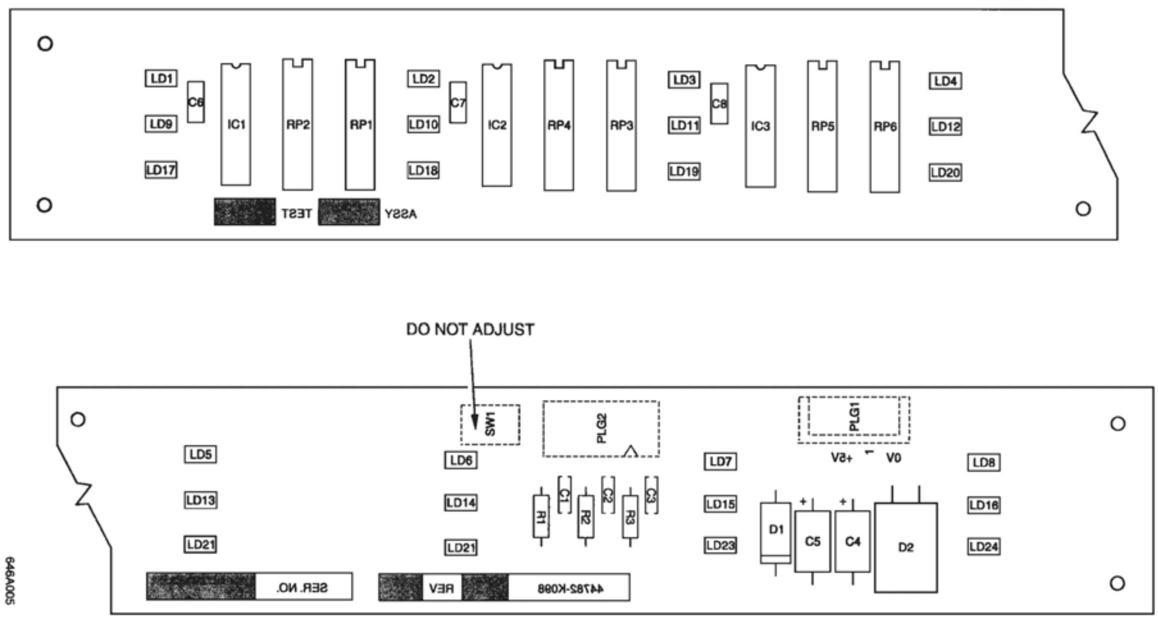



Figure 44: Eight Engine Status Display PCB

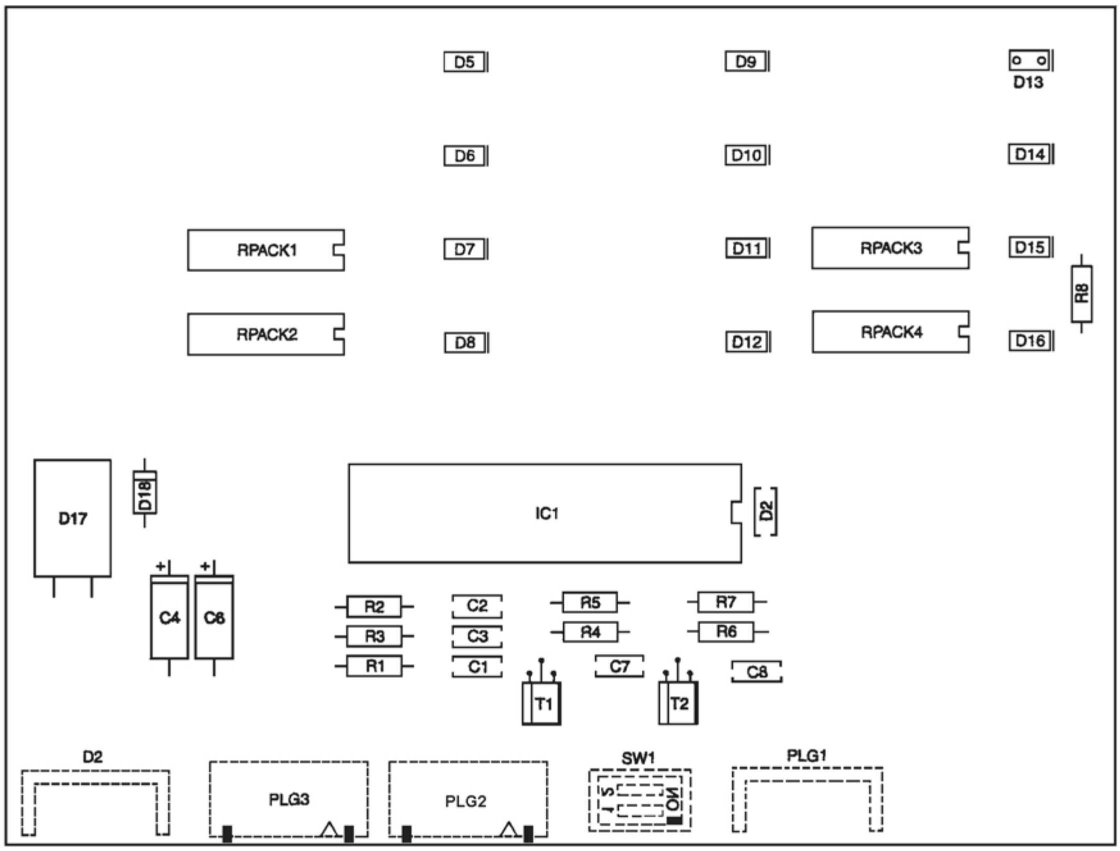



Figure 45: Status Display PCB

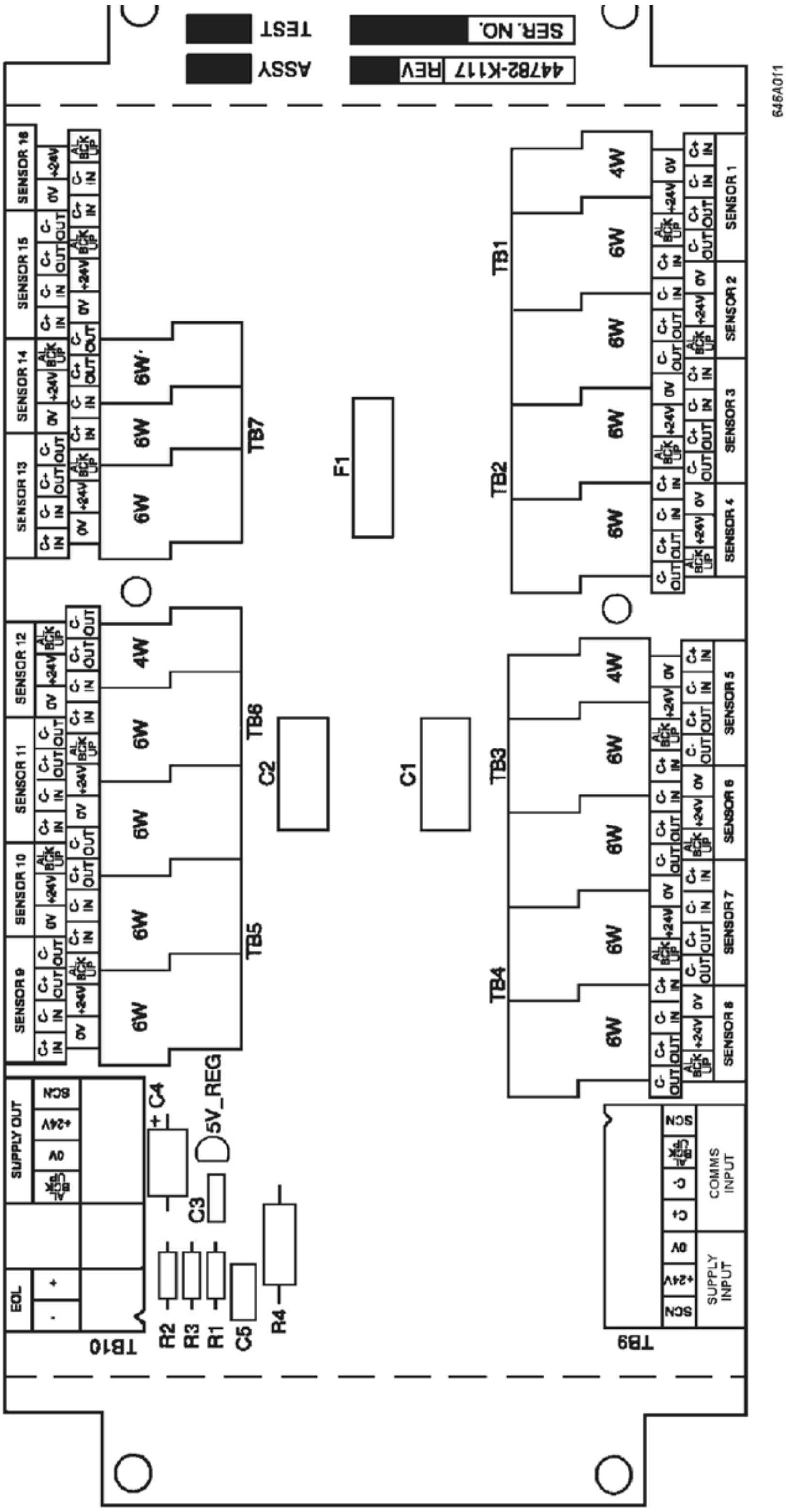



Figure 46: Junction Box PCB

Intentionally Blank

# CHAPTER 4

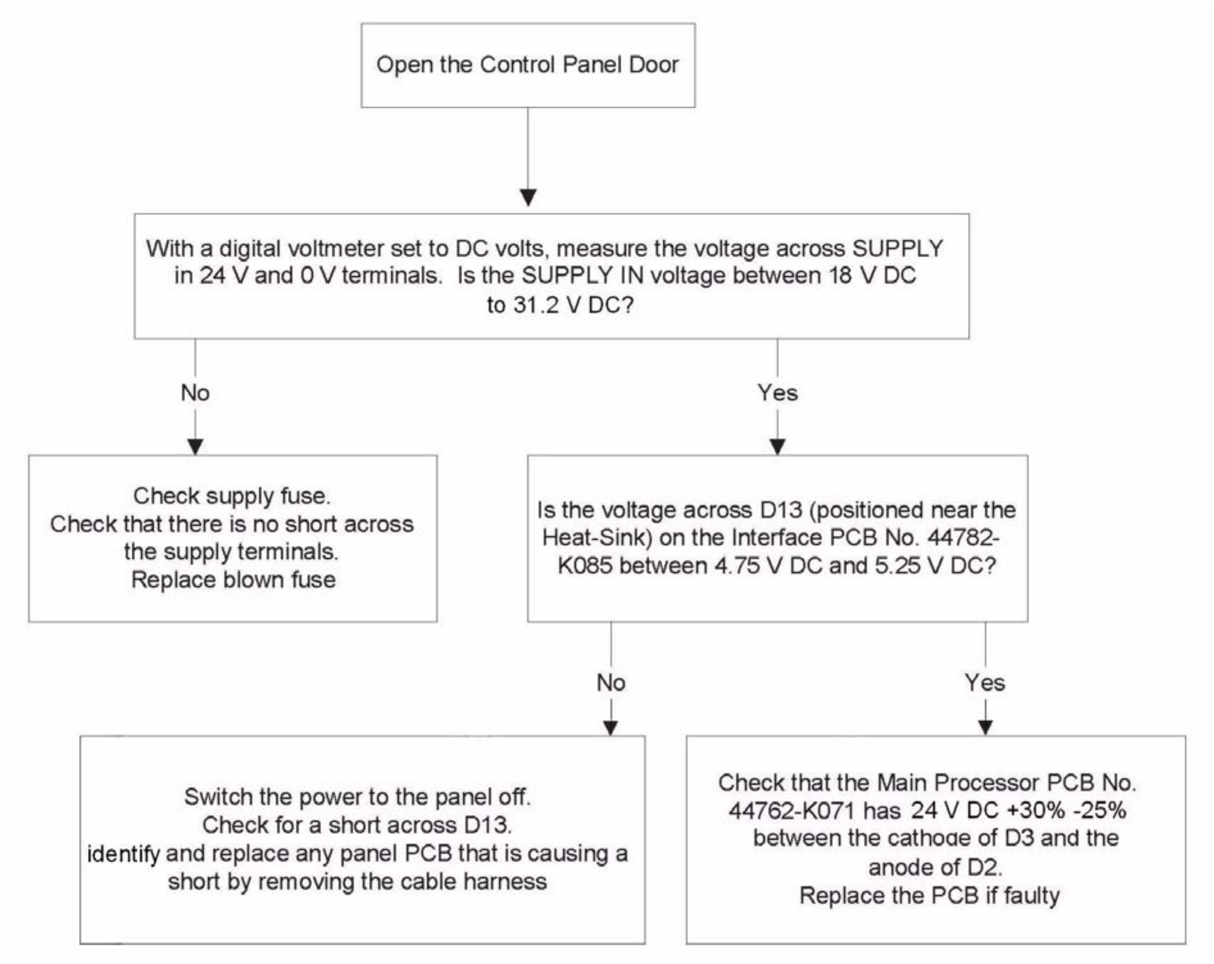
# FAULT FINDING

4.1 GENERAL

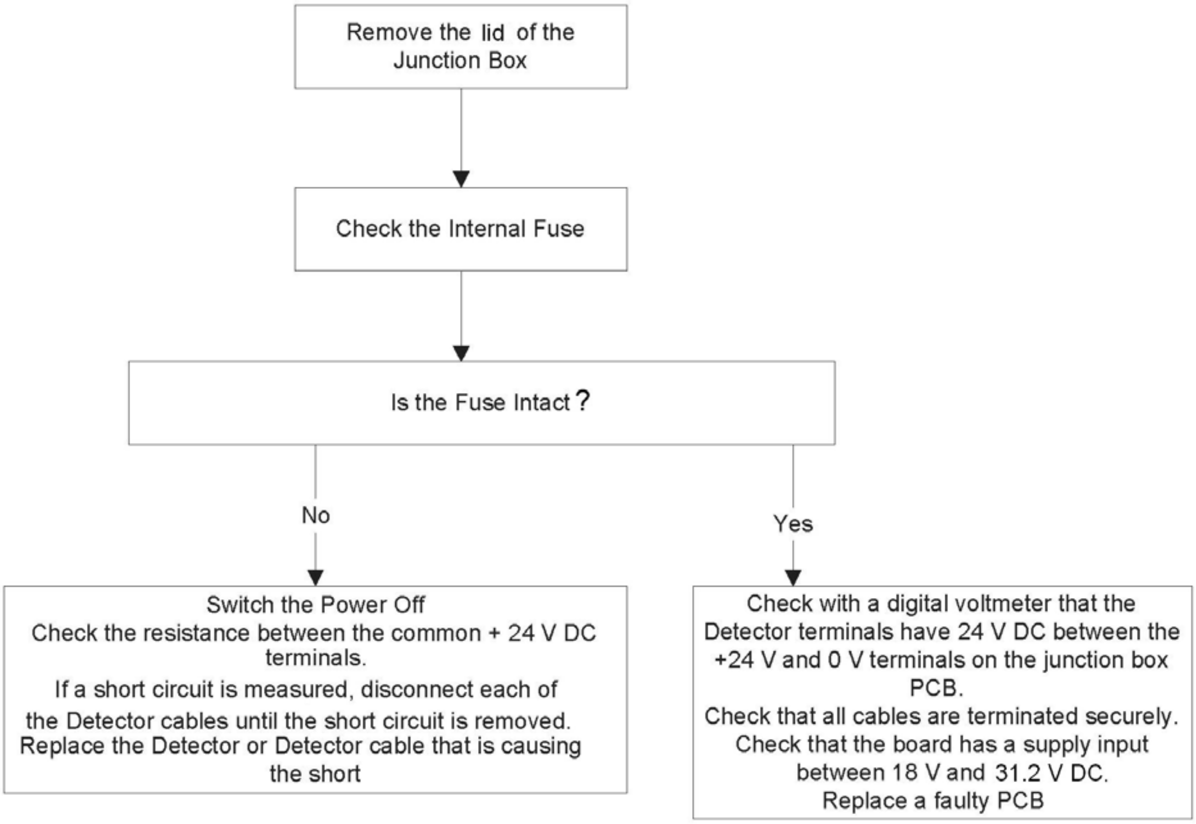
The table below lists a series of failure modes and the likely faults that would be indicated should that failure mode appear. Also listed are Actions, numbered 1 to 7, which should be followed if the associated fault appears. On the following pages, Actions 1 to 7 are shown as flow charts which will assist with fault finding on the Graviner Mk.6 OMD system.

| Failure Mode | Fault | Action |
|---|---|---|
| Control Unit power indicator is off and the display is blank | Supply Failure. | 1 |
| The Detector Power On (Green) indicator(s) are OFF on one engine | Junction Box fuse.<br>Faulty Detector. | 2 |
| The display shows COMMS FAULT e.g. Engine 1 det 3 | Incorrect Detector address setting.<br>Missing Detector Supply.<br>Faulty Detector. | 3 |
| The display shows FAN FAULT e.g. Engine 2 det 3 | Fan Failure. | 4 |
| The display shows LED Fault Engine 3 Det 1 | Detector circular oil mist cavity needs to be cleaned.<br>Failed LED. | 5 |
| The display shows Detector Fault e.g. Engine 4 det 2 | Blocked Detector aperture.<br>Damaged Detector light-guide.<br>Faulty Detector | 6 |
| False Deviation Alarm | Incorrect deviation alarm setting | 7 |

ACTION 1


Open the Control Panel Door
With a digital voltmeter set to DC volts, measure the voltage across SUPPLY in 24 V and 0 V terminals. Is the SUPPLY IN voltage between 18 V DC to 31.2 V DC?
No
Yes
Check supply fuse. Check that there is no short across the supply terminals. Replace blown fuse
Is the voltage across D13 (positioned near the Heat-Sink) on the Interface PCB No. 44782-K085 between 4.75 V DC and 5.25 V DC?
No
Yes
Switch the power to the panel off. Check for a short across D13. identify and replace any panel PCB that is causing a short by removing the cable harness
Check that the Main Processor PCB No. 44762-K071 has 24 V DC +30% -25% between the cathode of D3 and the anode of D2. Replace the PCB if faulty

ACTION 2


Remove the lid of the Junction Box
Check the Internal Fuse
Is the Fuse Intact ?
No
Yes
Switch the Power Off
Check the resistance between the common + 24 V DC terminals.
If a short circuit is measured, disconnect each of the Detector cables until the short circuit is removed.
Replace the Detector or Detector cable that is causing the short
Check with a digital voltmeter that the Detector terminals have 24 V DC between the +24 V and 0 V terminals on the junction box PCB.
Check that all cables are terminated securely.
Check that the board has a supply input between 18 V and 31.2 V DC.
Replace a faulty PCB

ACTION 3

Check that the Detector has been set to the correct address, e.g. Detector 3 is set to 03.
If not, remove label and set to correct address. Fit new label.
Is the GREEN Detector indicator ON?

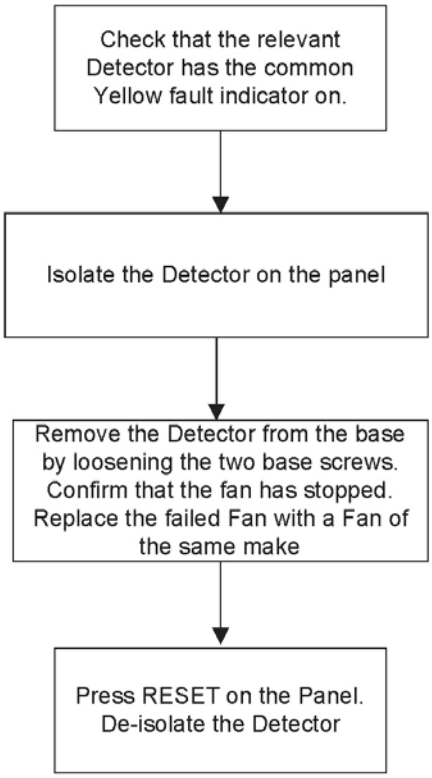
No

Remove the local Junction Box cover.
Check that the relevant Detector harness is securely and correctly terminated.
Measure with a DVM set to volts DC that the Detector has a supply of between 18 V to 32 V DC.
Replace a faulty Detector.

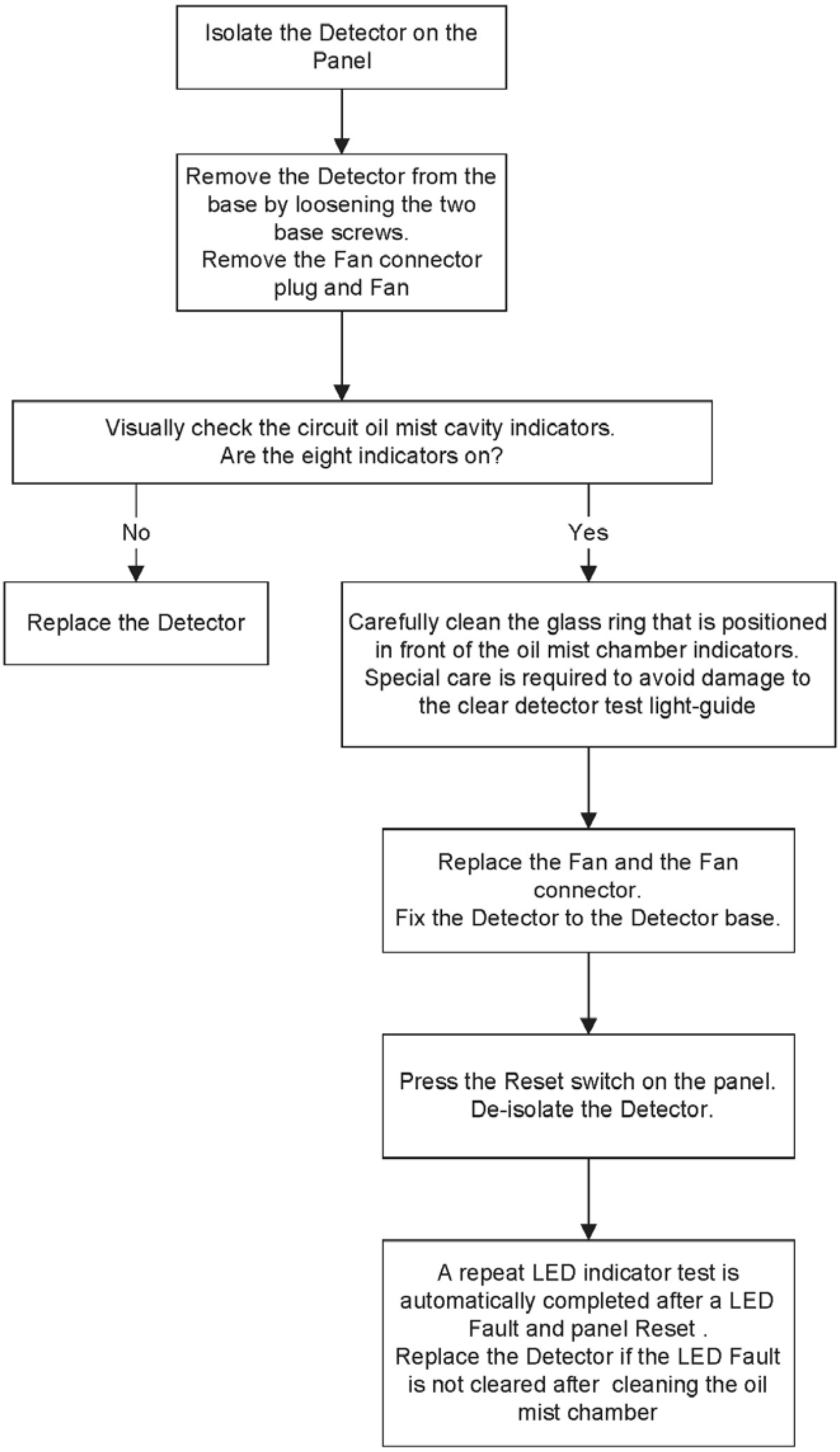
Yes

Remove the local Junction Box cover.
Check that the relevant Comms in and Comms out cable cores are terminated correctly.
Check that the last Detector comms out cable cores are terminated correctly in the End-of Line (EOL) terminals.
Replace a faulty Detector or cable harness.

ACTION 4


Check that the relevant Detector has the common Yellow fault indicator on.
Isolate the Detector on the panel
Remove the Detector from the base by loosening the two base screws. Confirm that the fan has stopped. Replace the failed Fan with a Fan of the same make
Press RESET on the Panel. De-isolate the Detector

ACTION 5


Isolate the Detector on the Panel
Remove the Detector from the base by loosening the two base screws.
Remove the Fan connector plug and Fan
Visually check the circuit oil mist cavity indicators.
Are the eight indicators on?
No
Yes
Replace the Detector
Carefully clean the glass ring that is positioned in front of the oil mist chamber indicators.
Special care is required to avoid damage to the clear detector test light-guide
Replace the Fan and the Fan connector.
Fix the Detector to the Detector base.
Press the Reset switch on the panel.
De-isolate the Detector.
A repeat LED indicator test is automatically completed after a LED Fault and panel Reset .
Replace the Detector if the LED Fault is not cleared after cleaning the oil mist chamber

ACTION 6

Isolate the Detector

Remove the Detector from the base by loosening the two base screws.
Remove the Fan connector and Fan

Visually check the small aperture hole at the top of the circular oil mist cavity

Is the aperture blocked with condensed oil?

**No**

Check that the Detector light-guide is intact.

Carefully clean any condensed oil from the light guide.

Replace the Fan and the Fan connector.
Fix the Detector to the Detector base.
De-isolate the Detector and press Reset.

Replace the Detector, if the Detector Fault is not cleared after the automatic retest.

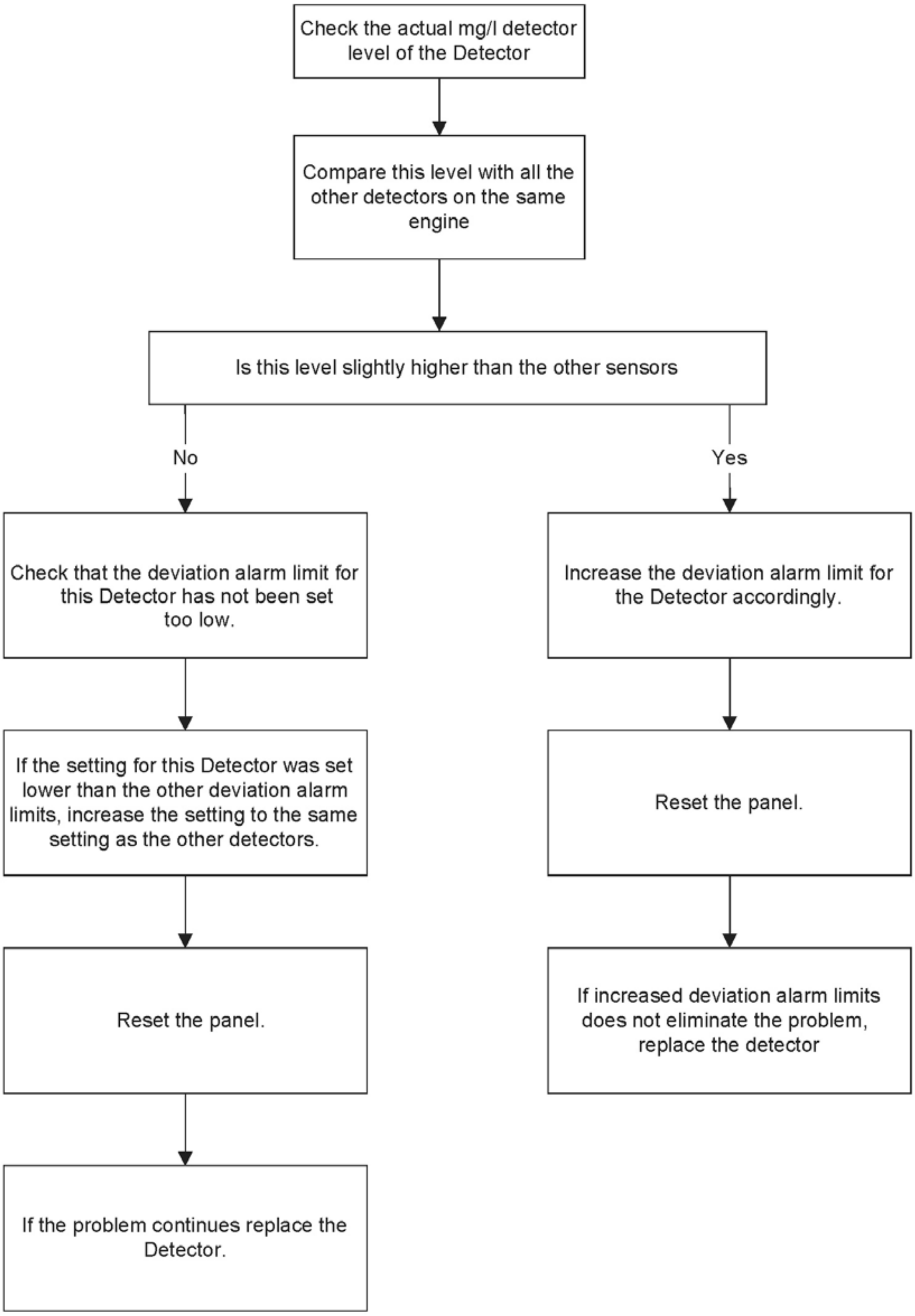
**Yes**

Carefully clean out the area above the aperture using material provided with the service kit.

Replace the Fan and Fan connector.
Fix the Detector to the Detector base.
De-isolate the Detector and press reset.

After pressing the Reset switch, another optics test is automatically completed.

ACTION 7


Check the actual mg/l detector level of the Detector
Compare this level with all the other detectors on the same engine
Is this level slightly higher than the other sensors
No
Check that the deviation alarm limit for this Detector has not been set too low.
If the setting for this Detector was set lower than the other deviation alarm limits, increase the setting to the same setting as the other detectors.
Reset the panel.
If the problem continues replace the Detector.
Yes
Increase the deviation alarm limit for the Detector accordingly.
Reset the panel.
If increased deviation alarm limits does not eliminate the problem, replace the detector

# CHAPTER 5

# SPARE PARTS

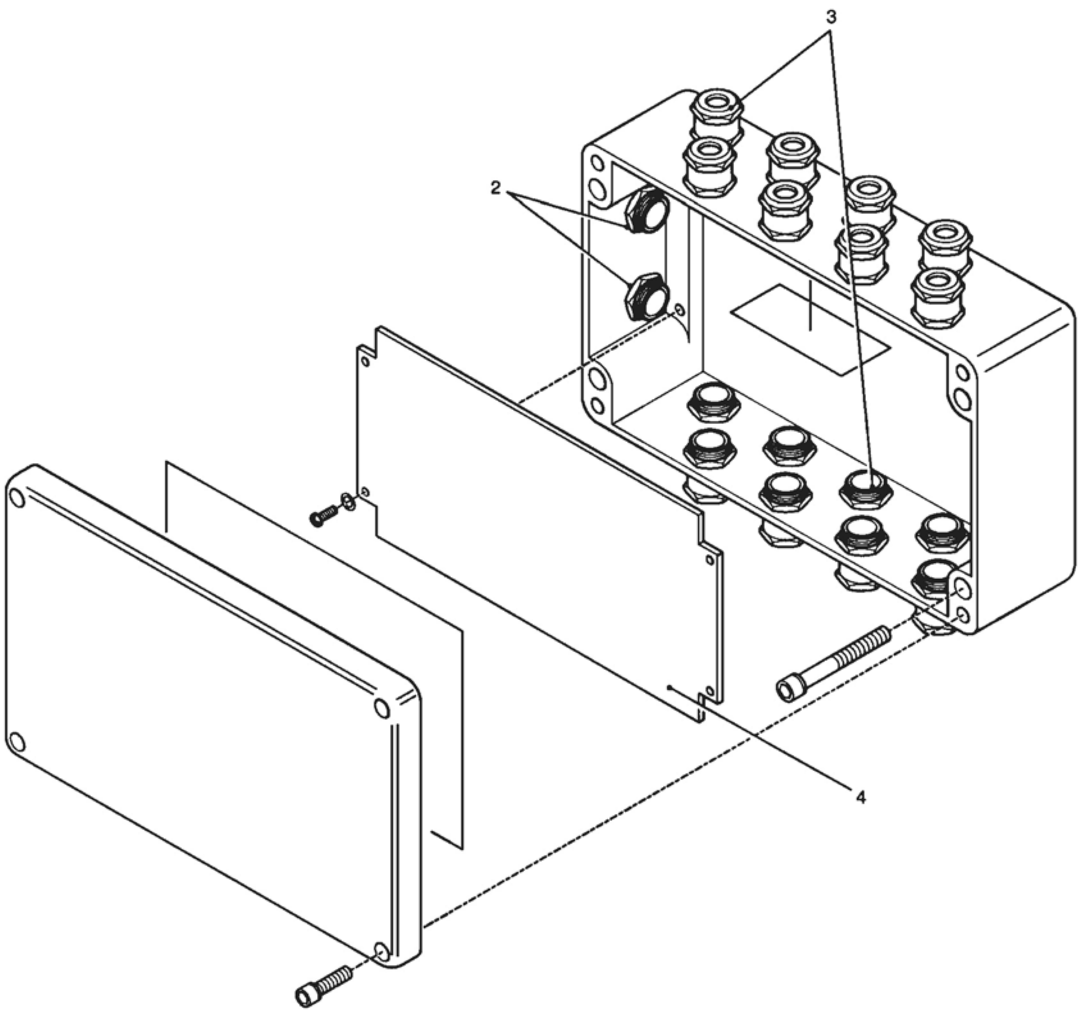



Figure 47: Junction Box - D4720-001-xx

| Item | Description | Part No. |
|---|---|---|
| | JUNCTION BOX | D4720-001-xx<br>xx = number of Glands |
| 2 | Nylon Gland M20<br>Nylon Gland M25 | B5151-013<br>21888-K042 |
| 3 | Metal Gland M20 | B5151-002 |
| 4 | Printed Circuit Board | 44782-K117 |
| | Fuse 4 amp, 20mm Slow Blow | 27411-K001 |

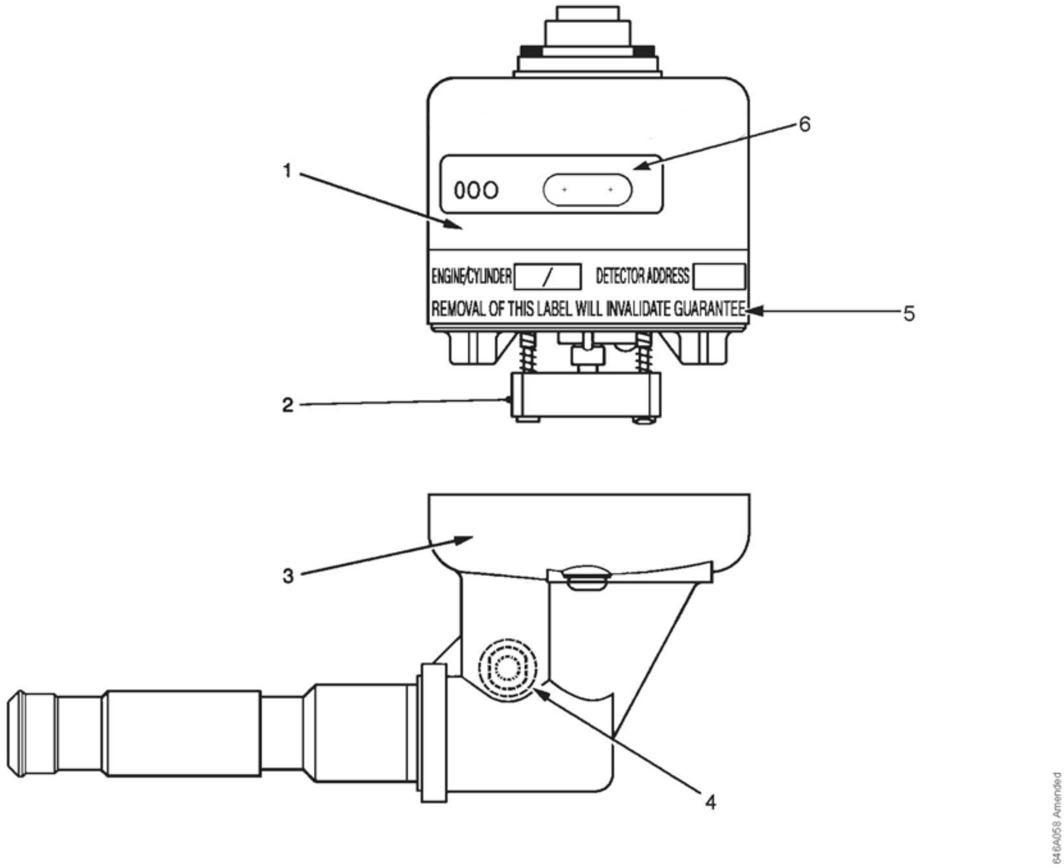



Figure 48: Detector Head E3561-301

| Item | Description | Part No. |
|---|---|---|
| 1 | Detector Head Assembly | D5622-001 |
| 2 | Fan Assembly (Papst) | D5622-005 |
| 2a<br>(alternative not shown) | Fan Assembly (Micronel) | D5622-005-02 |
| 3 | Base Unit Sub-Assembly | D5622-101 |
| 4 | Connector Push In | B5465-307 |
| 5 | Label Invalidate Guarantee | C9175-803 |
| 6 | Label Switch Window | C9189-801 |

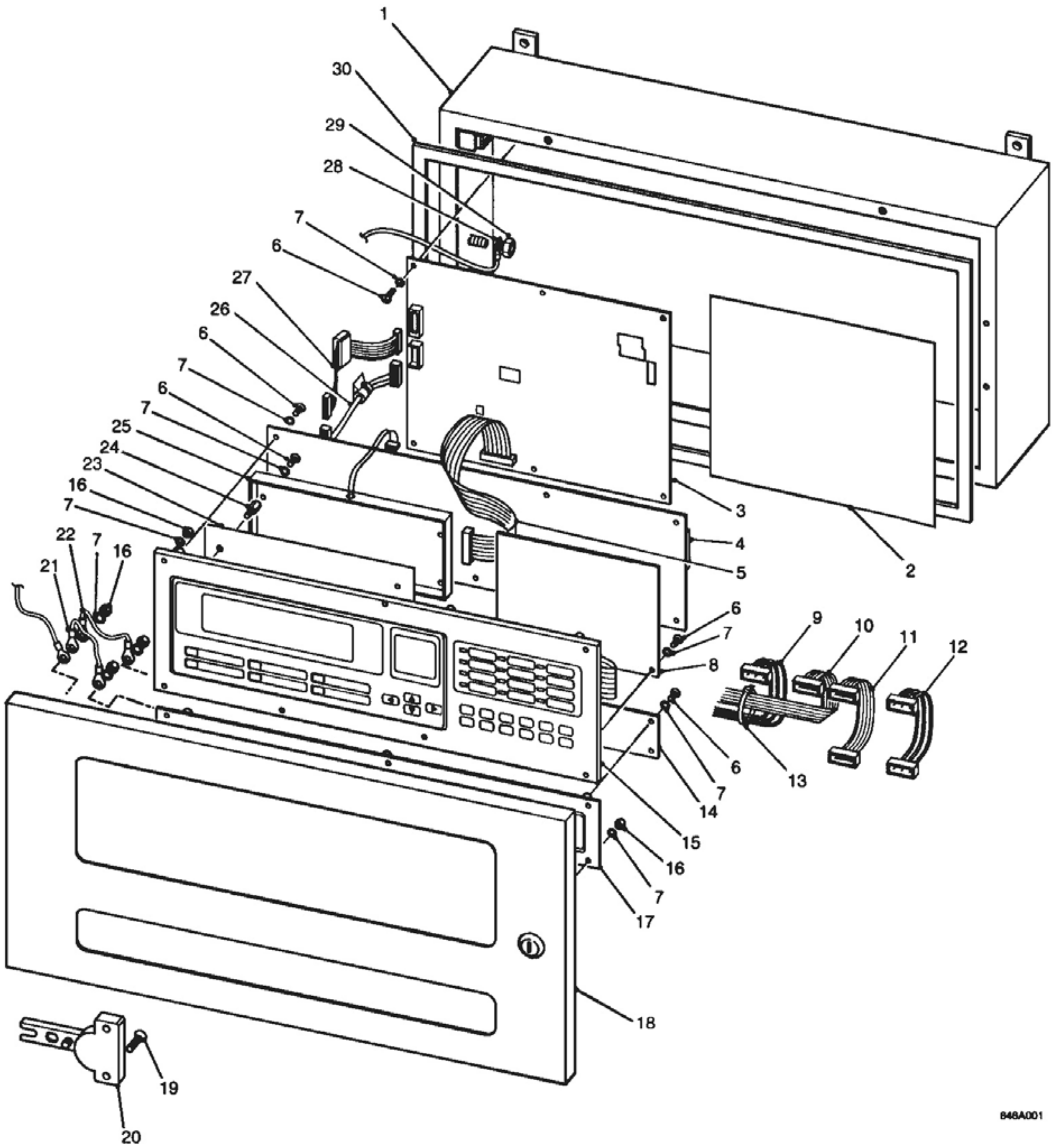



Figure 49: Graviner Mk 6 OMD Control Unit

| Item | Description | Part No. | Qty |
|---|---|---|---|
| 1 | Box | 35333-K053 | 1 |
| 2 | Label, Ext. Connections | 36215-K147 | 1 |
| 3 | PCB, OMD6 Interface | 44782-K085 | 1 |

| Item | Description | Part No. | Qty |
|---|---|---|---|
| 4 | PCB, Main Processor | 44782-K071-02 | 1 |
| 5 | Harness, MCP-LCD Display | 43682-K033 | 1 |
| 6 | Screw, M4 x 6 | 21883-D012 | 22 |
| 7 | Washer | 21177-164 | 36 |
| 8 | PCB Status Display | 44782-K102-01 | 1 |
| 9 | Harness, MCP-Status Display PSU | 43682-K067 | 1 |
| 10 | Harness, MCP-Status Display | 43682-K066 | 1 |
| 11 | Harness, MCP-Status | 43682-K030 | 1 |
| 12 | Harness, MCP-Status PSU | 43682-K036 | 1 |
| 13 | Tie Wrap | 22310-D001 | 2 |
| 14 | PCB, 8 Engine Display | 44782-K098 | 1 |
| 15 | Membrane Function Key | 39155-K040 | 1 |
| 16 | Nut, M4 | 21883-005 | 14 |
| 17 | Membrane Status Display | 39155-K039 | 1 |
| 18 | Door | 35300-K056 | 1 |
| 19 | Screw, M3.5 x 10 | 21833-D011 | 4 |
| 20 | Hinge | 27650-D009 | 2 |
| 21 | Earth Cable | 43682-K050 | 1 |
| 22 | Earth Cable | 43682-K050 | 1 |
| 23 | LCD Display | 43782-K120 | 1 |
| 24 | Spacer | 23700-K017 | 4 |
| 25 | LCD Cover | 15100-K091 | 1 |
| 26 | Harness, PCB-MBLC PSU | 43682-K037 | 1 |
| 27 | Harness, PCB-MBLC | 43682-K032 | 1 |
| 28 | Earth Cable | 43682-K015 | 1 |
| 29 | Nut, M6 | 21883-D014 | 1 |
| 30 | Seal | 13455-D033 | 1.5m |

| Spares Kit (Straight Cable Connector) D9221 024 consists of: | | |
|---|---|---|
| **Description** | **Part No.** | **Quantity** |
| Interface Board | 44782-K085 | 1 |
| Main Processor Board | 44782-K071-02 | 1 |
| Detector Head Assembly | D5622-001 | 1 |
| Switch Window Label | C9189-801 | 2 |
| Cable, 25 metres | 43682-K108-08 | 1 |

| Spares Kit (90 Degrees Cable Connector) D9221-025 consists of: | | |
|---|---|---|
| **Description** | **Part No.** | **Qty** |
| Interface Board | 44782-K085 | 1 |
| Main Processor Board | 44782-K071-02 | 1 |
| Detector Head Assembly | D5622-001 | 1 |
| Switch Window Label | C9189-801 | 2 |
| Cable, 25 metres | 43682-K109-08 | 1 |

| Commissioning Kit D9221-026 consists of: | | | |
|---|---|---|---|
| **Description** | **Part No.** | **Qty** | **Category** |
| Wipes, Wet & Dry | A7311-001 | 2 | Consumables |
| Smoke Test Oil - 30 ml | D9221-028 | 1 Bottle | Consumables |
| Wick - 150 mm | 17100-H06 | 3 | Consumables |
| | | | |
| Smoke Tester | D9221-029 | 1 | Tools |
| | | | |
| Materials Safety Data Sheet | - | 2 | Information |

| Service Kit D9221-027 | | | |
|---|---|---|---|
| **Description** | **Part No** | **Qty** | **Category** |
| Fan Retainer | B3741-902 | 5 | Spares |
| Compression Spring | B3721-006 | 5 | Spares |
| Base Moulding Seal | C1513-802 | 5 | Spares |
| Fan Connector Seal | C1413-801 | 5 | Spares |
| M3 Screw | 21833-H01 | 5 | Spares |
| | | | |
| Glass Cleaner 500ml | A7311-002 | 1 | Consumables |
| Air Duster | B6910-218 | 1 | Consumables |
| Foam Buds Pkts | B6910-217 | 2 | Consumables |
| | | | |
| 4mm Hexagon Key | B6910-219 | 2 | Tools |
| Pulling Tool | D9131-002 | 1 | Tools |
| | | | |
| Materials Safety Data Sheet | - | 2 | Information |

| Recommended Operational Spares | | |
|---|---|---|
| **Description** | **Part No.** | **Qty** |
| Spares Kit | D9221-024 or D9221-025 | 1 |
| Commissioning Kit | D9221-026 | 1 |
| Service Kit | D9221-027 | 1 |
| For systems with more than 14 detectors, it is recommended that additional detector head assemblies (D5622-001) are supplied. | | |

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